

**PLAYER BEHAVIOUR ANALYSIS AND
CHURN PREDICTION USING DATA
ANALYTICS AND MACHINE LEARNING: A
CASE STUDY INSPIRED BY HOYOVERSE
GAMES**

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ABSTRACT

This project presents a data-driven framework for analysing player behaviour and predicting churn, using Genshin Impact as a case study inspired by HoYoverse games. The study combines survey-based behavioural data with community-generated content collected from public platforms. Clustering techniques, including K-Means and Self-Organising Maps (SOM), are applied to identify different player segments, while classification models such as Logistic Regression and Random Forest are used to estimate churn likelihood. To improve transparency and understanding of the models, Explainable Artificial Intelligence (XAI) techniques, particularly SHAP-based analysis, are incorporated to explain how different behavioural factors influence churn predictions, with the aim of supporting clear and interpretable insights for player retention strategies in *Genshin Impact* (a free-to-play game).

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LIST OF ABBREVIATIONS

<i>AUC</i>	Area Under the ROC Curve
<i>CRISP – DM</i>	Cross-Industry Standard Process for Data Mining
<i>DT</i>	Decision Tree
<i>EDA</i>	Exploratory Data Analysis
<i>F2P</i>	Free-to-Play
<i>FYP</i>	Final Year Project
<i>FYP I</i>	Final Year Project I
<i>FYP II</i>	Final Year Project II
<i>LR</i>	Logistic Regression
<i>ML</i>	Machine Learning
<i>PCA</i>	Principal Component Analysis
<i>RF</i>	Random Forest
<i>SDGs</i>	Sustainable Development Goals
<i>SOM</i>	Self-Organizing Maps
<i>SHAP</i>	Shapley Additive Explanations
<i>SVM</i>	Support Vector Machine
<i>TF – IDF</i>	Term Frequency–Inverse Document Frequency
<i>XAI</i>	Explainable Artificial Intelligence
<i>XGBoost</i>	Extreme Gradient Boosting
<i>UGC</i>	User-Generated Content

CHAPTER 1

INTRODUCTION

1.1 Background

1.1.1 Game Industry Overview

The video games industry has undergone remarkable expansion over the last decade. In 2018, the video games industry's global market size was valued at over USD 150 billion, making it become one of the most lucrative segments of the entertainment and digital media sector (Roy, 2018). By 2022, the global games market had already exceeded USD 180 billion, with supported by a global player base of more than 3 billion (Newzoo, 2025). According to Newzoo's 2025 Global Games Market Report, the game industry is projected to generate approximately USD 360 billion in revenue, with the global player population also expected to reach 3.6 billion. This steady upward trend shows how important the gaming industry is to the economy and society.

However, alongside the growth of the player base, a critical challenge has arisen: player churn. Player churn refers to users discontinuing gameplay. Players may leave the game temporarily, which reduces play time. They may also exit permanently, which means they stop playing the game entirely (Viljanen et al., 2017; Xiong et al., 2023). As Roy (2018) notes, customer defection is unavoidable in freemium models. Sustaining retention is essential for business viability. This is particularly important because attracting new players to the game costs more. Likewise, Mulla et al. (2025) noted that churn is expected to impacts revenue streams. Moreover, churn also disrupts the community dynamics. Finally, it will further affecting the long-term sustainability of online games. Therefore, improving player retention has become a key concern for

developers, especially those operating within the free-to-play (F2P) model games, where long-term engagement is directly tied to their revenue generation.

1.2 Problem Statements

The rapid evolution of the gaming industry has revealed fundamental limitations in traditional approaches to analysing player behaviour. Earlier studies on mobile and online games commonly relied on survey-based methods, manual analysis, and basic segmentation techniques. While these approaches provided useful descriptive insights, they are limited in their ability to capture dynamic engagement patterns, scale effectively with large and complex datasets, or establish a direct connection with churn prediction.

These limitations were already evident in pre-2020 game models, such as MMORPGs and level-based progression games. However, the emergence of large-scale open-world titles after 2020 has further intensified these challenges. *Genshin Impact* represents a notable example, offering unprecedented freedom, cross-platform accessibility, and highly complex player interaction patterns. The resulting volume and diversity of player data exceed the analytical capacity of traditional methods, highlighting the urgent need for more advanced, data-driven approaches to player behaviour analysis and churn prediction.

First, surveys are widely used in gaming research to gather feedback and understand player experiences. However, surveys have the limitations in capturing real-time behavioural patterns. Roy (2018) points out that survey-based models often fail to reflect dynamic engagement factors. Such engagement characteristics are indispensable in regard to player retention and churn. It is essential to have a more data-driven solution for the limitation of behavioural patterns so that the hidden behaviour patterns may be displayed and thus knowledge relating to predicting churn may be strengthened considerably.

Second, manual analysis can provide patterns or relationship at a small scale of data. It will soon become inefficient and unsustainable when there is a large and

complex dataset. The exponential growth of gaming data today requires advanced methods to process and interpret information. According to Mulla et al. (2025) machine learning techniques can automate churn detection. These can also highlight complex patterns across various player characteristics. Offering a more efficient and accurate solution than traditional manual analysis.

Lastly, there is a need for data-driven systems that can segment players and provide valuable strategies for improving player retention rate. Yu et al. (2020) shows that using Self-Organizing Maps (SOM) with k-means clustering will improve and increase the efficiency of market segmentation in online games. However, most existing studies treat segmentation and churn prediction separately. This creates a gap of need for developing an integrated system that connects behavioural analysis, player clustering, and churn prediction together. By doing so, it would allow researchers to produce more comprehensive insights and targeted recommendations for maintaining player engagement.

In summary, existing methods for understanding player behaviour and churn prediction have key limitations of their own. Highlighting that there is a need for an integrated framework that is data-driven and combines behavioural analysis, player clustering, and churn prediction.

1.2.1 Case of Hoyoverse

Hoyoverse (formerly miHoYo) has developed a wide range of games since it started in 2009, starting with Fly Me 2 the Moon, followed by the Honkai series, Tears of Themis, Genshin Impact, Honkai: Star Rail, Zenless Zone Zero, Whisper of the Stars, and the most recent releases such as Petit Planet and Honkai: Fate of Spirits. So far, the company has released 11 games, and since Honkai Impact 3rd, all have been built on a long-term service model that relies on continuous updates and sustained player engagement.

Even the success of Hoyoverse's service-driven strategy, player churn is still an unavoidable phenomenon in the games. Previous studies have proposed there are

plenty of churn prediction frameworks such as, clustering, machine learning, and explainable AI methods. However, these efforts have not been specifically applied to Hoyoverse games, which having their own ways to catch player interested likes, exploration-driven gameplay, gacha monetisation, and event-based progression.

This study addresses this gap by focusing on Genshin Impact, a flagship online free-to-play game launched at 2020. Within this study, a data-driven system will be developed to identifying patterns of players disengagement and supporting strategies to improve player retention. The proposed framework is further tailored to the game's unique exploration-based and event-driven dynamics.

1.3 Objectives

Based on the previously identified problem statements, this study aims to develop a hybrid framework that combines behaviour analysis, churn prediction, and player segmentation within a data-driven environment. Therefore, the objective of this research designed to address the limitation that discussed at the previous problem statement section. To ensure that, developed system are not only detect the churns, but going to explain the root of cause and player's behavioural patterns that lead to churn. Accordingly, this study is guided by the following specific objectives, ensure the research remains on track.

1. To Develop a Player Behaviour Analysis and Churn Prediction System using python and machine learning techniques, that can automatically find or identify any hidden patterns or relationships from the data, instead of manual analysis.
2. To Develop Data-Driven Player Segmentation and Churn Prediction Models using clustering and classification algorithms, as these methods uncover hidden trends and allow targeted retention strategies.
3. To Evaluating the accuracy and effectiveness of the system using standard evaluation metrics. To ensuring the results are measurable, reliable, and practically useful.

1.4 Significant of Project

Significance of study can be viewed through four aspects below, such as academic contribution, practical application, social and personal relevance, and alignment with sustainable development goals (SDGs).

Academic significance. This study covers important areas of research. The integration of behavioural questionnaire and player survey in the study enhances research, application of clustering methodologies to find player segmentation and application of explainable AI to enhance the interpretability of the model.

Practical significance. Findings of this project, should provide useful information to the game developers. It will support HoYoverse, to evolve better retention strategies and contents. The proposed framework can also be applied for others free-to-play game, where it will help to maintains sustainability and innovation of business in a long term.

Social and personal significance. The project shows responsible and ethical use of data analytics to improve the digital entertainment ecosystem. It is benefits to both players and the company. At the same time, this project provides the hands-on experience in advanced analytics, machine learning, and explainable AI, which helps build stronger research and professional skills for the author in future academic and career growth.

1.4.1 Alignment with SDGs

SDG 8: Decent Work and Economic Growth. Supporting digital economy growth by providing frameworks that help sustain player retention and revenue streams in the gaming sector.

SDG 9: Industry, Innovation, and Infrastructure. Through the integration of AI and analytics, the project promotes innovation and strengthens technological applications within the creative industry.

1.5 Project Scopes

The scope of this project can be divided into four categories. These categories are data scope, methodological scope, target users or population and, time scope. Each category having a detailed description which is provided below.

1.5.1 Data Scopes

The primary dataset for this study is obtained through a self-designed survey targeting players of Genshin Impact. The survey not only collects demographic information such as age, gender, and play duration but also includes in-depth questions related to in-game behaviours. These questions are design to capture aspects such as gameplay frequency, exploration progress, quest completion, co-op and social interactions, in-game spending patterns, gacha (pulling) behaviour, cross-game involvement with other games by the Hoyoverse, and players' motivations for continuing or discontinuing the game. These survey responses serve as behavioural proxies, enabling the approximation of actual in-game actions and decision-making processes.

In order to ensure the sufficiently representativeness and statistical reliability for modelling, this research will collect at least 150 answers from the active player side. Following the previous behavioural and data driven churn research (Roy, 2018; Yu et al., 2020; Mulla et al., 2025), which normally contains several hundred of participant for ensuring sufficient diversity for clustering and predictive modelling.

Besides, this survey not only focus on active players. It also includes potential churners and churned players, those who have temporarily stopped playing or already leaves the games. These groups provide crucial insights into the reasons behind discontinuation, as well as the factors that might encourage re-engagement or explain to us why they are leaving.

In addition to the survey, secondary data sources are incorporated to enrich the dataset and validate the findings. It will include player reviews and discussions collected from platforms such as HoYoLAB, Bilibili, YouTube, and Google Play to provide further insights to us or developer into player sentiment and retention issues.

1.5.2 Methodological Scope

This study applies a combination of data analysis and machine learning techniques to examine player behaviour and predict churn. The methodology consists of two components.

Clustering and Classification. Clustering techniques such as, K-means and Self-Organizing Maps (SOM) are applied to segment players. Based on their gameplay patterns, rate of frequency engagement, and retention characteristics. Classification Models of Logistic Regression and Random Forest, will be employed to predict churn probability and to find the most influential factors behind player discontinuation.

Explainable AI (XAI). Methods such as SHAP or feature importance analysis are applied to improve interpretability. It will enable and providing more clearer insights into why churn are happening.

1.5.3 Target Users/Population

This section specifies the population under study and the stakeholder who are benefit from the finding. Clearly identifying the target user and target respondents make sure that research outcome remains relevant and applicable in both academic and industry contexts.

Target population. The primary respondents of this study are Genshin Impact players, classified into three groups. First, active players who still remain active in the game. Second, churned players who have discontinued or temporarily stopped playing. Third, potential return players who may consider returning to the game. Through the previous segmentation, it allows that the dataset captures the full lifecycle of engagement and disengagement. As a result, this representation providing a balanced perspective for churn analysis.

Focus on active players. In addition to cross-group comparisons, this study places particular emphasis on analysing the active player segment. The survey design to collect detailed information on recent play frequency, daily playtime, exploration progress, social and co-op activities, spending and pulling behaviour, and motivational

factors for continued play. These dimensions enable the study to identify not only risk factors for churn but also the key drivers of long-term retention among players who remain engaged.

Intended users. The insights derived from this study are intended for multiple stakeholders, including game developers and analysts at HoYoverse, who can utilise the findings to improve retention strategies and game design. Besides, academic researchers can also use the results of this study the theoretical foundation of the analysing of player behaviour. To the wider gaming industry, they can gain from the lessons learned in this case, and these can be applied to their own free-to-play models of games.

1.5.4 Time Scope

This project will be conducted across two long semesters in accordance with follow the Final Year Project (FYP) structure at University of Technology Sarawak (UTS).

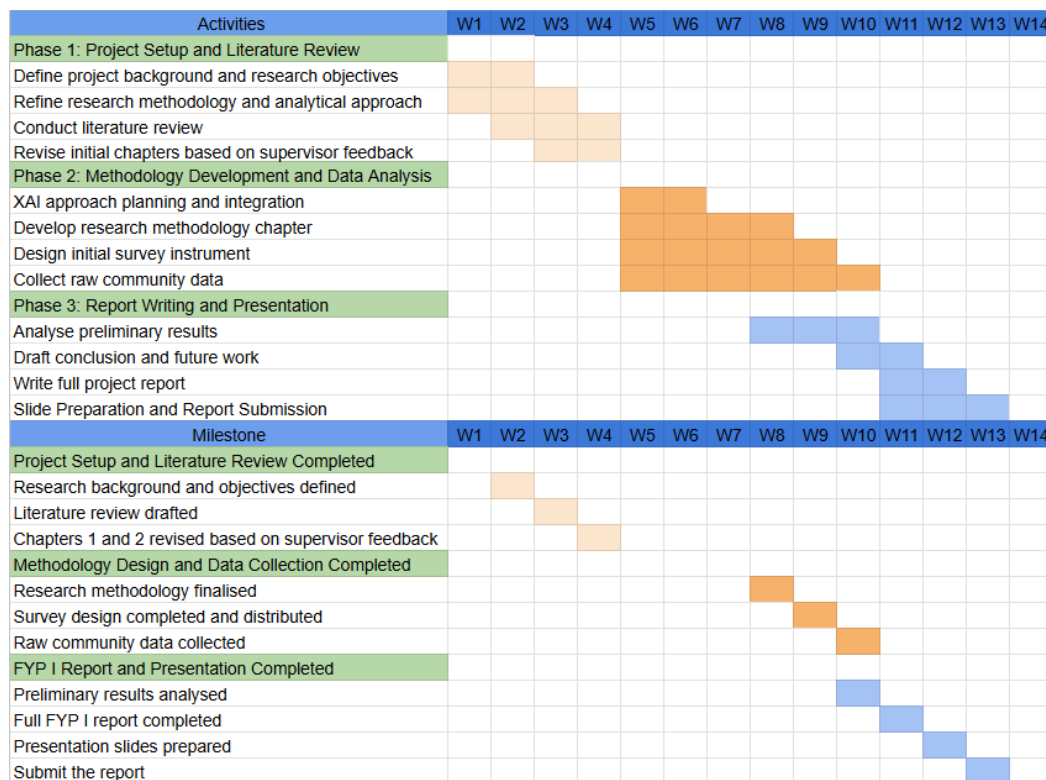


Figure 1.1: Gantt Chart and Milestone of FYP I

As shown in Figure 1.1, the first semester (FYP I) will concentrate mainly on proposal writing, literature review, preliminary models, research framework preparation and research instrument preparation such as the survey questionnaire. At this stage, research requirements and ethical considerations are confirmed, and the groundwork for data collection is established.

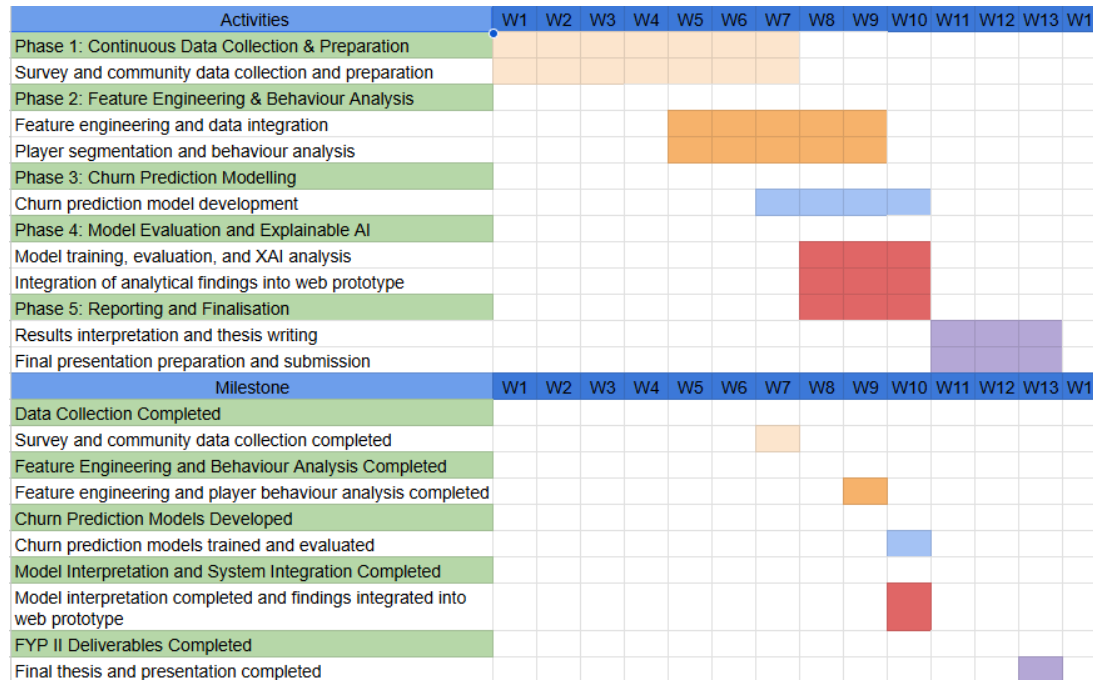


Figure 1.2: Gantt Chart and Milestone of FYP 2

The second semester (FYP II) will cover the data collection, analysis and system implementation. This will include the preparing and testing of the machine learning models, the evaluation of the predictive performance, and the integration of findings from both survey and review datasets. The final submissions will be made at this stage, which will include full thesis report, project brochure, exhibition poster and the developed analytical framework presented through a prototype web application.

1.6 Summary of Chapter 1

This chapter describes the context and motivation for the research. It begins with a discussion of the rapid growth of the global gaming industry and the increasing importance of player retention, more specifically with reference to free-to-play (F2P) business models.

The problem statements addressed important shortcomings in traditional behavioural analysis approaches such as dependency on survey collection, manual analysis and no integration between player segmentation and churn prediction. The study formulates clear objectives to overcome these constraints, with particular focus on a data-driven developmental framework for analysing player behavioural patterns and predicting churn.

In summary, Chapter 1 introduces the background of the study, outlines the problem statements, and defines the objectives and scope of the research. It explains the purpose of the study and what the research aims to achieve. This chapter also provides an overview of the thesis structure and guides the reader to the subsequent chapters, including the literature review, methodology, and system implementation.

CHAPTER 2

LITERATURE REVIEWS

2.1 Introduction

This chapter reviews the relevant literature to establish the theoretical and methodological foundation for the system. It outlines the progression from early statistical approaches (Roy, 2018) to more recent machine learning frameworks for player churn prediction (Mulla et al., 2025). In addition, this chapter discusses the development of Explainable Artificial Intelligence (XAI) methods that help clarify the factors influencing player churn (Xiong et al., 2023). Studies on clustering and player segmentation are also reviewed to support the analysis of player groups with similar behavioural patterns (Yu et al., 2020). Finally, psychological research is examined to provide further insight into players' motivations and the factors that influence continued gameplay.

2.2 The Game Industry & Player Behaviour

The global video games industry has undergone remarkable expansion in the past decade. According to Newzoo's 2025 Global Games Market Report, the video game market was expected to generate over USD 360 billion in revenue, also with a global player base of more than 3.6 billion. This expansion is largely driven by free-to-play (F2P) and freemium business models since 2020. In these business models, players enjoy games at no cost to them, while revenue from games is primarily generated through the use of microtransactions or gacha mechanisms, and in-game events. This

suggest that survival of long-term engagement is even more important because revenues are derived from retention as opposed to one-time purchases (Roy, 2018).

From a psychological point of view, the investigations conducted by Arbeau et al. (2020) demonstrate that online gaming is not merely a source of entertainment, but also a source of socio-emotional advantages. For example, online gaming provides social rewards, support identity development, and reduce stress levels. These experiential attributes reflect the reason why people continue playing games even beyond the effect of the core mechanics of game are still remain unchanged. Apart from that, from a data driven point of view, Roy (2018) has shown how measurable behavioural attributes, for instance play-time, mission progress, and in-game achievements, which serve as strong prediction indicators of longer period retention. The two aspects of this issue lead to the conclusion that both subjective motivations and objectively measureable behaviours need to be taken into account when analysing the phenomena regarding player engagement.

The case of HoYoverse games show these patterns more clearly. As a leading developer in F2P open-world titles, HoYoverse designs games such as Genshin Impact with exploration-driven gameplay, social collaboration, gacha-based monetisation, and event-driven progression. These features create complex behavioural patterns that traditional models often fail to capture. For example, some players remain active because of mechanical incentives such as resource farming or quest completion. Others also still continue playing due to community-driven motivations such as cooperative gameplay or participation in seasonal events. These multifaceted behaviours is essential and necessary for building an effective also a data-driven churn prediction frameworks which are discussed in the following section.

2.3 Churn Prediction in Games

Player churn is widely defined as a temporary or permanent discontinuation of playing games of any kind. Player churn has become one of the major issues of the gaming industry today. In both free-to-play and freemium models, both of them depending largely on long-term engagement of players, a mass appearance of player dropout leads

directly to a loss of revenue and reduces also the sustainability of community management. Player churn not only adversely affects immediate revenue streams from micro-transactions but as Roy (2018) articulated, it also leads to a reduction in the active playing population. Such reductions will also negatively influence potential future long-term growth of the games. Hence, churn prediction modelling is an important tool in order to improve player retention which is deemed so important to the continued profitability of online games.

Various methods have been used to predict churn. Early studies used statistical models such as logistic regression and survival analysis. Logistic regression, in particular, it has been valued for its simplicity and interpretability, widely used as a common baseline for analysing behavioural relationships (Roy, 2018). Because of traditional statistical models are limited in capturing complex and non-linear relationships, recent studies have increasingly employed machine-learning frameworks to address these shortcomings. Mulla et al. (2025) specifically proved that combined methods such as Random Forest achieved higher accuracy compared to Logistic Regression alone. This result highlights the potential of advanced algorithms especially when handling with big-data, especially the game data which become larger and more complexity in recent years.

Furthermore, great predictive accuracy does not guarantee better decision-making by company. As contended by Xiong et al. (2023), churn prediction systems must identify the players that are at risk, but are also needed to provide explanations regarding the causes of their churn. To overcome this limitation, their study introduced an integrates framework that combined with explainable AI (XAI) techniques such as SHAP and Anchors. To provides transparent, feature-level explanations, enabling developers and analysts to understand why specific players disengage from the game. Allow game developers to interpret model outputs, helping them design targeted improvement by pointing out critical factors that contribute to churn, also share these informations to the management of the company.

Recent studies have generally followed a data-mining pipeline inspired by the CRISP-DM framework, which provides a systematic structure for churn-prediction

research in gaming analytics (Chapman et al., 2000; Mulla et al., 2025). The detailed workflow adopted in this study will be further described in Section 2.12. Building upon these foundational studies, the following section reviews the machine-learning approaches that underpin modern churn-prediction frameworks

2.4 Machine Learning Approaches

Machine learning has become a central approach in churn prediction research, providing more robust performance than traditional statistical methods. Logistic Regression has frequently been used as a baseline due to its interpretability and ability to estimate the probability of churn from behavioural variables. Roy (2018), for example, applied logistic regression and demonstrated its effectiveness in linking engagement indicators such as playtime and mission completion to churn likelihood. Logistic regression remains limited in its capacity and cannot effectively capture any patterns or relationship of a modern gaming datasets, which containing various of linear and high-dimensional patterns.

To overcome these limitations, researchers have adopted more sophisticated models. For example, Decision Trees, Random Forests, and other ensemble methods are effectively applied in their research. Mulla et al. (2025) showed that Random Forest achieved superior predictive accuracy compared to logistic regression. It can effectively capture complex interactions between behavioural variables. Similarly, survival analysis techniques, including Kaplan–Meier estimation and Cox proportional hazards models, have been employed to offer temporal insights into player lifecycles (Roy, 2018). However, survival analysis requires exact time series data, which often are difficult to collect through the survey.

Although advanced models such as Random Forest achieving higher predictive accuracy, but they also having the problem of lacking transparency. These ensemble methods are often regarded as “black-box” models, making developers and analysts cannot having any real understanding at all of how does the model works to get the prediction (Mulla et al., 2025). This lack of transparency limits their viability for using in the real-world application. Besides, in a real-life practice, games analyst needs more

than the accurate forecasting, they must know that behavioural factors that drive player to churn. In this regard, recent studies have suggested using explainable artificial intelligence (XAI) techniques of SHAP and LIME to retune the problem of interpretability and predictive ability (Xiong et al., 2023). These XAI-based technical methods will give a better decision support by help with the giving transparent details of the features and these feature importance in churn prediction. Lastly, it allows the churn-prediction models can be transformed into actionable analytical tools. Therefore, this work builds on this premise by integrating interpretable and predictive models into the proposed framework.

2.4.1 Advanced Algorithms and Model Evaluation

On the other hands, comparing with the previous methods, there are also other advanced algorithm that are applied in the recent churn prediction studies to improve accuracy and generalisation.

Support Vector Machine (SVM) are the one mostly adopted method for modelling churn in a high-dimensional behavioural datasets. It works by finds optimal boundary that maximises the distance between the classes of churned and active-players, to separate the churned from the active-players. This capability allows the SVM performing well even the datasets features are non-linear data or there was a complex dataset. However, SVM are requires the careful parameter tuning, which may lead to the longer training time, and the higher computational cost for a large or complex datasets (AlShourbaji et al., 2023).

There is another popular algorithm is XGBoost (Extreme Gradient Boosting), an ensemble method that uses optimisation and regularisation to reduce overfitting. XGBoost are performing well especially on a large and imbalanced churn dataset, which achieving a higher accuracy and recall (Krishna et al., 2024). It boosting process strengthens weak learner step by step, making it a stable and widely used model in churn prediction at industry.

Table 2.1 Comparison of Classification Algorithms in Player Churn Studies

Algorithm	Strengths	Limitations	Application
Logistic Regression	● Highly interpretable linear model ● Direct estimation of churn probability.	● Relies on linear assumptions. ● Cannot capture complex or non-linear patterns.	Roy (2018) – Freemium game churn analysis
Decision Tree	● Intuitive “if-then” decision rules. ● Capable of handling non-linear data.	● Easily overfits small datasets. ● Lower predictive accuracy than ensemble methods.	Roy (2018) – Exploratory segmentation
Random Forest	● High predictive accuracy and robustness. ● Provides feature importance measures.	● Reduced interpretability compared to single models. ● Higher computational cost for large datasets.	Mulla et al. (2025) – Candy Crush study
Support Vector Machine	● Effective for high-dimensional and non-linear data. ● Strong generalisation ability.	● Sensitive to parameter and kernel selection. ● Computationally expensive for large-scale data.	AlShourbaji et al. (2023) – Churn classification
Extreme Gradient Boosting	● Strong performance on large and imbalanced datasets. ● Incorporates regularisation to reduce overfitting.	● Limited interpretability (“black box”). ● Requires extensive hyperparameter tuning.	Krishna et al. (2024) – Industrial churn prediction pipeline

Table 2.1 summarises the main classification algorithms using in the churn analysis. As shown, the Ensemble Methods such as Random Forest and, XGBoost having a higher accuracy on a complex dataset, but it was difficult to interpret. In contrast, simpler models like Decision Tree and Logistic Regression offer clearer insights for explaining churn behaviour

Apart from that, there are five common metrics are using to evaluate the performance of models. For example, Accuracy, Precision, Recall, F1-score, and Area Under the ROC Curve (AUC).

- Accuracy measures the overall correctness of predictions.
- Precision and Recall showing how well the models identifies churners without too much false alarms or missed cases.
- F1-Score combines both Precision and Recall to provide a balanced evaluation.
- AUC reflects the model's ability to distinguish churners from non-churners across different thresholds.

Confusion Matrix	
TP=136	FN=3
FP=3	TN=258
Precision=0.98 Recall=0.98 F1=0.98 Accuracy=0.98	

Figure 2.1 Confusion matrix of Random Forest model for player churn prediction.

Figure 2.1 illustrates the confusion matrix used to evaluate the performance of churn-prediction models. The matrix presents the distribution of correctly and incorrectly classified players, where True Positives (TP = 136) and True Negatives (TN = 258) represent accurate predictions of churners and non-churners, respectively, while False Positives (FP = 3) and False Negatives (FN = 3) indicate misclassifications. The model achieved a precision of 0.98, recall of 0.98, F1-score of 0.98, and an overall accuracy of 0.98 as reported in prior studies. These results demonstrate a strong predictive ability consistent with findings from Mulla et al. (2025), confirming the Random Forest's robustness in churn classification tasks.

In churn prediction, where churners typically represent the minority class, relying solely on Accuracy can produce misleading results due to class imbalance. Therefore, more balanced evaluation metrics such as Macro-F1 and Weighted-F1 are preferred, as they account for unequal class distributions and provide fairer performance assessments

Macro-F1 treats both classes equally by averaging their individual F1-Scores, while Weighted-F1 assigns proportional importance based on the number of instances in each class. Mulla et al. (2025) reported that a Weighted-F1 of 0.91 provided a more realistic evaluation when churners accounted for less than 25% of all players. Furthermore, AUC (Area Under the ROC Curve) remains a crucial indicator of discriminatory power, where values above 0.9 signify excellent separation between churners and non-churners, and values between 0.8–0.9 indicate good performance (Roy, 2018).

As a whole, these metrics ensure that both the predictive accuracy and the practical usefulness of the churn-prediction model are properly evaluated, mitigating bias and enhancing reliability in imbalanced datasets.

2.4.2 Dimensionality Reduction and Temporal Models

Dimensionality reduction and temporal modelling are critical for enhancing the effectiveness of churn prediction models, especially when dealing with complex survey data.

Behavioural data derived from detailed questionnaires, such as the one designed for this study, are often high-dimensional. For example, collecting exploration progress, quest completion, and satisfaction ratings across multiple in-game regions (e.g., Mondstadt, Liyue, Inazuma, etc.) generates a large number of correlated features for each player. These high-dimensional and often noisy variables can decrease model efficiency and predictive accuracy. Techniques like Principal Component Analysis (PCA) are essential for this purpose. PCA transforms correlated variables into a smaller set of orthogonal components that preserve most of the

variance, enabling faster model training and mitigating the risk of overfitting. As Mulla et al. (2025) demonstrated that the application of PCA can reduce model training time by 18% and improve accuracy from 0.87 to 0.89, proving its value in enhancing computational efficiency and model performance. Similarly, feature selection based on Random Forest feature importance can also be applied to identify the most influential variables, helping researchers build more interpretable and targeted models.

Separately, temporal modeling offers an alternative approach to churn analysis by estimating when a player might leave. Methods like the Kaplan-Meier estimator and Cox Proportional Hazards models have been used to estimate player survival time based on factors like spending or social activity (Roy, 2018). While this study's survey-based, cross-sectional design is less suited for survival analysis (which requires precise time-to-event data), these temporal methods remain a relevant consideration for future research using longitudinal game logs. Therefore, dimensionality reduction methods like PCA are a core component of the proposed methodology, supporting the classifier by improving input data quality and computational efficiency.

2.5 Limitations of Previous Models in Churn Prediction

For significant progress in machine learning approaches for player churn analysis, several limitations remain that restrict their practical use in real-world settings. Earlier studies have largely focused on improving predictive accuracy, often giving less attention to model explainability, interpretability, and decision support. However, these aspects are critical for enabling game developers and stakeholders to understand model outputs and apply them effectively in practice.

Traditional models such as Logistic Regression and Decision Tree are relatively interpretable and provide direct statistical relationships between behavioural variables and churn probability. However, their simplicity often comes at the cost of representational power, as they assume linear relationships and struggle to capture the complex, multi-dimensional interactions typical of player behaviour in open-world and event-driven games (Roy, 2018). On the other hand, ensemble and boosting methods such as Random Forest, Support Vector Machine, and XGBoost achieve

superior predictive performance on high-dimensional and imbalanced datasets (Mulla et al., 2025; AlShourbaji et al., 2023; Krishna et al., 2024). Yet these models operate largely as “black boxes,” making it difficult to determine why a player is classified as at-risk or what behavioural factors most strongly drive churn. This opacity reduces managerial trust and hinders the translation of analytical results into actionable retention strategies.

Another critical limitation lies in the lack of feature-level interpretability. Most prior studies merely report aggregate feature importances or overall model weights, which cannot explain how individual players differ in their risk profiles. As noted by Xiong et al. (2023), without transparent player-level explanations, developers may identify churners but remain unable to design precise interventions. Moreover, most models provide static predictions that do not integrate temporal or motivational variables, thereby neglecting dynamic aspects of player engagement such as periodic activity drops or social influence effects.

Taken together, prior studies suggest an ongoing trade-off between predictive accuracy and model interpretability. While advanced algorithms often improve prediction performance, their limited transparency can reduce their practical usefulness in game analytics. As a result, recent research has increasingly explored Explainable Artificial Intelligence (XAI) techniques, such as SHAP, LIME, and Anchors, to address this issue. These methods provide more interpretable insights by attributing model predictions to behavioural, social, and psychological factors, supporting clearer understanding and more informed decision-making in churn analysis.

The next section therefore introduces Explainable AI (XAI) as a methodological advancement designed to overcome these interpretability challenges. Through transparent feature attribution and instance-level explanations, XAI transforms churn prediction models into practical tools for decision-making in game retention analysis.

2.6 Explainable AI (XAI)

As machine learning models become increasingly complex, their lack of interpretability has emerged as a critical limitation in practical applications. In churn prediction, high predictive accuracy alone is insufficient if developers cannot understand why certain players are at risk of leaving. Explainable AI (XAI) seeks to address this challenge by providing tools and frameworks that clarify how input features contribute to model outputs. This interpretability transforms churn prediction from a predictive task into a decision-support process, and also enabling actionable strategies for improving player retention.

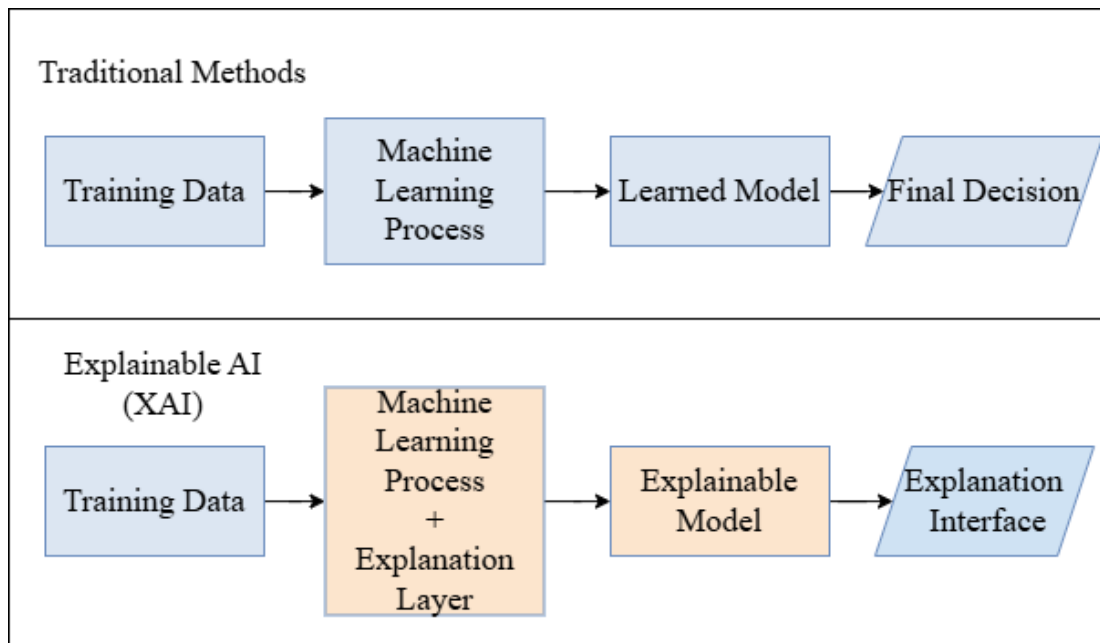


Figure 2.2 Comparison between traditional machine learning and Explainable AI (XAI) frameworks.

Figure 2.2 illustrates the structural difference between conventional machine learning workflows and the Explainable AI (XAI) framework. While traditional models generate predictions through opaque processes that offer little insight into how outcomes are derived, XAI introduces an additional Explanation Layer and Interface that clarify the influence of each feature on model decisions. This enhancement transforms the analytical process from a “black-box” prediction pipeline into a transparent and interactive decision-support system (Barredo Arrieta et al., 2020).

Building upon this foundation, several explainability techniques have been developed to quantify and visualise these feature contributions. Among them, SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) are the most frequently applied methods for understanding player-behaviour-based churn models. SHAP values provide both global and local interpretations of feature importance, revealing how specific behavioural factors affect churn risk, whereas LIME and Anchors generate rule-based explanations for individual predictions. These models help to highlighting the key conditions under which a player may be classified as “at risk.” Xiong et al. (2023) demonstrated the utility of these techniques. Their framework embedded SHAP and Anchors into their churn prediction framework. As a results, it showing how explainability can guide targeted interventions such as incentive design or content adjustments

SHAP Feature Importance (Impact on Churn Risk)

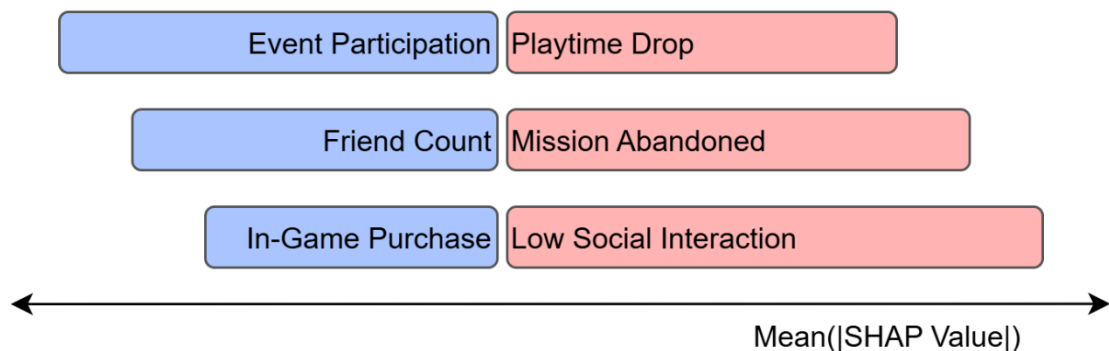


Figure 2.3 SHAP feature importance showing key factors influencing player churn (adapted from Xiong et al., 2023)

Figure 2.3 visualises how SHAP analysis identifies the relative influence of behavioural features on churn prediction. Positive SHAP values (red) indicate behaviours that increase churn probability, such as playtime reduction, mission abandonment, and reduced social interaction. Negative SHAP values (blue) represent retention factors, including event participation, friend interaction, and in-game purchases. The horizontal axis, $\text{Mean}(|\text{SHAP Value}|)$, measures the average magnitude of each feature’s contribution to the model’s output. A higher mean value implies

greater importance in the churn-prediction decision (Yu et al., 2020; Xiong et al., 2023).

Despite these contributions, the adoption of XAI in gaming remains relatively limited. Most existing studies either prioritise accuracy over interpretability or apply interpretability methods in isolation without integrating them into a broader predictive pipeline. This study addresses these limitations by embedding XAI into its proposed churn prediction framework, ensuring that predictions are both accurate and actionable. By combining predictive models with explainable outputs, the research aims to provide HoYoverse developers with interpretable insights that can inform player retention strategies more effectively.

2.6.1 Applications of XAI in Gaming Analytics

Explainable AI has recently been implemented in commercial-scale games to enhance decision transparency and bridge the gap between analytics and design. As an example, Xiong et al. (2023) employed TreeSHAP and an augmented Anchors framework in Justice Online, a large-scale MMORPG developed by NetEase. Their analysis used gameplay logs from over 200,000 players and represented one of the most wide-spread implementations of XAI in gaming so far.

Their framework demonstrated that the explainable AI could function on different levels of analysis. At the individual level, SHAP values produced explanations as to why specific players were predicted to churn. Examples of churn triggers would be reduced guild activity, stalling progress, or frequent task failures. At group level, the clustering of SHAP values allowed the identification of distinct churn segments, discriminating between progression-driven players and social-driven players. At the global level, generalised feature importance demonstrated systemic churn triggers, such as excessive cutscene viewing.

Importantly, these were not purely theoretical insights. The researchers recorded metrics so that the results could be easily turned into user interventions by doing an A/B test. By create a button of “skip cutscenes”, the experimental group score

a reduced churn of 1.6 per cent against the control group and significant increases in average playtime. As a result, this becomes a permanent option within the game.

This demonstrated that the XAI while having the ability to increase model interpretability, could also be used as decision supporting devices between analytics and design. A similar interpretive approach could be implemented in Genshin Impact where whether social interaction, pulling spending or exploration overall progress is the strongest behavioural marker with player retention.

2.6.2 Ante-hoc vs Post-hoc Explainability

Explainable AI (XAI) methods can be categorised as ante-hoc or post-hoc, depend on how and when interpretability is achieved in the modelling process (Barredo Arrieta et al., 2020). Understanding this distinction is important because it determines how model transparency and predictive accuracy can be balanced in churn prediction research.

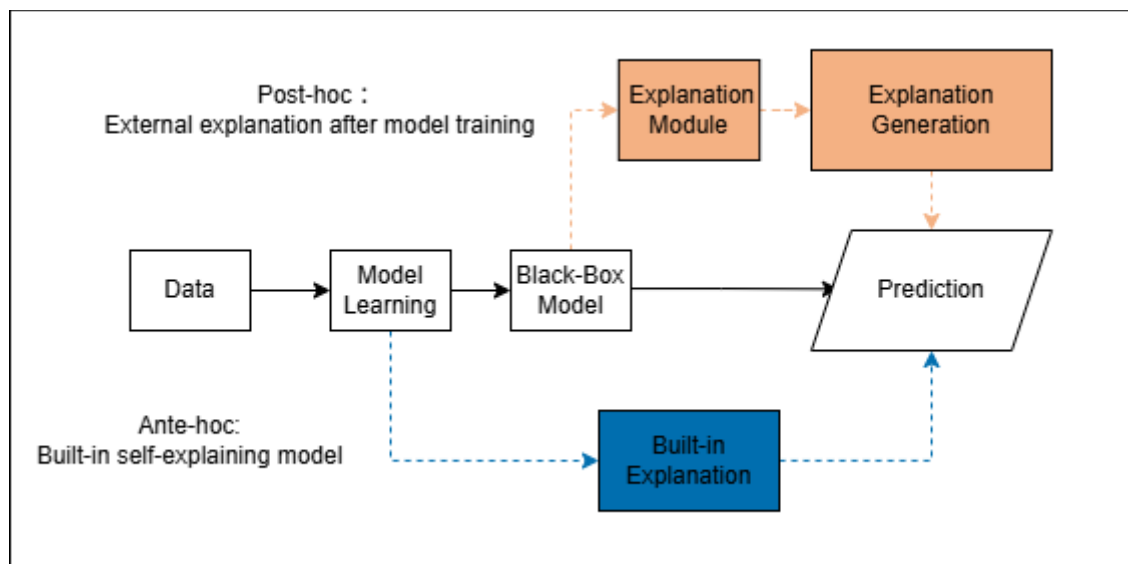


Figure 2.4 Visual comparison between ante-hoc (built-in) and post-hoc (external) explainability approaches.

Figure 2.4 illustrates the conceptual distinction between models that are self-explaining by design (ante-hoc) and those requiring external interpretation after training (post-hoc), as discussed in the XAI literature (Barredo Arrieta et al., 2020).

Ante-hoc models are inherently interpretable, as their internal logic and parameters can be directly examined. Typical examples include Logistic Regression and Decision Trees, where each variable's contribution to the outcome can be explicitly measured through coefficients or decision rules. These models are particularly valuable when transparency and ease of interpretation are prioritised. In churn analysis, Logistic Regression enables direct estimation of how behavioural and motivational factors affect the probability of player churn, for example play frequency or satisfaction score. However, such models often assume linear relationships and may struggle to capture the nonlinear and high-dimensional interactions found in large-scale player datasets (Barredo Arrieta et al., 2020; Molnar, 2022).

In contrast, **post-hoc** approaches explain complex or ensemble models after they have been trained. Methods such as SHAP (SHapley Additive exPlanations), LIME, and Anchors are commonly applied to interpret the predictions of high-performing algorithms like Random Forests (Ribeiro et al., 2016; Lundberg & Lee, 2017). These techniques quantify the contribution of each feature to a specific prediction, making it possible to understand why a model has classified a certain player as “likely to churn.” For example, a SHAP analysis may show that low login frequency and declining event participation jointly increase a player's churn probability by a measurable margin, thus revealing feature-level decision logic even in black-box models.

For example, Xiong et al. (2023) have compared ante-hoc and post-hoc methods in a large-scale study of Justice Online, reporting that TreeSHAP required approximately 0.006 seconds per instance, nearly 40 times faster than Anchors, also maintaining the explanation accuracy. Such computational efficiency supports near-real-time player-risk monitoring in live game environments. As a result, combining simple, interpretable baseline models such as Logistic Regression with SHAP-based explanations for Random Forest ensembles can provide a balance between interpretability and predictive reliability.

In summary, ante-hoc models provide inherent transparency but limited complexity, while post-hoc methods enhance interpretability in advanced predictive systems. Accordingly, this study employs Logistic Regression as an interpretable baseline and Random Forest, complemented by SHAP explanations, to balance accuracy and interpretability.

While Explainable AI (XAI) enhances the transparency of predictive models, understanding the underlying behavioural structures among players is equally important. Hence, clustering and segmentation methods are incorporated to identify distinct player profiles prior to churn prediction and interpretability analysis.

2.7 Clustering & Segmentation

Following the XAI discussion above, segmentation serves as a complementary exploratory tool before classification. Clustering and segmentation are common applied in gaming research. They allows players to be segmented into groups according to their behaviours, motivations, and spending patterns. Segmentation also will provide valuable insights for designing personalised experiences as well as tailoring retention strategies. For instance, Roy (2018) used clustering to distinguish between engaged and disengaged players. Showing that segmentation can significantly improve the recognition of high-value users. By grouping players into meaningful categories, clustering will help game developers in focusing on players retention strategies and improving resource allocation more effectively.

There are various of clustering algorithms have been used in this field. The most popular is probably K-means, because its efficiency and simplicity, but it often struggles with its inability to work well with non-linear boundaries and its sensitivity to initialisation. To address these issues, Yu et al. (2020) suggested a hybrid method that combines Self-Organising Maps (SOM) with k-means clustering. This was well received and worked efficiently in segmenting players within the Taiwanese online game market. Their findings demonstrate that hybrid clustering models can uncover more stable and meaningful segments than single-method approaches, and it is useful effective in analysing games where player behaviour is diverse and dynamic.

Although these above improvements available, but challenges still remain in the process of integrating clustering results with predictive churn models. Many studies treat segmentation as an independent process, and do not relate cluster membership to churn probability or retention strategies. The current research addresses these gaps by building on these shortcomings by using clustering as a focal point in its churn prediction pipeline, highlighting distinct behavioural groupings in advance of prediction modelling. This provides the foundation for integrating social and motivational factors in the next section.

2.8 Integrating Social and Behavioural Dimensions

Recent studies emphasize that integrating player motivation, particularly social factors, with behavioural data substantially improves churn segmentation. While technical Social Network Analysis (SNA) using log data can map player connections, research based on psychological perspectives (Yee, 2006; Arbeau et al., 2020) argues that a player's perceived social integration is a powerful retention driver. Factors such as having a fixed group of friends, the frequency of co-operative gameplay, and whether friends still playing acts as a primary motivator are critical dimensions of engagement.

This study's methodology aligns with this perspective by collecting data on these specific social motivators directly via the survey questionnaire. These social variables will be used alongside gameplay behaviours (like exploration progress and challenge completion) as inputs for clustering. A practical workflow will connect these clusters with supervised models such as Random Forest or XGBoost, to forming a Clustering To Classification Pipeline. This hybrid process first groups players based on their combined behavioural and motivational profiles, then predicts churn risk for each group, yielding actionable insights for retention strategies that integrate both what players do and why they do it.

This integration not only enhances segmentation granularity but also bridges quantitative behaviour with qualitative motivation, a key limitation identified in prior churn studies.

2.9 Psychological & Experiential Perspectives

Beyond behavioural and social factors, psychological and experiential dimensions further explain the underlying motivations of player engagement. Although quantitative modelling dominates churn prediction research, but psychological and experiential perspectives provide essential insights into the reasons and motivations behind why players still playing or leave games. Arbeau et al. (2020), using a phenomenological approach, found that online gaming experiences are closely tied to socio-emotional factors such as social rewards, identity development, personal growth, and stress reduction. These findings determine that player retention cannot be purely explained in terms of measurements of gameplay behaviour but also has a meaning in terms of gamers' interpretations of their gaming experiences.

Other frameworks in game psychology lend support to this perspective. For example, Yee (2006) proposed his powerful model showing that there are three major motivational drivers for online gaming, such as achievement, social interaction and immersion. These motivational help to explain why direct engagement continues among gamers even when the core mechanics of a game have relatively been stable. In the context of freemium and open-world titles such as Genshin Impact, immersion in narrative and exploration, and with strong social community bonds, can sustain long-term engagement and reducing the probability of churn.

Despite their relevance, psychological insights are often not included in data-oriented research into customer churn, which naturally insists on behavioural indicators of business activity over experiential significance. This omission inhibits the ability of churn prediction models to correctly grasp player motivations. The present research aims to remedy this shortcoming by utilising survey data which embraces behavioural frequency and motivational variables. This compensates for the traditional focus of churn prediction on players' behavioural patterns in games, by including the psychological foundations of why players participate in games.

By combining psychological motivations with behavioural data, this study aligns with the XAI goal of producing interpretable and human-centred insights into player churn.

2.10 Gaps & Challenges

Despite the progress made in the literature, several gaps and challenges remain.

Firstly, a large number of existing churn prediction studies prioritised predictive accuracy over interpretability. Black box models including Random Forests show good performance (Mulla et al., 2025) but fail to provide interpretable insights into the reasons for player churn. Without interpretability, developers may know who is likely to churn, but may lack the understanding of why this is the case, which reduces the practical usefulness of such models.

Second, psychological and data-driven perspectives are often studied in separately. Psychological research (e.g., Arbeau et al., 2020; Yee, 2006) highlights motivational drivers such as achievement, social interaction, and immersion, whereas data science studies (Roy, 2018) focus on quantifiable behavioural metrics such as playtime, mission completion, and in-game purchases. This separation creates a gap: churn models tend to capture surface-level behaviours without fully addressing the motivational and experiential reasons behind them.

Third, although surveys and in-game behavioural logs are widely used, few studies incorporate large-scale player comments and community-generated content as complementary data sources. Player reviews from platforms such as HoYoLAB, Bilibili, TikTok, and Google Play provide rich qualitative insights into sentiment, satisfaction, and engagement trends, yet these remain underutilised in churn prediction frameworks. The lack of integration between structured behavioural data and unstructured player-generated content limits the comprehensiveness of current approaches.

Lastly, while most churn studies focus on mobile or casual titles, few have examined complex open-world RPGs games like Genshin Impact, a context where behavioural, social, and motivational factors interact more dynamically.

2.11 Research Adoption – XAI

In light of the above-discussed gaps, this study adopts an Explainable AI (XAI)-driven hybrid framework that integrates behavioural, psychological, and community-based data within a unified, interpretable churn-prediction process. The framework is designed not only to improve predictive accuracy but also to enhance human-centred interpretability in analysing player retention and disengagement.

Phase 1 – Data Collection.

The framework typically involves survey-based data collection to capture players' behavioural patterns, motivational factors, and social influences. This structured data may be complemented by community-generated content, such as reviews and discussions from platforms including HoYoLAB, YouTube, Google Play, and Bilibili, which provide additional sentiment and experiential insights related to player engagement.

Phase 2 – Behavioural Segmentation and Prediction.

To identify underlying player patterns, clustering algorithms (K-means and Self-Organising Maps) are applied to segment players into groups sharing similar behavioural and motivational profiles. These cluster features are then used as inputs to classification models such as Logistic Regression and Random Forest, which estimate churn probabilities for each segment. This dual-stage process ensures that both behavioural diversity and motivational context are preserved throughout prediction.

Phase 3 – Explainability and Decision Support.

To address the interpretability–accuracy trade-off identified in prior research, churn prediction frameworks often incorporate Explainable AI (XAI) methods, such as SHAP value analysis and feature-importance visualisation. These post-hoc explainability techniques are commonly used to help interpret how individual features, such as social interaction, exploration progress, and spending behaviour which might relate to churn risk. By linking model predictions to interpretable behavioural and

psychological factors, XAI methods support the translation of predictive outputs into insights that can inform decision-making in player retention strategies.

In summary, the proposed XAI-driven hybrid framework combines data diversity, predictive modelling, and interpretability enhancement into a single analytical pipeline. It therefore enables not only the identification of who is likely to leave but also an explanation of why, aligning analytical precision with managerial relevance in the context of large-scale open-world RPGs such as Genshin Impact.

2.12 Theoretical and Methodological Frameworks

This study's development is principally premised on one cohesive approach, namely the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework, which provides a coherent and systematic workflow for data-analytics projects (Chapman et al., 2000). The CRISP-DM model integrates six phases, such as Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment. Making sure that each component of the analytical process follows a reproducible and structured methodology. Which serving as the core foundation for the data-mining pipeline applied in this study.

The general workflow of this study, illustrated in Figure 2.5, adapts the CRISP-DM model to the context of player-behaviour analysis and churn prediction. It outlines the sequential stages of data collection, preprocessing, feature selection, model training, evaluation, and explainability. This process ensures that each analytical phase follows a structured and reproducible design consistent with data-mining best practices.

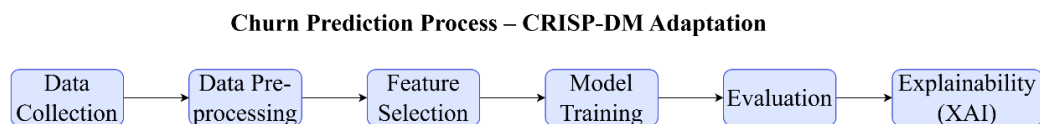


Figure 2.5 General churn-prediction pipeline (adapted from CRISP-DM and Mulla et al., 2025)

There are six sequential stages in the churn prediction pipeline.

1. **Data Collection**, gather all the information from our surveys, and community reviews.
2. **Data Pre-processing**, clean the missing values and convert categorical variables into numeric ones and normalize the data.
3. **Feature selection**, reduces the redundancy and dimensionality by applying PCA or Random Forest based importance.
4. **Model Training**, using a lot of algorithms including Logistic Regression, Decision Tree, Random Forest, SVM and XGBoost to classify churn.
5. **Evaluation**, we measure how well our models perform using Accuracy, Precision, Recall, F1-Score, and AUC metrics.
6. **Explainability (XAI)**, we apply SHAP and LIME to understand what drives players' behaviour that leads to churn and give actionable insights on which player retention strategies will be most effective.

To complement CRISP-DM, the study incorporates an Agile and Prototyping approach to enhance the iterative development of analytical outputs and web-based visualisation interfaces. Rather than focusing on software engineering lifecycles, Agile principles are applied here as a research-oriented strategy to encourage continuous improvement and user-centred refinement.

Each iteration involves designing and presenting interactive web dashboards and visualisation interfaces, receiving feedback from the project supervisor (Dr. Tanalachimi A/P Ganapathy), and refining the interpretability and clarity of visual outputs based on the comments received. These short feedback cycles allow the study to adapt its models and presentation methods to ensure both analytical accuracy and effective communication of findings.

The integration of Agile principles within the CRISP-DM framework therefore ensures that the analytical process remains flexible, responsive, and stakeholder-driven. The prototype functions as an interactive medium for demonstrating machine-learning results and explainable AI insights to both technical and non-technical audiences. This iterative and participatory design approach bridges data analysis and user experience, ensuring that the insights derived from machine-learning models are transparent, interpretable, and visually understandable to support decision-making in game analytics.

2.13 Summary

This chapter reviewed key literature on player behaviour, churn prediction, machine learning techniques, clustering and segmentation, explainable artificial intelligence (XAI), and psychological perspectives in gaming research.

Overall, the literature shows a clear progression in churn prediction studies, moving from early statistical approaches (Roy, 2018) to more advanced machine learning models with improved predictive performance (Mulla et al., 2025), and more recently towards explainability-oriented frameworks that seek to support interpretation and decision-making (Xiong et al., 2023). Research on clustering and player motivation (Yu et al., 2020; Arbeau et al., 2020; Yee, 2006) further suggests that player engagement is influenced by a combination of behavioural patterns, social interactions, and psychological factors.

Despite these developments, several gaps remain evident. Much of the existing research continues to prioritise predictive accuracy, often at the expense of interpretability. In addition, psychological motivations are frequently examined in isolation from behavioural data, limiting their practical integration into churn prediction models. Community-generated content, such as player reviews and discussions on platforms like HoYoLAB, TikTok, and Bilibili, is also underutilised, despite providing valuable qualitative insights into player sentiment and satisfaction. Furthermore, complex open-world role-playing games, such as Genshin Impact, remain relatively under-explored, particularly in terms of long-term retention dynamics within exploration-driven environments.

In response to these limitations, this study proposes a hybrid analytical framework guided by the CRISP-DM process and informed by Agile and Prototyping principles. The framework integrates behavioural, psychological, and community-based data sources, applying clustering techniques (K-means and Self-Organising Maps) to identify player segments prior to churn classification. Logistic Regression and Random Forest models are then employed for churn prediction, with explainable artificial intelligence techniques, including SHAP and LIME, used to enhance model transparency. By combining predictive performance with interpretability, the study aims to generate actionable and human-centred insights to support player retention strategies in Genshin Impact.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter outlines the research methodology employed in this study, describing the overall framework, design, data-collection methods, analytical techniques, and ethical considerations adopted throughout the research process. This chapter will provide a structured plan for how to analysing player behaviour, identifying key churn factors, developing predictive models, and communicating the research findings.

3.2 Research Approach

This study adopts pragmatic research approach, focusing on use of methods that are most suitable for addressing the research objectives. It is well-suited for analysing the player behaviour and churn factors in Genshin Impact, because understanding player retention requires both behavioural data and contextual understanding of player motivation or social dynamics.

Accordingly, the study adopts a mixed-methods approach, with a quantitative survey serving as the main method, supported by the qualitative insights from community discussions to provide additional context. Based on the structured survey, this study examines four key behavioural and psychological factors, namely the **Motivation and Satisfaction, Social Influence, Engagement and Spending**, and **Retention Intention**. These factors are translated into measurable indicators suitable for statistical and machine-learning modelling. In addition, the qualitative component

draws on secondary community-generated content from platforms such as HoYoLAB, GooglePlay, YouTube, and Bilibili. These qualitative evidences is used to support the quantitative findings by illustrating player sentiment, experiences, and perceptions related to the game.

Methodologically, the research follows a **deductive** reasoning process. The research is guided by theories, or the findings from previous studies on player motivation and churn prediction, as discussed in Chapter 2. These established relationships are examined using the survey-based data and predictive modelling techniques. Besides, limited inductive analysis is applied when interpreting the community-generated content and model explainability outputs, such as SHAP feature-importance results. This allows patterns and insights emerging from player feedback and model interpretation to further support and refine the understanding of player disengagement.

Overall, this research approach provides a clear and structured methodological framework for the study. It ensures that each stage of the analysis, starting from data collection and player segmentation to churn prediction and explainable AI interpretation are methodologically sound and also aligned with the research objectives. As a result, to generate a meaningful and practical insights into player retention in Genshin Impact.

3.3 Research Design

This study is designed to conduct two main analytical tasks, namely “Community Side Analysis” and “Survey Based Analysis”. Data analytics and machine learning techniques are applied to understand player behaviour and the factors causing churn from both behavioural and community context.

Community Side Analysis

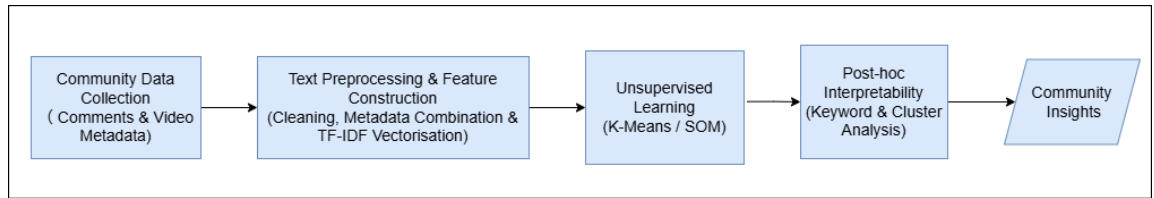


Figure 3.1 : Workflow of the Community Side Analysis

Figure 3.1 illustrates overall workflow of the community side analysis. Player comments and video metadata related to the Genshin Impact will be collected first and analysed through preprocessing, unsupervised machine learning, and post-hoc interpretation to identify community patterns and discussion themes.

Survey Based Analysis

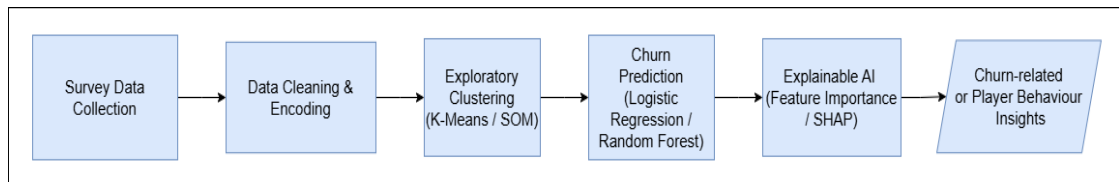


Figure 3.2 : Workflow of the Survey Side Analysis

Figure 3.2 illustrates the overall workflow of the survey side analysis. Survey responses collected from active, or the churned players are analysed using clustering and churn prediction models, followed by explainable AI (XAI) techniques to explain the prediction results and identify key factors associated with player churn.

In a nutshell, this research design combines community and survey side analysis to examine player behaviour and churn factors, also the information from community discussions that help explain the background and reasons behind player behaviour.

3.4 Population and Sampling

Target population of this study refers to individuals who have played the game, including both current and former players.

Respondents are categorised into two groups, active players and churned players, who have played or not played game within the past 30 days. Further distinctions within the active players group are derived from survey responses related to recent login frequency and gameplay activity.

Participants were eligible if they had played Genshin Impact for at least one month and voluntarily agreed to participate. A minimum target of 150 valid responses set to ensure sufficient data for analysis at the undergraduate level.

3.5 Data Collection Methods

This section describes the methods and instruments used to collect both primary and secondary data for this study. The research employed a mixed-method data collection strategy, combining a structured online survey and secondary textual data from gaming communities to achieve a comprehensive view of player behaviour and churn patterns in Genshin Impact.

3.5.1 Primary Data: Google Form Survey

The Google Forms survey used for collecting player responses included both active players and players who had dropped out (churned) of Genshin Impact. The questionnaire asked about respondents' gaming activity within the past 30 days and

utilised branching logic to direct participants to either the active player or churned player sections based on their responses.

For **active players**, the survey collected data including, recent login frequency at the range of 7 days and 30 days, average playtime of one day, current gameplay progress, content engagement, co-operative experience, challenge participation such as the “Spiral Abyss and Imaginarium Theater”, their spending behaviour and the future play intention.

For **churned players**, the questionnaire focused on reasons for disengagement, continued exposure to game-related previews or livestreams, social influence, participation in other HoYoverse or non-HoYoverse games, and any conditions that may encourage a return to the game.

Most questions were closed-ended using multiple-choice and Likert-scale formats, while selected open-ended questions were included to capture supplementary qualitative feedback from respondents.

To reach a sufficient number of respondents, the survey was actively distributed across multiple channels, including *Facebook*, and was also promoted through SurveyCircle, an academic survey distribution platform. Evidence of these distribution efforts is provided in Appendix B (Figures B.4 and B.5).

It should be noted that participants were recruited through gaming community platforms and personal networks, which may introduce self-selection bias, as highly engaged players are more likely to respond to a gaming-related survey. This may limit the generalisability of findings to the broader player population.

3.5.2 Secondary Data: Player Reviews and Discussions

Qualitative community data were collected through non-participant observation from open-access community platforms to provide the community understanding about player discussions and reactions. The purpose of collecting secondary data was to support triangulation with the survey findings, and also capture recurring themes and

sentiment patterns, which related to satisfaction with new content, complaints about gameplay difficulty, or perceived imbalance in rewards.

On Bilibili, both official content released by the game publisher and user-generated content (UGC) related to Genshin Impact are collected. For official videos, engagement indicators such as the number of likes and comments are recorded, together with a sampled subset (approximately 10%) of user comments. For UGC videos, metadata including video title, content creator, tags, number of likes, favourites, and comment counts are collected to reflect community participation and interest.

A similar data collection approach is applied to YouTube, focusing on official videos and relevant user-generated content, together with associated engagement metrics such as views, likes, and comment counts.

For HoYoLAB, data collection is limited to official posts and selected player discussion channels, such as community forums and discussion sections, where players actively share opinions and experiences related to the game.

Other social media platforms, such as X or Facebook, are excluded from the current study due to their large data volume and the difficulty of systematic data collection within the project scope.

It is important to acknowledge that community-generated content collected from platforms such as Bilibili, YouTube, and HoYoLAB may be subject to sampling bias. Users who actively post or comment on these platforms tend to be more engaged or vocal than the average player, meaning their opinions may not be representative of the broader player base. For instance, players who have already churned are less likely to remain active on game-related community platforms, which may underrepresent disengagement perspectives in the secondary data. Therefore, the community data in this study is treated as supplementary and contextual evidence rather than a statistically representative sample, and findings derived from it should be interpreted with this limitation in mind.

3.6 Data Analysis Techniques

This section describes the analytical procedures and tools applied to process, model, and interpret the collected data. Followed the CRISP-DM process to ensure a structured, replicable, and data-driven workflow. Both quantitative survey data and secondary textual data were analysed to identify behavioural patterns, predict churn likelihood, and generate explainable insights for the study.

3.6.1 Data Preparation

Before modelling, all responses exported from Google Forms were cleaned and standardised using Python (pandas, NumPy).

The preparation process included:

- **Handling Missing Values:** Replacing or removing null entries in continuous variables like, playtime, spending.
- **Encoding Categorical Variables:** Converting nominal responses such as gender, device type, or player group into numerical form via one-hot or label encoding.
- **Outlier Detection:** Identifying extreme values in variables like monthly spending or playtime to prevent skewed analysis.
- **Normalisation and Scaling:** Applying min-max scaling to ensure consistency between features with different ranges.

For secondary textual data, basic cleaning (removing emojis, URLs, and duplicate comments) and tokenisation were applied prior to frequency and sentiment analysis.

3.6.2 Descriptive and Exploratory Analysis

Descriptive statistics summarised the overall dataset to reveal general player characteristics such as age distribution, gameplay frequency, and spending habits.

Visualisations were created using Tableau to present survey-based findings, including:

- Heatmaps of age and play duration distributions across active and churned players
- Stacked bar charts of event participation patterns against retention intention
- Heatmaps of NodKrai satisfaction ratings by play duration
- Horizontal bar charts of self-reported motivations for staying and desired improvements
- Horizontal bar charts of top churn reasons and conditions for return
- Donut chart presenting the three churn archetypes (Watching & Waiting, Passively Churned, Permanently Lost)

These analyses provided an initial understanding of behavioural diversity between active and churned player groups, forming the descriptive foundation for subsequent machine learning and clustering analysis.

3.6.3 Clustering Analysis

To identify hidden behavioural segments, unsupervised machine-learning techniques were applied:

- **K-Means Clustering:** Used to group players based on quantitative indicators such as login frequency, playtime, exploration level, and spending behaviour. Based on the Elbow Method and Silhouette Score, $K=2$ was selected as the optimal number of clusters, resulting in two distinct engagement profiles: a high-engagement segment and a low-engagement segment.
- **Self-Organising Maps (SOM):** Employed to visualise multi-dimensional similarity among players and uncover non-linear behavioural patterns. Given the sample size ($n=126$), the SOM results are treated as an exploratory topological visualisation to complement the K-Means findings.

Features were standardised using StandardScaler and missing values imputed with column-wise median before clustering. The optimal K was selected by evaluating Elbow Method and Silhouette Score across $K=2$ to $K=6$, with a minimum cluster size of 5 enforced. For SOM, an 8×8 grid was trained over 5,000 iterations. K-Means labels were then mapped onto SOM winner coordinates to generate a cross-validation agreement matrix, confirming consistency between both methods.

These cluster assignments were subsequently used as an analytical lens for post-hoc SHAP interpretation, enabling comparison of churn-driving factors across different player segments.

3.6.4 Classification and Prediction Models

After clustering, supervised learning models were used to predict if a player would churn. The cluster results from K-Means and SOM were not used as input features, but instead served as a way to analyze the model's decisions. Specifically, SHAP values were compared across different groups to show how various factors affect churn risk

for each type of player. This group-based approach links player segmentation with predictive modeling, helping to create more effective retention strategies.

- **Logistic Regression (LR):** Selected for its interpretability and ability to estimate the probability of churn based on behavioural attributes.
- **Random Forest (RF):** An ensemble method used to enhance prediction accuracy and handle non-linear interactions among variables.

Two separate models were built: a churn model using 5 shared demographic features across both groups (n=170), and a retention model using full behavioural features from active players only (n=126). An 80/20 train-test split was applied for evaluation. A second Random Forest trained on the full dataset was used solely for SHAP interpretation, as this is standard practice, which means SHAP is an explainability tool, not an evaluation metric.

While Survival Analysis (Kaplan–Meier estimation or Cox Proportional Hazards model) has been explored in prior studies as an auxiliary temporal modelling approach to estimate player retention duration (Viljanen et al., 2017), it was not applied in this study as the survey-based data does not provide the continuous time-series records required for such analysis. Instead, model performance was evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix, with Stratified 5-fold cross-validation applied to ensure the stability and reliability of predictive outcomes.

3.6.5 Explainable AI and Interpretation

To improve interpretability, the study incorporated Explainable AI (XAI) through the SHAP (Shapley Additive Explanations) framework.

A LinearExplainer was applied for Logistic Regression, and TreeExplainer for all Random Forest models. Feature importance was summarised as mean absolute SHAP values, with signed mean SHAP values used to distinguish features that increase versus decrease retention probability.

SHAP values quantify each feature's contribution to the churn prediction, enabling transparent explanation of the model's decision process. Visual explanations, such as SHAP summary plots and feature-importance charts, are integrated into the prototype dashboard to help users and supervisors understand why certain factors such as, play frequency, social influence, spending level are strongly influence churn outcomes.

3.6.6 Toolchain and Environment

All analyses in this study were carried out using Python within the Anaconda environment, with Jupyter Notebook serving as the development and execution platform.

The analytical pipeline was implemented across four Python scripts, each serving a distinct purpose: raw survey responses were first processed through feature engineering, applying ordinal, binary, one-hot, and multi-label binarisation encoding; multi-select question data were then restructured into long format to support Tableau visualisation; the encoded dataset was subsequently used for player segmentation; and finally, churn prediction and explainability analysis were conducted using machine learning and SHAP techniques.

The core libraries used across these scripts included pandas and NumPy for data manipulation, scikit-learn for machine learning, clustering, and feature encoding tasks, Matplotlib and Seaborn for data visualisation, SHAP for model explainability and feature-importance analysis, MiniSom for Self-Organising Map implementation, and openpyxl for structured Excel output formatting.

Given the survey dataset scale (n=170), pandas provided a sufficient and efficient environment for all preprocessing, modelling, and exploration tasks without requiring distributed computing frameworks.

Lastly, analytical outputs and selected visualisations were integrated into the project's website prototype for presentation and interactive exploration.

3.7 Ethical Considerations

Ethical considerations are essential to ensure the credibility, transparency, and integrity of this research. All procedures involving human participants and publicly available data were conducted in accordance with the University of Technology Sarawak (UTS) Research Ethics Guidelines and general academic standards for responsible research conduct.

3.7.1 Informed Consent and Voluntary Participation

All participants were informed about the purpose, duration, and confidentiality of the study before completing the Google Form questionnaire. The survey introduction clearly stated that participation was voluntary, responses would remain anonymous, and no personally identifiable information would be collected. Participants had the right to withdraw at any stage by choosing not to submit their responses. By proceeding with the questionnaire, each respondent provided informed consent to contribute their data for academic use only.

3.7.2 Privacy and Anonymity

No personally identifiable information, such as names, email addresses, or user IDs, was collected during data collection. Only non-identifiable metadata, including response time and timestamp, were recorded solely for data integrity and verification purposes. All survey responses were anonymised and coded numerically prior to analysis. The raw survey data will not be shared or distributed beyond this study, and access is restricted to the researcher only.

3.7.3 Use of Secondary and Public Data

The secondary data used in this study were derived exclusively from publicly accessible online platforms such as HoYoLAB, YouTube, Bilibili, and Google Play. Only open comment sections under official announcements, version trailers, and

review posts were selected. No usernames, profile links, or private messages were collected, ensuring that all content complied with public-domain fair-use standards. This approach respects both the intellectual property and privacy rights of online users while providing genuine community insights.

3.7.4 Avoidance of Harm and Data Integrity

The study posed no physical or psychological risks to participants. All survey questions were designed to focus on gameplay experience and personal opinions, avoiding any sensitive or invasive topics. Data were handled ethically throughout the process, including collection, cleaning, analysis, and presentation, make sure that no manipulation or misrepresentation occurred during interpretation or visualisation

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Introduction

This chapter presents the analysis and finding across five key components. Using a combination of survey responses and scraped community data, the study profiles active and churned players through descriptive dashboards. This research implements unsupervised learning for player segmentation, followed by machine learning with SHAP explainability to identify retention drivers. Finally, community-generated content is used to contextualize these findings, providing a multi-dimensional perspective on player behaviour in *Genshin Impact*.

4.2 Data Collection and Sampling Result

For primary data collection, an online survey was designed and distributed via Google Forms targeting both active and churned *Genshin Impact* players. A total of 170 valid responses were collected, 126 of respondents identified as active players and 44 as churned players, yielding a ratio consistent with the broader free-to-play player population where a segment of the active base eventually discontinues play.

In parallel, or secondary data collection, community-generated content was retrieved from four publicly accessible platforms: *Bilibili*, *YouTube*, *HoYoLAB*, and *Google Play*. Data collection covered game versions 4.8 through 6.4, covering both the *Natlan Arc* (v4.8–v5.8) and the *Luna Arc* (v6.1–v6.4). This period was chosen to compare how community sentiment and engagement changed across two major story

chapters, to providing longitudinal coverage across two major content arcs with distinct narrative and design characteristics.

4.3 Customer Profile Analysis

This section using Tableau dashboards to profile the survey sample, comparing active and churned players. Structured across six dashboards, the analysis addresses **Research Objective 1** by exploring player demographics, engagement patterns, and motivations. The narrative focuses on three core questions: **who plays, how they play, and why they leave.**

4.3.1 Active Player Demographics

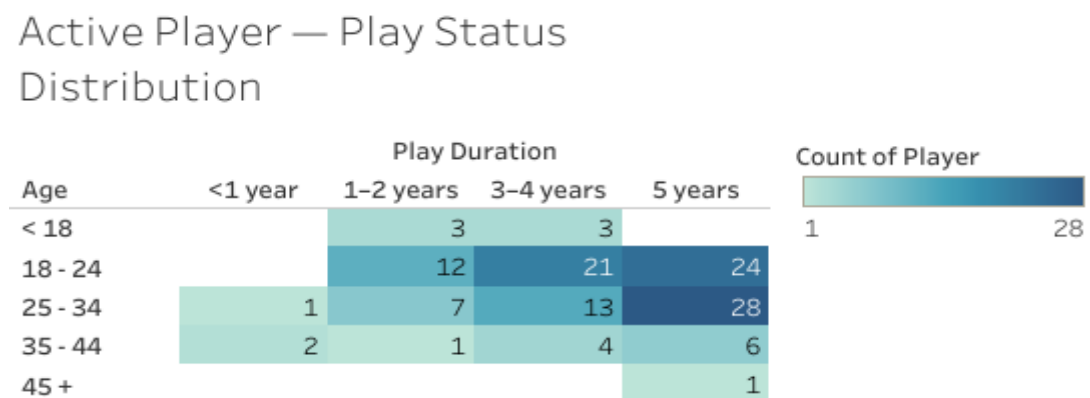


Figure 4.1: Active Player Demographics

Figure 4.1 presents the age and play duration distribution of active respondents. The heatmap shows that the active player base is largely made up of players aged 18 to 34. The highest concentration falls in the 25 to 34 age group with five or more years of playtime (n=28), closely followed by the 18 to 24 group also with five or more years (n=24). These two cells together account for a substantial portion of all active responses, suggesting that *Genshin Impact* has been able to hold on to players who have been with the game since its early days. Across all age groups, the five-year duration bracket consistently shows the most respondents, pointing to a stable long-

term player base that has remained engaged despite the natural churn risks that come with any live-service game.

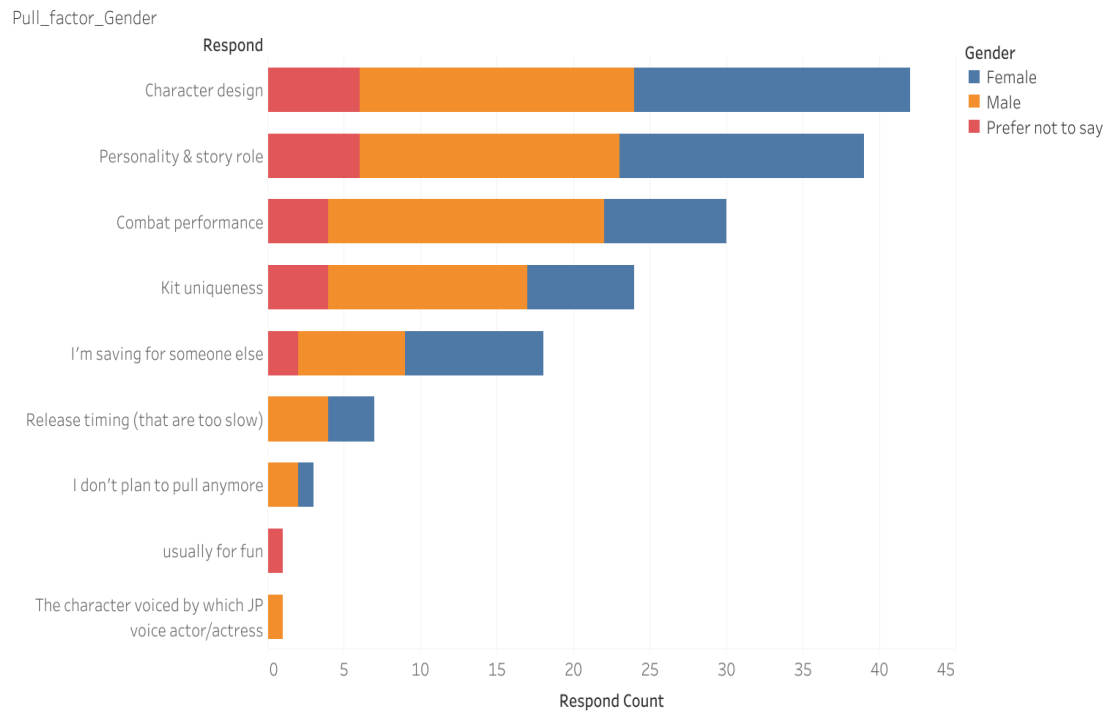


Figure 4.2: Pull Factor by Gender

Figure 4.2 shows what factors influence active players when deciding to pull for a character, broken down by gender. **Character Design** ranks first with 42 total responses, making it the single most influential pull factor across all gender groups. **Personality and Story Role** comes in second at around 39 responses, followed by **Combat Performance** (~30) and **Kit Uniqueness** (~24). Notably, these top four factors are all character-related rather than system-related, which suggests that players are drawn to specific characters rather than mechanical advantages. **Saving for Someone Else** (18 responses) also appeared as a notable consideration, reflecting how some players plan their Primogem budgets around upcoming characters they care about. Factors like **Release Timing Being Too Slow** and **I Don't Plan to Pull Anymore** ranked much lower, showing that most active players are still engaged with the gacha system.

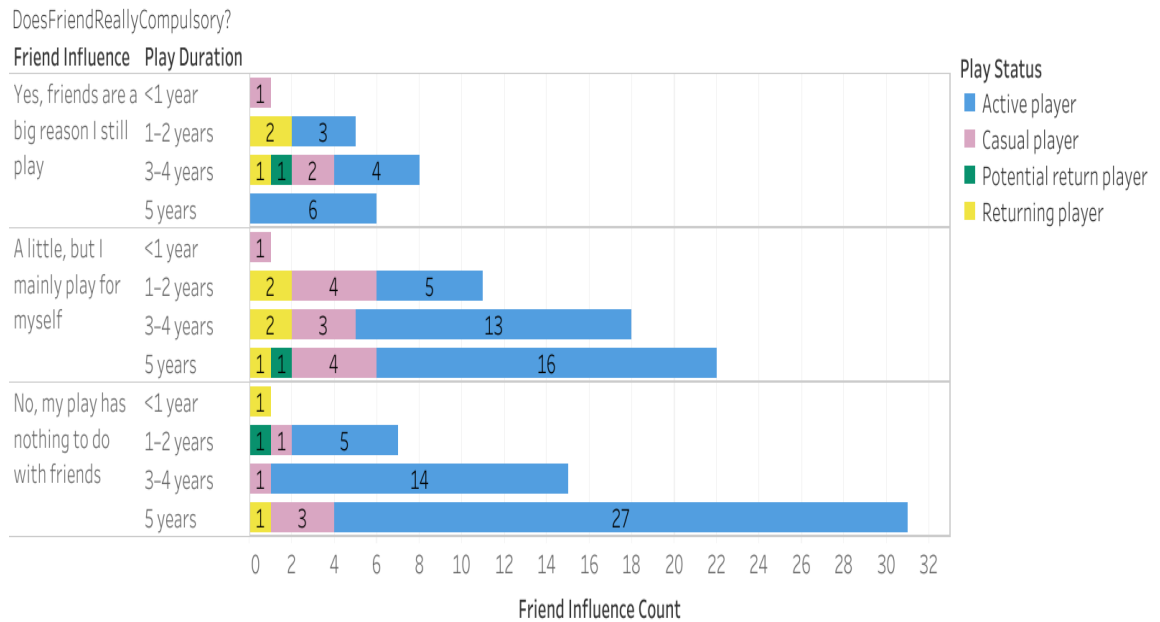


Figure 4.3: Does Friend Influence Really Matter

Figure 4.3 explores whether having friends still playing the game is a reason that keeps players around, broken down by how long they have been playing. A clear pattern emerges across play duration groups: the longer someone has been playing, the less they rely on friends as a motivation to continue. Among players with five or more years of playtime, the most common response is ***"No, my play has nothing to do with friends"*** (n=27 active players), compared to just 6 who said friends are a big reason they still play. In contrast, among players with fewer years of playtime, social influence plays a slightly bigger role. This trend suggests that long-term players have developed a self-sustained motivation to play that goes beyond social pressure, they stay just because they genuinely enjoy the game, not because their friends are around.

4.3.2 In-Game Behaviour

Task_Completion

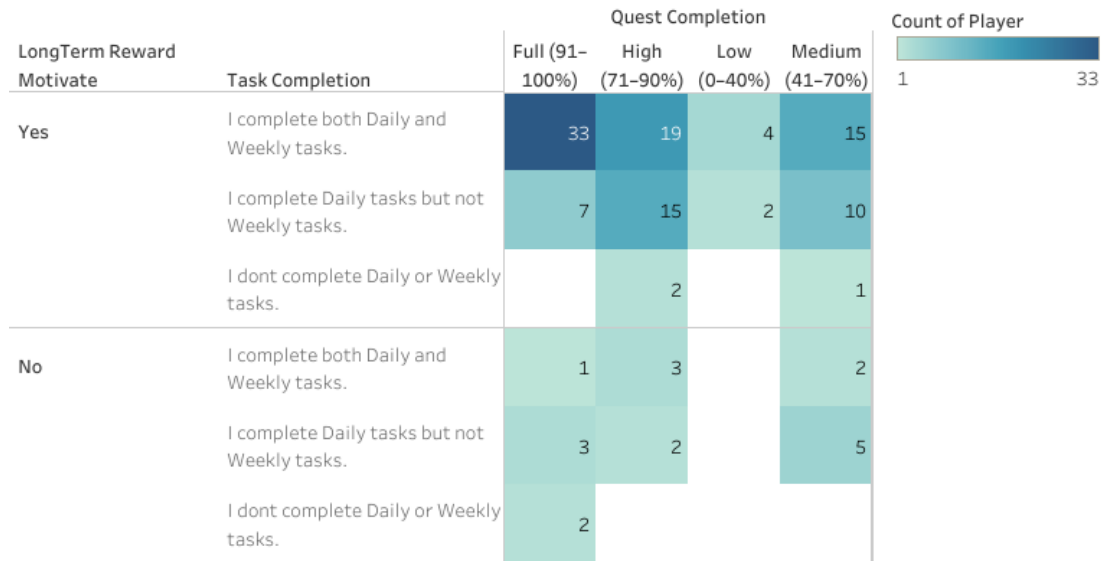


Figure 4.4: Task Completion Heatmap

Figure 4.4 presents the relationship between long-term reward motivation and in-game task completion behaviour. Among players who said they are motivated by long-term rewards, the highest count ($n=33$) falls in the intersection of **completing both Daily and Weekly tasks** and **Full quest completion (91–100%)**. This is the darkest cell in the heatmap, indicating the strongest behavioural pattern. Players who are driven by long-term progression tend to be more consistent in their daily routines and complete more of the game's quest content. In contrast, players who are not motivated by long-term rewards show lower and more scattered task completion behaviour across all categories. This proving that reward structure directly shapes how disciplined players are in their daily habits, a finding that becomes relevant when thinking about how changes to the reward system could influence overall player engagement.

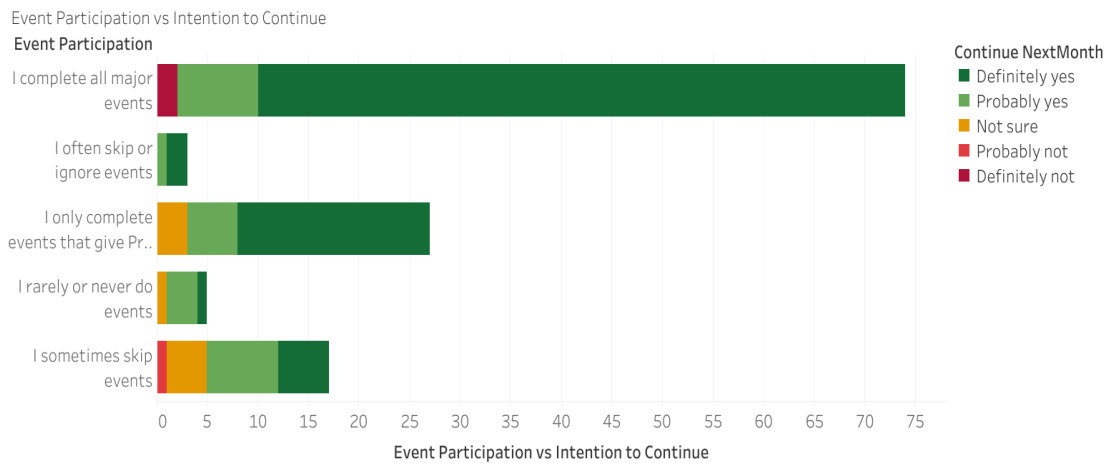


Figure 4.5: Event Participation vs Intention to Continue

Figure 4.5 shows the relationship between how actively players participate in events and whether they plan to keep playing. The pattern is very clear: players who **complete all major events** show an overwhelmingly strong intention to continue playing, with the vast majority selecting "**Definitely yes**". The bar for this group extends to around 74 responses, by far the longest in the chart. Players who **often skip or ignore events**, on the other hand, have a much smaller count and a more mixed intention distribution. Players who **only complete events that give Primogems** fall somewhere in between, showing that their engagement is more transactional in nature. Overall, this chart provides strong support for event participation being one of the most reliable indicators of player retention, consistent with what the SHAP analysis in Section 4.5 later confirms.

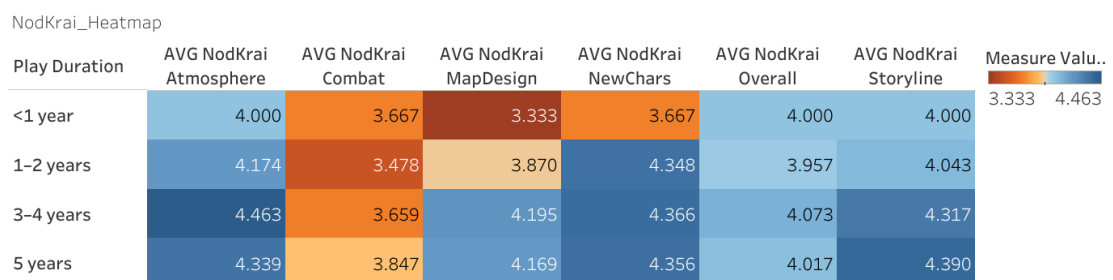


Figure 4.6: Nod-Krai Region Satisfaction by Play Duration

Figure 4.6 presents the average satisfaction ratings across six dimensions of the Nod-Krai region, broken down by how long players have been playing. A consistent pattern appears across all play duration groups: **Combat** receives the lowest ratings across the board, ranging from 3.478 (1 to 2 years) to 3.847 (5 years or more). **Atmosphere** and **Storyline**, on the other hand, consistently receive higher ratings across all groups. Interestingly, longer-term players (3 to 4 years and 5 years) tend to rate **Atmosphere** and **Storyline** higher than newer players, with the 3 to 4 year group giving **Atmosphere** the highest average of 4.463. This may partly reflect the fact that Nod-Krai is a smaller region compared to areas like Sumeru or Fontaine, making it easier for veterans to fully explore and collect resources, which translates into a more satisfying overall experience. The consistently low combat scores across all groups, however, indicate a long-standing issue that does not improve with play duration, making it a design pain point worth addressing.

4.3.3 Motivations and Spending Behaviour

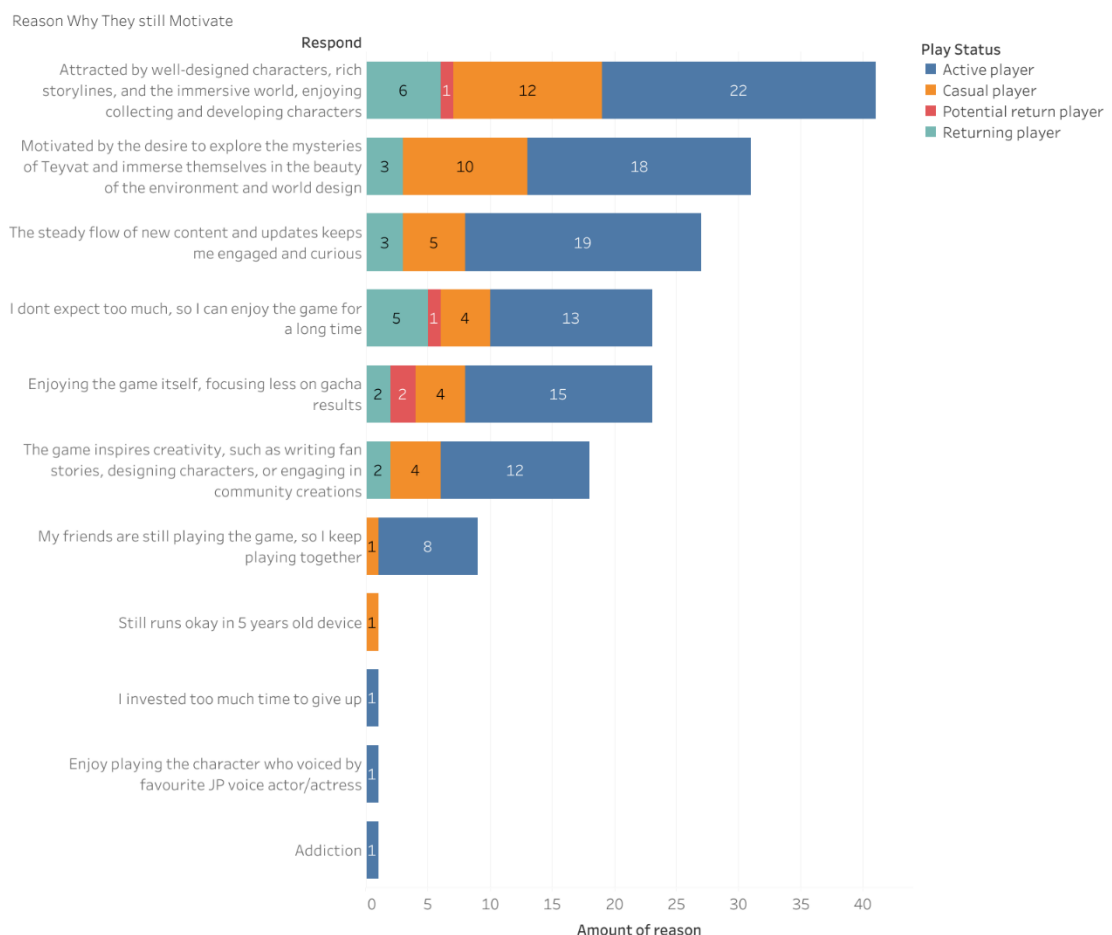


Figure 4.7: Reasons Why Players Stay Motivated

Figure 4.7 shows the self-reported reasons why active players continue playing *Genshin Impact*. The top response, selected by 41 respondents, is ***"Attracted by well-designed characters, rich storylines, and the immersive world, enjoying collecting and developing characters"***. This is significantly higher than the second-ranked reason, ***"Motivated by the desire to explore the mysteries of Teyvat and immerse themselves in the beauty of the environment and world design"*** (n=31), and the third, ***"The steady flow of new content and updates keeps me engaged and curious"*** (n=27). Further down, ***"I don't expect too much, so I can enjoy the game for a long time"*** and ***"Enjoying the game itself, focusing less on gacha results"*** (both around 23 responses) highlight that a portion of the player base maintains engagement by managing their own expectations rather than chasing every new release. Social motivation, represented by ***"My friends are still playing, so I keep playing together"***, ranks notably lower at 9 responses, which aligns with the finding in Figure 4.3 that long-term players are less friend-dependent. Overall, character design and narrative immersion are the core reasons players stay, rather than competitive or social factors.

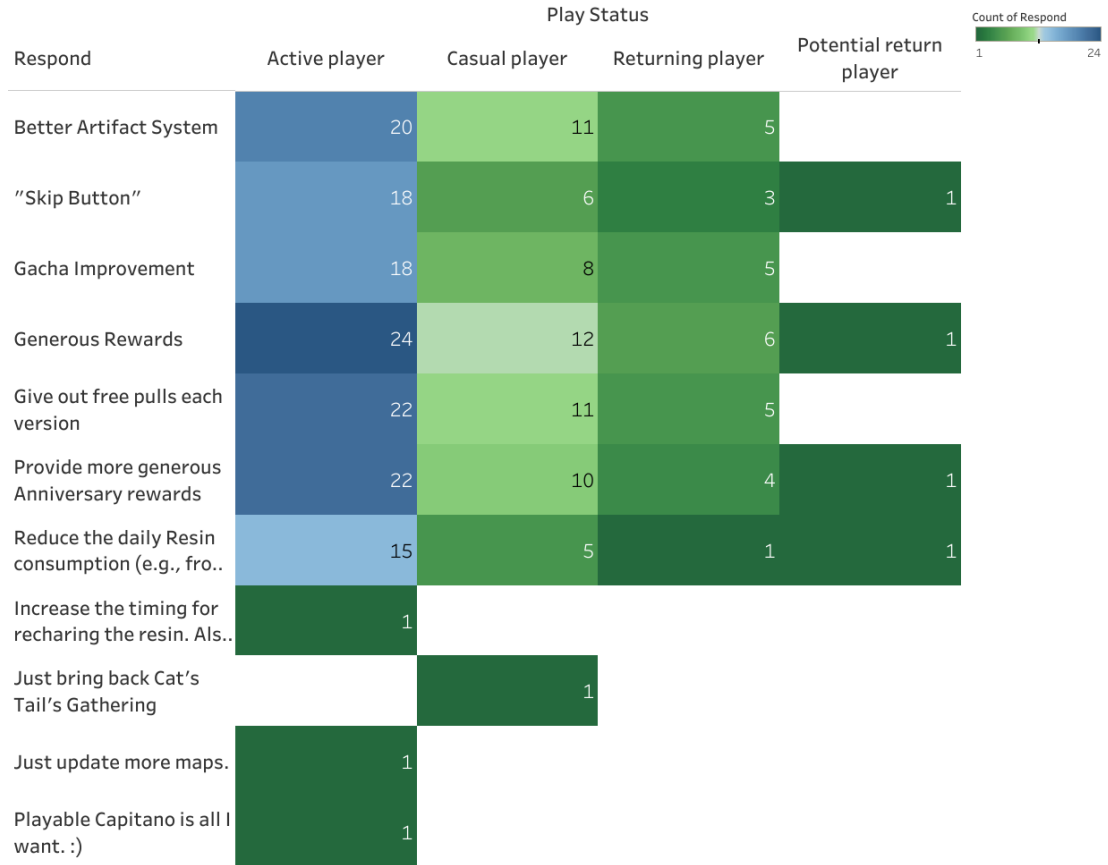


Figure 4.8: Desired Game Improvements by Play Status

Figure 4.8 presents what improvements active players would most like to see, broken down by their current play status (Active, Casual, Returning, and Potential Return players). Across all groups, **Generous Rewards (More free pulls/Primogems)** ranks first with 43 total mentions (24 active + 12 casual + 6 returning + 1 potential return), making it the most widely requested improvement by a clear margin. **Give out free pulls each version** (38 total) and **Provide more generous Anniversary rewards** (37 total) follow closely, both of which are also reward-related. **Better Artifact System** (36 total) comes next, followed by the **"Skip Button" for dialogue or faster dailies** (28 total) and **Gacha Improvement** (26 total). The fact that the top three most-requested improvements are all reward-related sends a clear message: players across all engagement levels feel the reward system is not generous enough. This connects directly to the churn findings in Section 4.3.6, where the same reward concerns emerge as the main barrier to bringing churned players back.

Spending Behaviour

In terms of spending, the majority of active respondents are free-to-play, with 41% reporting no spending during the Natlan patch cycle (v5.0–v5.8). Among those who did spend, most were light spenders, with around 23% spending between USD \$4 and \$11, and 13% between USD \$22 and \$44, while only about 12% were heavy spenders at USD \$110 or above. This distribution confirms that free in-game rewards and events are what keep most of the player base engaged day-to-day, and that HoYoverse's revenue from this segment comes primarily from a small group of higher-spending players.

4.3.4 Churned Player Demographics and Segmentation

Churn Age × Play Duration Heatmap

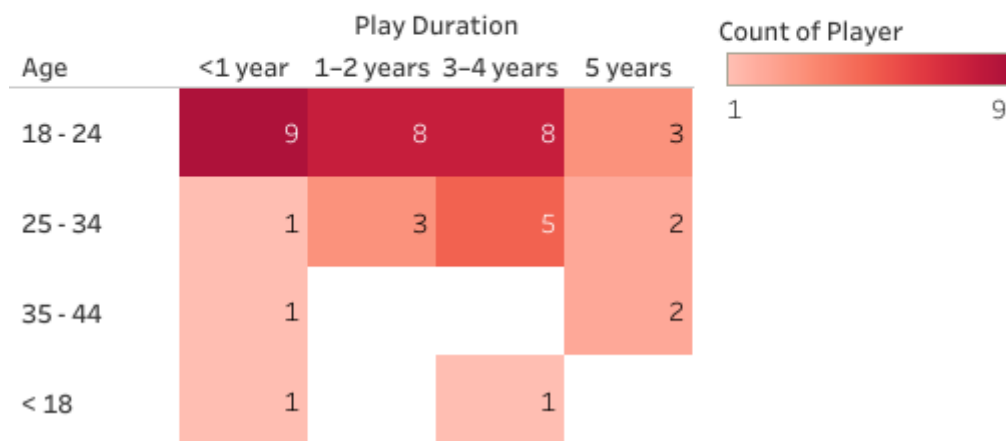


Figure 4.9: Churn Age × Play Duration Heatmap

Figure 4.9 shows the age and play duration distribution for the 44 churned respondents. Churned players are concentrated in the 18 to 24 age group, with the highest count in the **under 1 year** play duration cell (n=9). The **1 to 2 years** and **3 to 4 years** cells within the 18 to 24 group are also high (n=8 each), while the **5 years or more** cell drops to just 3. The 25 to 34 age group also shows some churn, particularly in the 3 to 4 year range (n=5), though the numbers are lower overall. This pattern

suggests that players who are newer to the game tend to churn more easily, as they have not yet built up as deep a connection to the characters and story as long-term players have. Players who have stayed for five years or more, on the other hand, are far less likely to churn, likely because their investment in the game in terms of both time and money makes it harder to walk away.

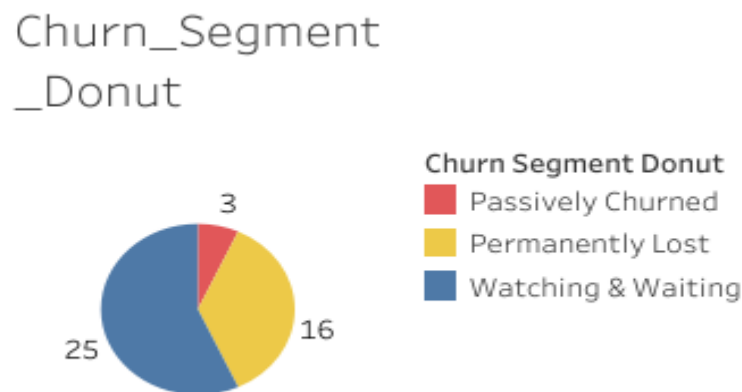


Figure 4.10: Churn Segment Distribution

Figure 4.10 presents the churn segment breakdown for all 44 churned respondents using a donut chart. The largest group is **Watching and Waiting** (n=25, approximately 56.8%), representing players who have stopped playing but are still following game news, watching previews, or staying connected to the community. The second group is **Permanently Lost** (n=16, approximately 36.4%), made up of players who have fully disengaged and show little to no interest in returning. The smallest group is **Passively Churned** (n=3, approximately 6.8%), consisting of players who drifted away gradually without a single clear reason.

The dominance of the **Watching and Waiting** segment is an encouraging sign from a business perspective. More than half of churned players are not completely gone, as they are still paying attention to the game and could potentially be brought back with the right content or reward improvements. Together, **Watching and Waiting** and **Passively Churned** make up about 63% of all churned respondents, suggesting that a meaningful portion of the churned population remains reachable through targeted re-engagement efforts.

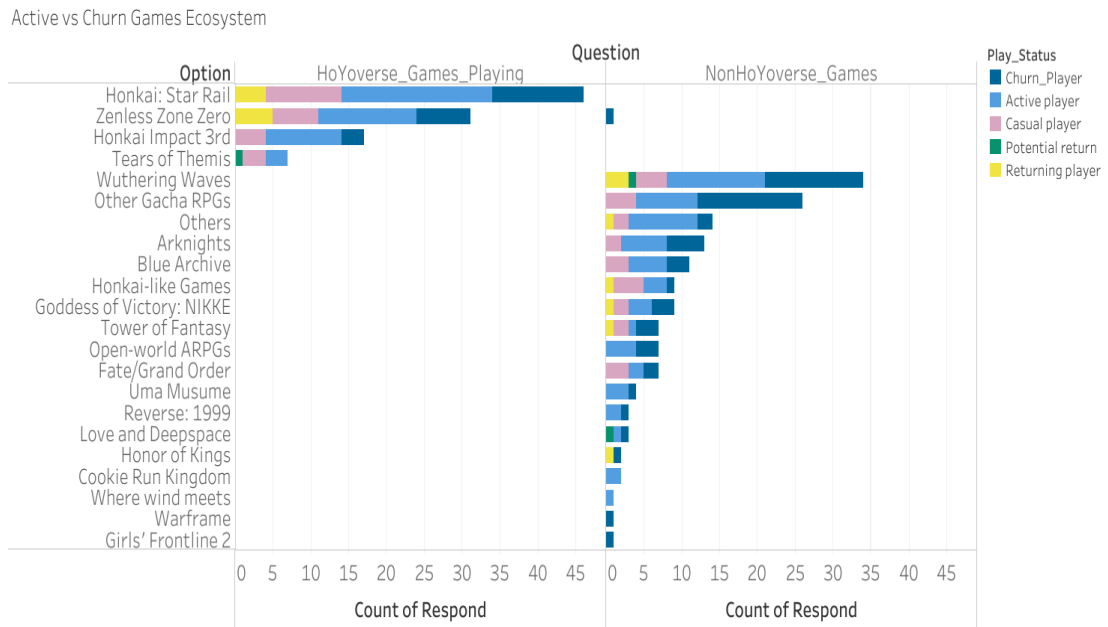


Figure 4.11: Games Ecosystem Comparison: Active vs Churned Players

Figure 4.11 provides a side-by-side comparison of the games that both active and churned players are playing, split into HoYoverse titles (left panel) and non-HoYoverse titles (right panel).

On the HoYoverse side, *Honkai: Star Rai** is the most commonly played title alongside *Genshin Impact* with around 45 total responses, followed by *Zenless Zone Zero* (~30) and *Honkai Impact 3rd* (~16). This shows that cross-game engagement within the HoYoverse ecosystem is fairly common among active players.

On the non-HoYoverse side, *Wuthering Waves* stands out clearly as the top destination for churned players, with its bar dominated by the dark blue (Churn_Player) colour at around 34 responses. This is significantly higher than the next option, *Other Gacha RPGs* (~25 responses), and well above games like *Arknights*, *Blue Archive*, and *Honkai-like games* (~11 to 12 each). The fact that *Wuthering Waves*, a structurally similar open-world action RPG with gacha mechanics, is where churned players are going suggests that players are not leaving the genre but rather switching to a competitor. This makes it a direct competitor for the churned player segments identified in this study, and is discussed further in the HoYoverse ecosystem analysis in Section 4.5.7.

4.3.5 Top Churn Factors

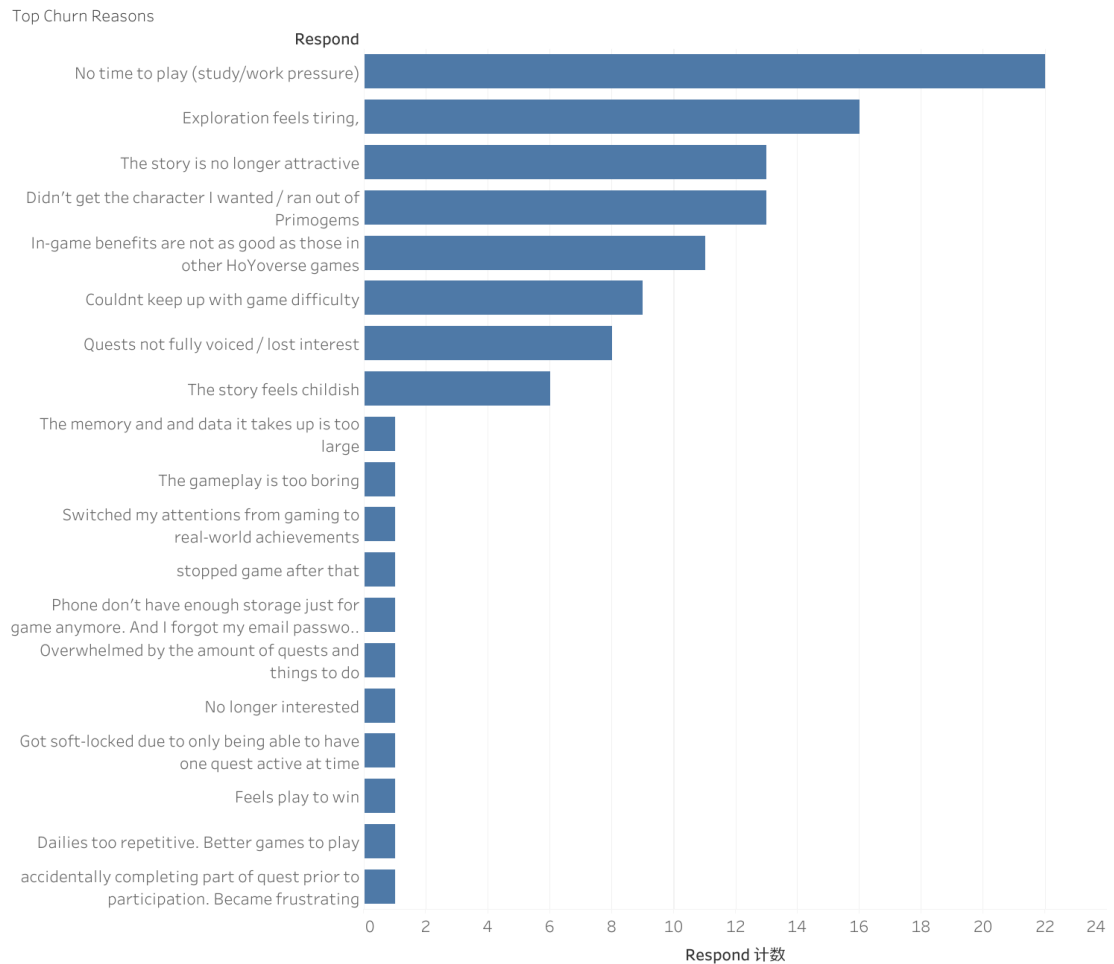


Figure 4.12: Top Churn Factors

Figure 4.12 presents the ranked churn reasons reported by the 44 churned respondents. The reasons can be grouped into two broad types, which helps to understand the nature of player dropout more clearly.

The first type is **situational churn**, referring to players who stopped because of real-life circumstances rather than dissatisfaction with the game itself. *No Time to Play (study/work pressure)* leads this group with 22 mentions, making it the single most cited churn reason overall. Players in this category did not necessarily leave because the game failed them; life simply got in the way. This group is generally more recoverable, as their disengagement is tied to temporary conditions rather than a permanent change in attitude toward the game.

The second type is **design-driven churn**, where the game's own structure pushed players away. *Exploration Feels Tiring* (16 mentions) and *The Story Is No Longer Attractive* (13 mentions) are the leading reasons here. *Didn't Get the Character I Wanted / Ran Out of Primogems* (13 mentions) and *In-Game Benefits Are Not as Good as Those in Other HoYoverse Games* (11 mentions) point to reward dissatisfaction as a compounding factor, a theme already established in Section 4.3.3 through active players' improvement requests, and one that cuts across both player groups.

Together, the mix of situational and design-driven reasons suggests that a meaningful share of churned players left under circumstances that could be addressed, either through re-engagement during quieter real-life periods, or through changes to the game's exploration pacing and reward structure. These pain points form the starting point for understanding what conditions would bring players back, which is examined in the following section.

4.3.6 Conditions for Return

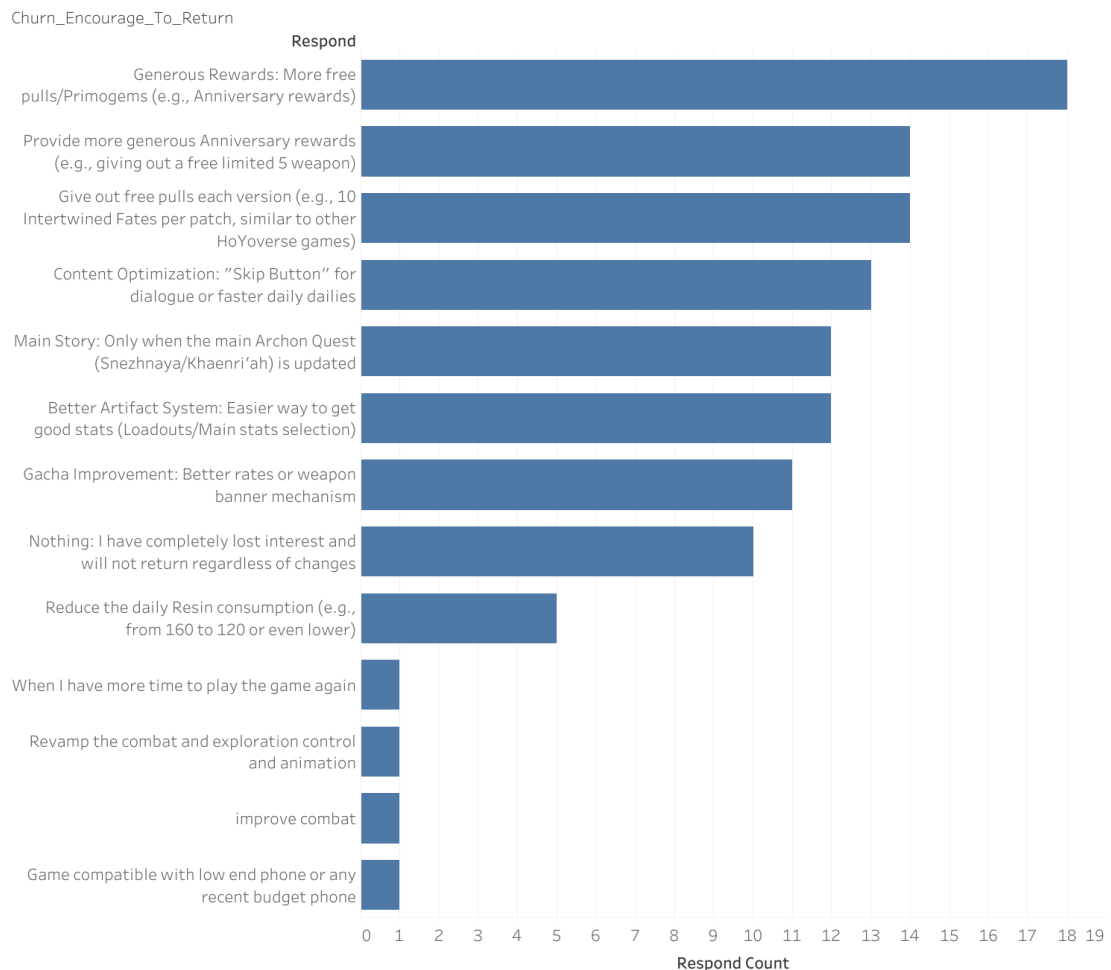


Figure 4.13: Conditions for Encouraging Return

Figure 4.13 shows what changes churned players say would most likely bring them back to *Genshin Impact*. **Generous Rewards: More Free Pulls/Primogems** ranks first with 18 mentions, followed jointly by **Provide More Generous Anniversary Rewards** and **Give Out Free Pulls Each Version** (14 mentions each). **Content Optimisation: Skip Button for Dialogue or Faster Dailies** (13 mentions) and **Main Story Update for Snezhnaya/Khaenri'ah** (12 mentions) round out the top five.

What stands out is how closely these return conditions mirror the desired improvements reported by active players in Section 4.3.3. Both groups, those still playing and those who have already left, point to the same core issues: reward generosity and content accessibility. This consistency is one of the more meaningful

findings in the survey data. Improving the reward system is not just a retention tool for players who are still around; it is also the most cited re-engagement lever for players who have already walked away.

Notably, 10 respondents selected *"Nothing: I have completely lost interest and will not return regardless of changes"*, which aligns with the **Permanently Lost** segment identified in Section 4.3.4. However, the majority of churned respondents did identify at least one condition that would encourage their return, suggesting that the window for re-engagement remains open for most of this group.

Taken together, Sections 4.3.5 and 4.3.6 establish a clear picture from the players' perspective: the game's exploration pacing, reward generosity, and content update frequency are the areas most in need of attention. The following sections examine whether this picture is confirmed at the model level through machine learning and SHAP analysis.

4.4 Player Segmentation

This section reports the results of unsupervised learning applied to the survey dataset, identifying natural groupings within the active and churned player populations. Two methods are used: K-Means clustering and Self-Organising Maps(SOM). Results from both methods are compared to check consistency.

4.4.1 Optimal K Selection



Figure 4.14: Optimal K Selection

Figure 4.14 presents the results of the Elbow Method and Silhouette Score analysis across K=2 to K=6 for the active player dataset. The Elbow curve shows a clear bend at K=2, after which the improvement becomes much smaller. The Silhouette Score supports this, K=2 gives the highest score of approximately 0.23, with all later values going down. Although 0.23 is on the lower end, it is the best result possible given the small number of respondents (n=126) and the complex nature of the survey features. K=2 was therefore chosen as the best cluster count, giving a clear and useful two-group structure for the active player population.

4.4.2 Active Player K-Means Clusters



Figure 4.15: Active Player K-Means Clustering (K=2)

Figure 4.15 presents the PCA-reduced two-dimensional view of the K-Means clustering result for active players (n=126). The two clusters show some separation in the reduced feature space.

Cluster 0 is characterised by higher scores in engagement-related features, *more frequent logins, higher event participation, and stronger task completion*. This group is labelled the *High Engagement segment*. Cluster 1 shows lower scores in the

same areas, representing a *Lower Engagement segment* that are still active, shows patterns that suggest lower intention to keep playing. The SHAP per-cluster analysis in Section 4.4.6 looks further into how retention risk differs between these two groups.

4.4.3 Churned Player K-Means

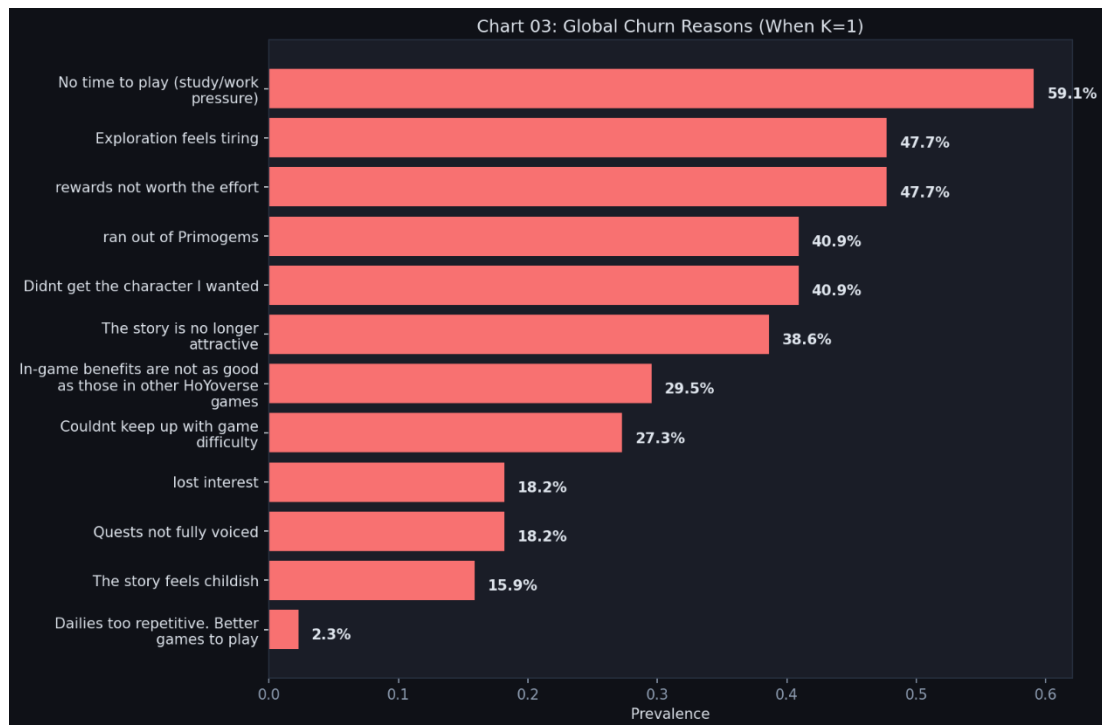


Figure 4.16: Churned Player K-Means Clustering (K=1)

Figure 4.16 presents the K-Means output for the churned player dataset (n=44). Due to the small sample size, the best K for the churned group turned out to be K=1, meaning the churned respondents cannot be split into separate sub-groups in a meaningful way. Rather than forcing an artificial multi-group structure, K=1 is kept and the single-cluster output is treated as an overall picture of churn reasons across all churned players.

The chart shows that *No Time to Play* (59.1%) is the most common churn reason, cited by more than half of all churned respondents. *Exploration Feels Tiring* (47.7%) and *Rewards Not Worth the Effort* (47.7%) follow jointly in second place, with *Ran Out of Primogems* (40.9%) and *Didn't Get the Character I Wanted* (40.9%)

in joint fourth. These five leading reasons represent the most widespread churn drivers and directly support the findings in Section 4.3.5. Reasons such as *Lost Interest* (18.2%) and *The Story Feels Childish* (15.9%) appear less frequently, suggesting they are secondary rather than main churn drivers for most respondents. The only reason at the bottom, *Dailies Too Repetitive / Better Games to Play* (2.3%) — is the only one that directly mentions competing games, indicating that switching to another game is a minor stated factor compared to push factors such as time pressure and reward dissatisfaction.

4.4.4 Self-Organising Map and Consistency Check

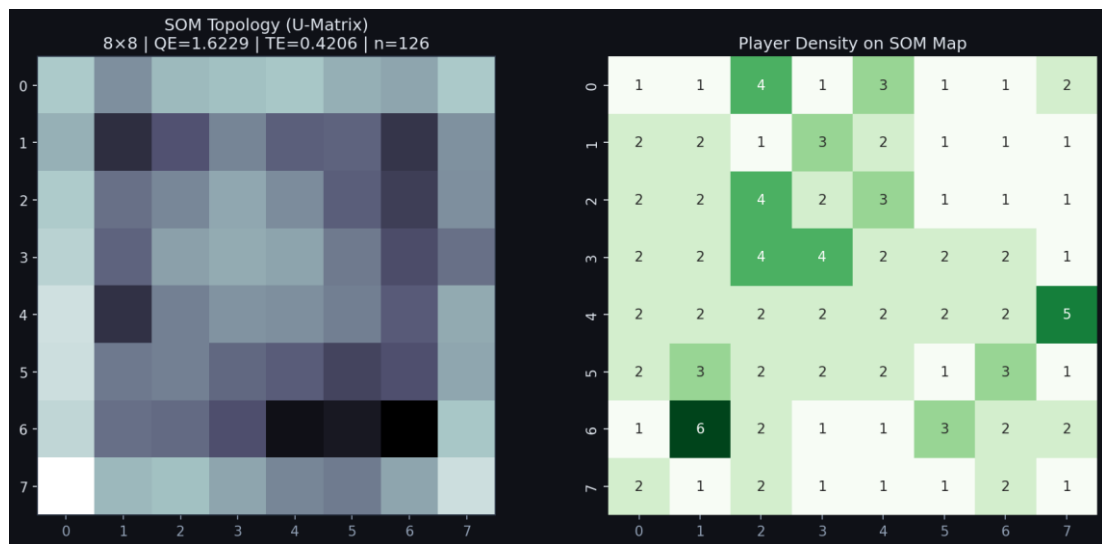


Figure 4.17: Active Player Self-Organising Map

Figure 4.17 presents the SOM topology for the active player dataset. The 8x8 grid produced a Quantisation Error (QE) of 1.6229 and a Topographic Error (TE) of 0.4206, both of which are higher than ideal. This is largely because the 64-node grid is too large for the relatively small sample of n=126 active players. Given this limitation, the SOM output is treated as a supplementary exploratory tool rather than a standalone result, with its main purpose being to cross-check the K-Means findings using a different method.

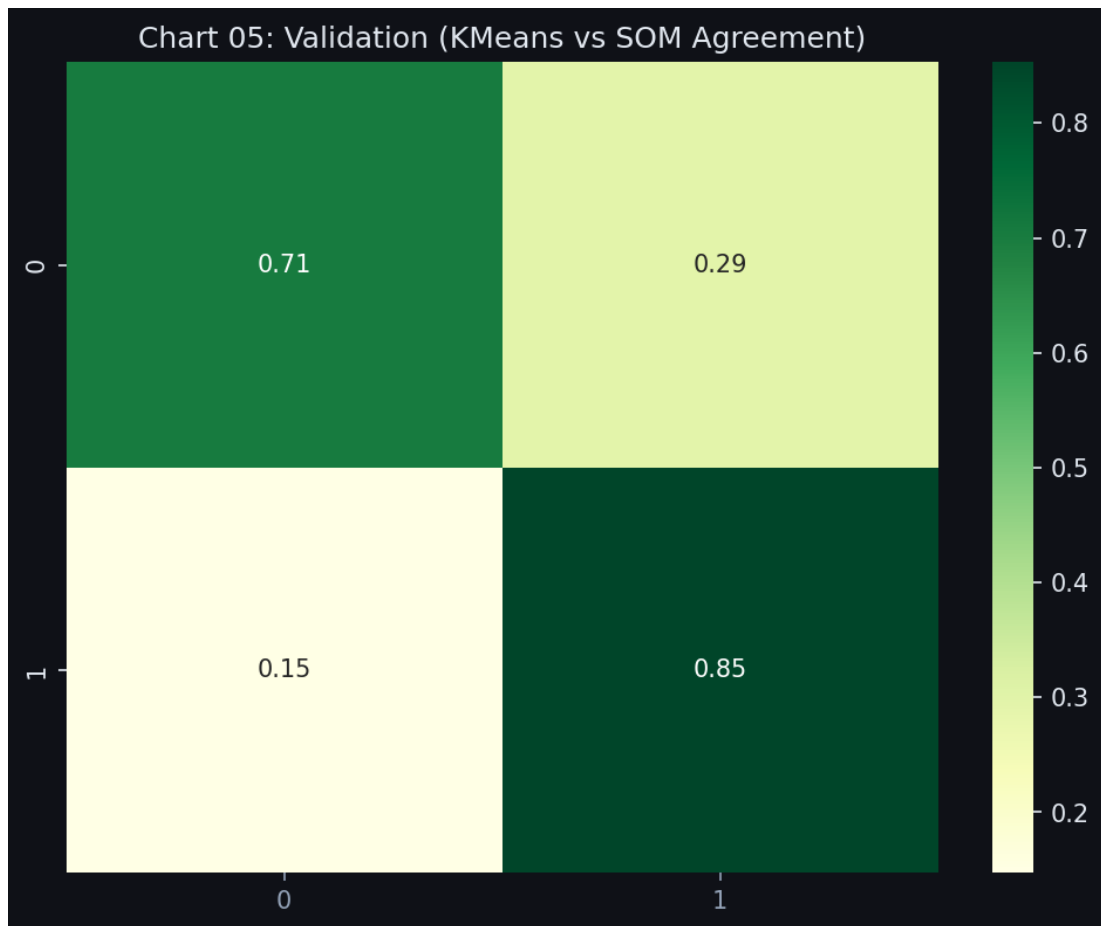


Figure 4.18: K-Means vs SOM Agreement

Figure 4.18 presents the comparison between K-Means cluster assignments and SOM node groupings. The agreement matrix confirms that the two-group structure identified by K-Means is broadly supported by the SOM, with most respondents placed in the same group by both methods. This consistency suggests that the High Engagement versus Lower Engagement distinction is not simply a product of how each algorithm works, but reflects a genuine pattern in the data that both methods are independently picking up.

4.4.5 Per-Cluster SHAP Insights

Figures 4.19 and 4.20 present the SHAP feature importance profiles broken down by K-Means cluster for active and churned players respectively. This analysis connects

the segmentation findings of this section to the SHAP prediction analysis in Section 4.5.

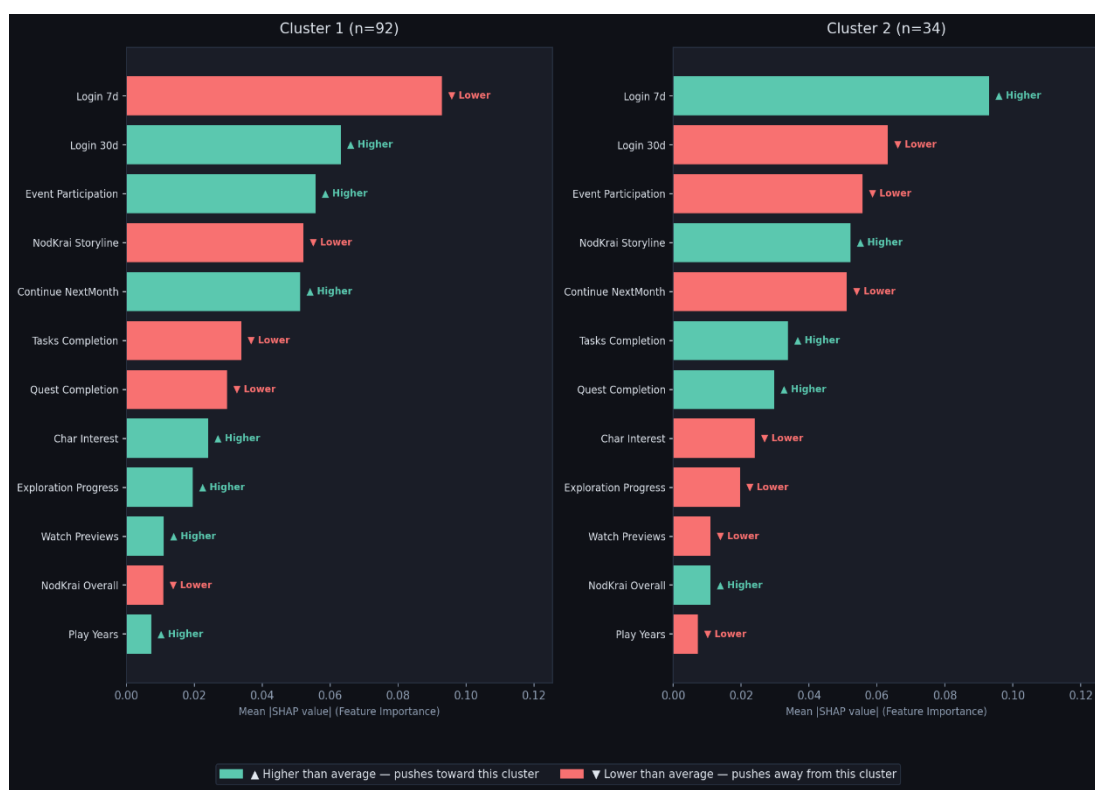


Figure 4.19: Active Player Per-Cluster SHAP Feature Importance

Figure 4.19 presents the SHAP feature importance profiles broken down by K-Means cluster for active players. The per-cluster profiles show that the features driving retention risk differ between the two segments. For the Lower Engagement cluster, login frequency and event participation carry higher SHAP contributions, suggesting that a drop in regular in-game activity is the more immediate warning sign for this group rather than dissatisfaction with content itself. This has practical implications for re-engagement: players in the Lower Engagement cluster may respond well to activity-based incentives such as daily login bonuses or event reminders, while players in the High Engagement cluster are likely more sensitive to content quality.



Figure 4.20: Why Stay vs Why Leave

Figure 4.20 provides a supporting view for churned players, showing that retention and churn are driven by different dimensions. Interest in upcoming characters, login frequency, and event participation are the strongest signals on the retention side, while time pressure and reward dissatisfaction dominate on the churn side. These patterns are consistent with the survey findings in Section 4.3.5 and are examined further at the model level in Section 4.5.

4.5 Churn Prediction and SHAP Analysis

This section reports the results of the machine learning models and SHAP-based explainability analysis. Building on the behavioural patterns identified in Section 4.3, the models here test whether those patterns hold up at a statistical level and identify which features carry the most predictive weight.

4.5.1 Model Performance

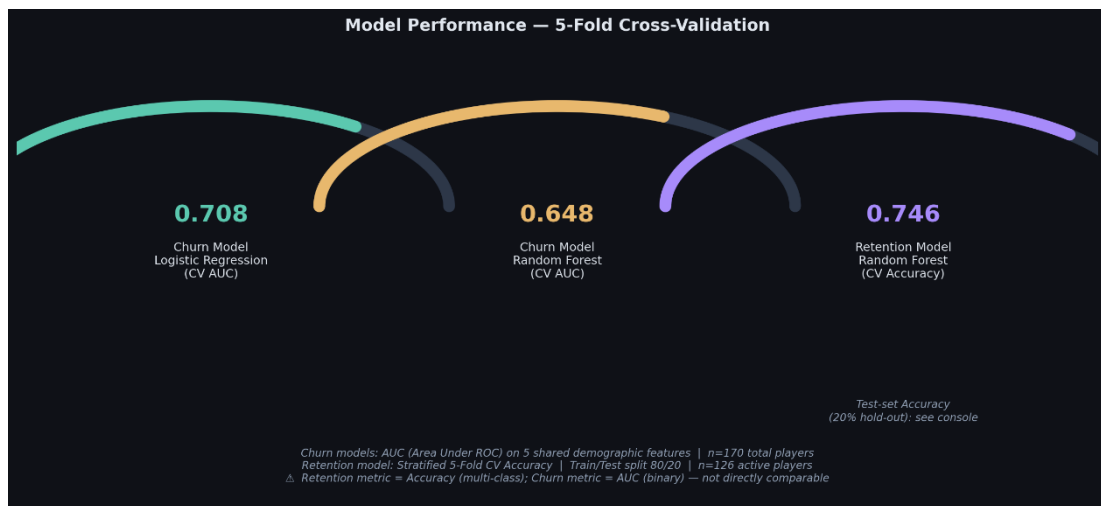


Figure 4.21: Model Performance Summary

Figure 4.21 presents the model performance summary for both tasks. For the churn prediction task, Logistic Regression (LR) achieved a 5-Fold Cross-Validation AUC of 0.708, outperforming Random Forest (RF), which yielded an AUC of 0.648. This unexpected result, where a simpler model outperforms an ensemble and it is consistent with the known behaviour of Random Forest in low-dimensional feature spaces. The churn model was limited to five basic demographic features available across both active and churned respondents, which puts Random Forest at a disadvantage compared to Logistic Regression.

Churn model – LR CV AUC: 0.708 RF CV AUC: 0.648				
Retention Model – 5-Fold CV Accuracy: 0.746				
Retention Model – Classification Report (TEST set, out-of-sample):				
	precision	recall	f1-score	support
2	0.00	0.00	0.00	3
3	0.50	0.29	0.36	7
4	0.73	1.00	0.84	16
accuracy			0.69	26
macro avg	0.41	0.43	0.40	26
weighted avg	0.58	0.69	0.62	26

Figure 4.22: Retention Model Classification Report

For the retention prediction task, the Random Forest model achieved a 5-Fold Cross-Validation Accuracy of 0.746 on the full active player dataset (n=126). The test set accuracy on the 20% hold-out (n=26) was 0.69, with a weighted F1-score of 0.62 (see Figure 4.15). These results should be read carefully because the data is unevenly distributed: the lowest retention intent classes (0 and 1) have too few samples to appear in the test split, and the Class 2 F1-score of 0.00 reflects the model's inability to identify this minority class. The 5-Fold CV accuracy of 0.746 is considered more reliable, as it averages across the full dataset and is less affected by the small test set.

4.5.2 Retention Correlation Analysis

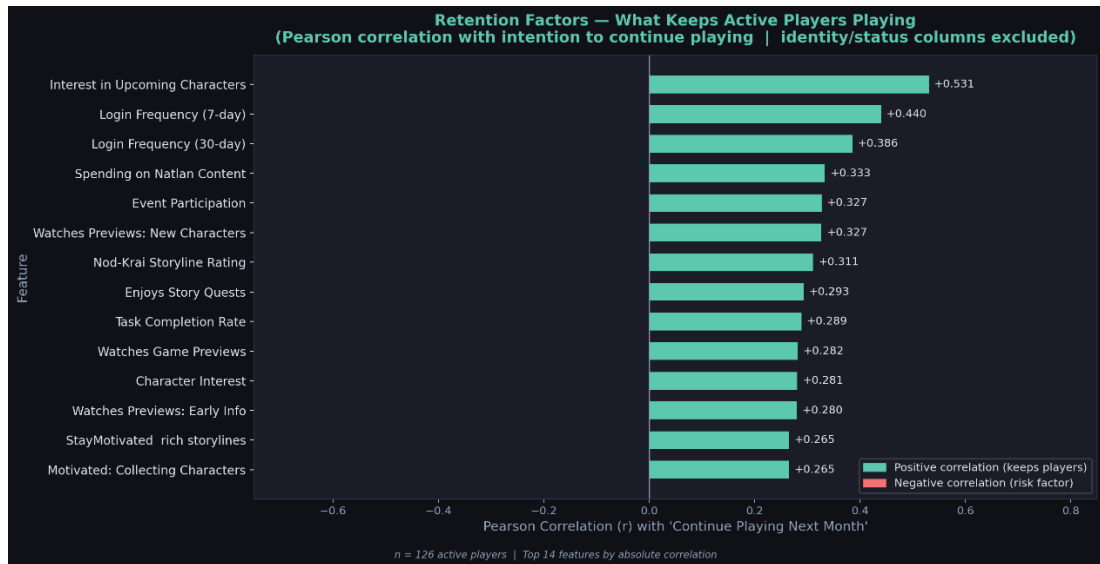


Figure 4.23: Retention Feature Correlation

Figure 4.23 presents the Pearson correlation coefficients between the top 14 behavioural features and the retention target, with identity and status columns excluded before analysis. All 14 features shown carry a positive correlation with retention intent, meaning higher values in these features are associated with a stronger intention to keep playing.

It is worth noting that Login Frequency (7-day, $r=+0.440$) and Login Frequency (30-day, $r=+0.386$), while showing high correlation values, are better understood as indicators of retention rather than drivers of it. A player who logs in frequently is already retained by definition, so these features reflect the outcome rather than explaining what caused it. The more meaningful features to examine are those that represent genuine behavioural motivations.

Among those, *Interest in Upcoming Characters**shows the strongest link ($r=+0.531$), suggesting that anticipation for future content is the most reliable signal of whether a player intends to stay. *Spending on Natlan Content* ($r=+0.333$) and *Event Participation* ($r=+0.327$) follow, pointing to active investment in the game, both financially and in terms of engagement, as meaningful retention drivers. Further down, *Nod-Krai Storyline Rating* ($r=+0.311$), *Enjoys Story Quests* ($r=+0.293$), and *Task*

Completion Rate ($r=+0.289$) highlight that satisfaction with narrative content and consistent in-game habits also play a role. Once these values start to drop, it is a signal that the player is gradually beginning to disengage from the game.

4.5.3 Key Retention Features - SHAP Analysis

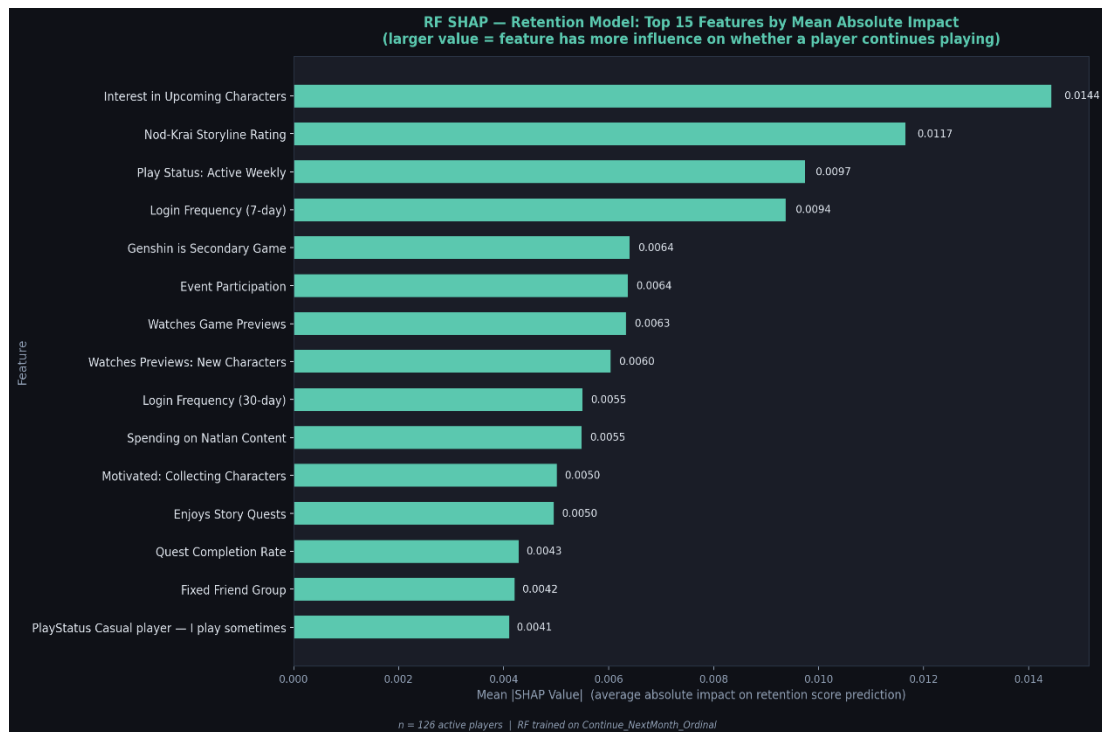


Figure 4.24: Top 15 Retention Features

Figure 4.24 presents the Random Forest SHAP mean absolute feature importance for the top 15 retention features, averaged across all 126 active player predictions. **Interest in Upcoming Characters** ranks first with a mean absolute SHAP value of 1.4432, making it the strongest single predictor of retention. This shows that how excited a player is about future character releases plays the biggest role in whether they continue playing. **Nod-Krai Storyline Rating** (SHAP=1.1659) ranks second, followed by **Play Status: Active Weekly** (0.9746) and **Login Frequency (7-day)** (0.9368). These top four features cover two key areas, content satisfaction and regular play habits, which is consistent with the findings in Section 4.5.2. The extended view to 15 features further includes **Fixed Friend Group** and **Play Status: Casual Player**,

confirming that social context and play style also play a secondary role in the model's predictions beyond content and login habits.

4.5.4 Feature Direction Analysis

Figures 4.25 and 4.26 show whether each feature pushes retention up or down, and how this plays out at the individual level.

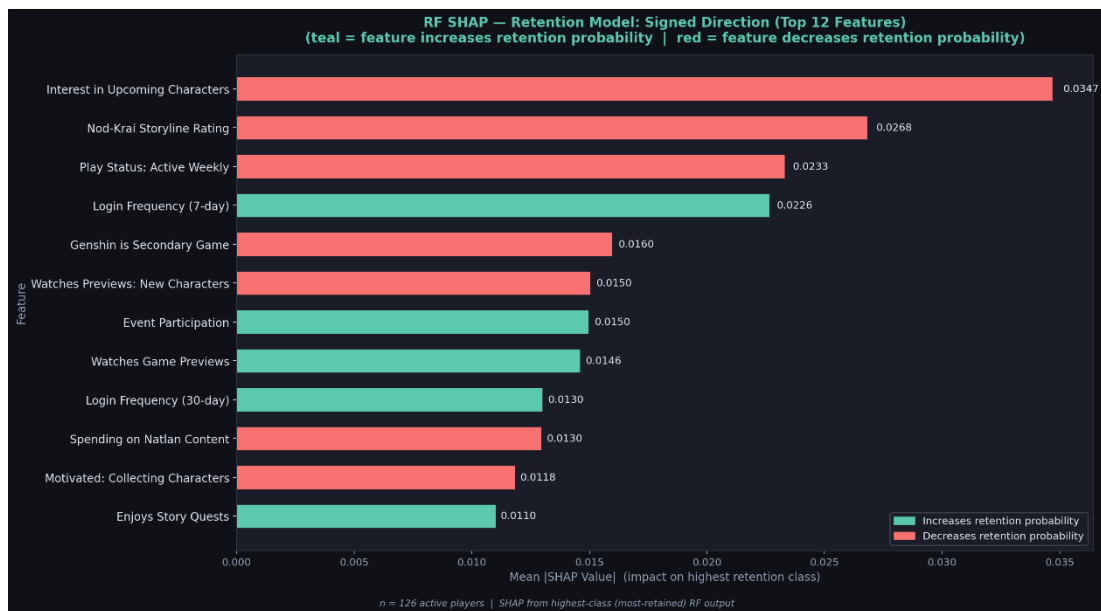


Figure 4.25: RF SHAP Directional Attribution

Figure 4.25 shows the signed SHAP values, where teal bars indicate features that increase predicted retention and red bars indicate features that push predictions toward lower retention. Features including *Login Frequency (7-day and 30-day)*, *Event Participation*, *Watches Game Previews*, and *Enjoys Story Quests* show positive direction. Conversely, *Interest in Upcoming Characters*, *Nod-Krai Storyline Rating*, *Play Status: Active Weekly*, and *Spending on Natlan Content* show negative signed values. This is a result of how the prediction categories are set up: for high-engagement features, the main effect may be toward middle rather than top retention classes, producing a signed value that appears negative. This chart should therefore be read together with the importance charts rather than on its own.

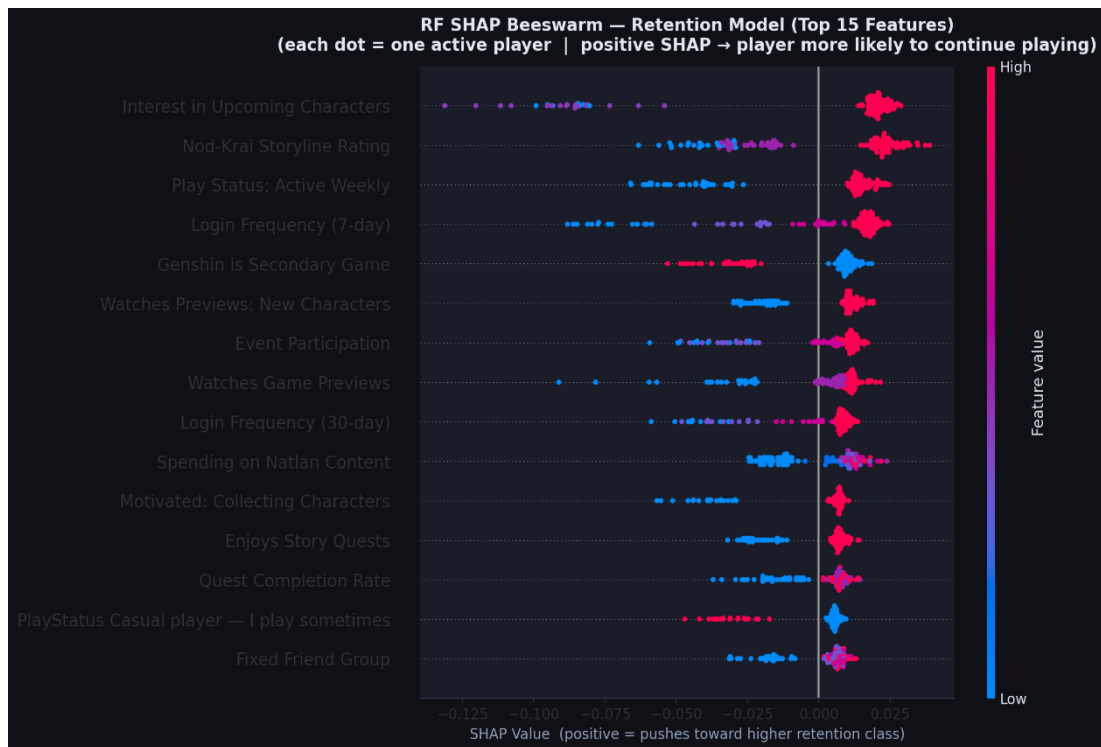


Figure 4.26: SHAP Beeswarm Plot - Individual-Level Retention Feature Distributions

Figure 4.26 presents the beeswarm plot, where each point represents an individual player and colour shows the feature value. The beeswarm confirms that players with high interest in *upcoming characters* and strong *Nod-Krai story ratings* tend toward higher SHAP contributions. Social and behavioural features including *Event Participation* and *Fixed Friend Group* also appear here, confirming their role in individual-level retention prediction.

4.5.5 Churn Risk Factor Distribution

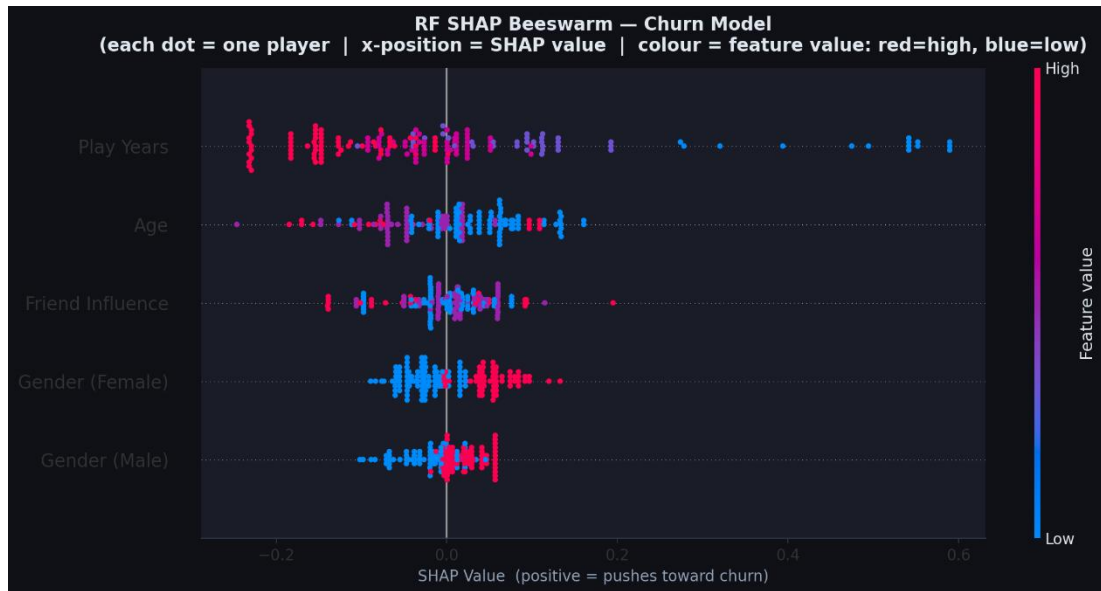


Figure 4.27: SHAP Beeswarm Plot - Churn Model Feature Distribution

Figure 4.27 presents the *SHAP beeswarm plot* for the binary churn model. As noted in Section 4.5.1, the churn model was trained on five basic demographic features (n=170), which limits how much can be interpreted from the SHAP results compared to the full retention model. The feature distributions show that *play duration* and *age* are the main contributors within this limited feature set, consistent with the customer profile finding that younger players with shorter playtime show the highest churn rates.

4.5.6 Churn Reasons and Return Conditions - ML Perspective

The survey-based churn reasons and return conditions are provided in full in Appendix B (Figures B.1, B.2, and B.3) as reference. Rather than repeating the descriptive findings from Sections 4.3.5 and 4.3.6, the focus here is on what the ML analysis adds: confirmation that the behaviours players say caused their churn are the same behaviours the model identifies as statistically significant predictors.

No Time to Play and *Exploration Feels Tiring*, the top two stated churn reasons, correspond directly to the low login frequency and low event participation features that the SHAP analysis (Section 4.5.3) ranks among the highest-impact predictors. Similarly, the return conditions centred on reward generosity align with the spending and pull-related variables that show up consistently in the feature importance outputs.

This cross-validation between self-reported reasons and model-identified predictors is an important methodological checkpoint. It shows that what players say is driving their behaviour is reflected in what the model detects from their behavioural data. The non-return reasons in Figure B.2, including complete loss of interest and negative views on gacha spending, correspond to the Permanently Lost segment in Section 4.3.4, where re-engagement is least likely regardless of game changes.

4.5.7 HoYoverse Ecosystem Analysis

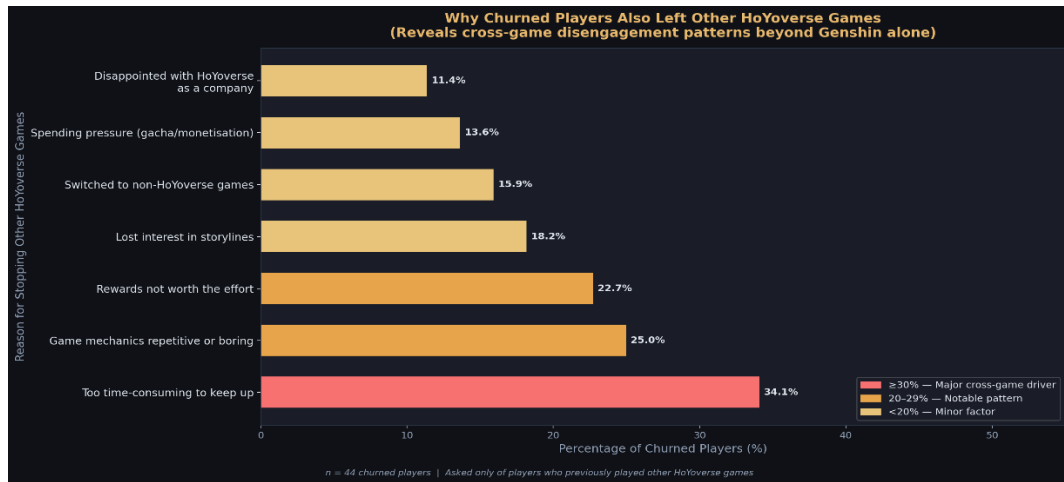


Figure 4.28: HoYoverse Ecosystem Exit Reasons

Figure 4.28 presents the reasons given by respondents who stopped playing all *HoYoverse* titles, not only *Genshin Impact*. The analysis shows that *gacha spending pressure* and *content fatigue* are the main reasons for leaving the ecosystem entirely, with *Wuthering Waves* (*Kuro Games*) identified as the most common game that players switched to. The rise of *Wuthering Waves* as a competitor is relevant here, it is a structurally similar game (open-world action RPG with gacha) that offers an alternative to players unhappy with *Genshin Impact*'s reward system, making it a direct destination for the churned player segments identified in this study.

This competitive pressure, alongside the exploration and reward pain points identified throughout Section 4.3, provides the backdrop for examining whether community-level data reveals the same patterns, and whether any design response from HoYoverse is visible in the community signals across the two content arcs.

4.5.8 Bridge Analysis - Churn Warning Signals

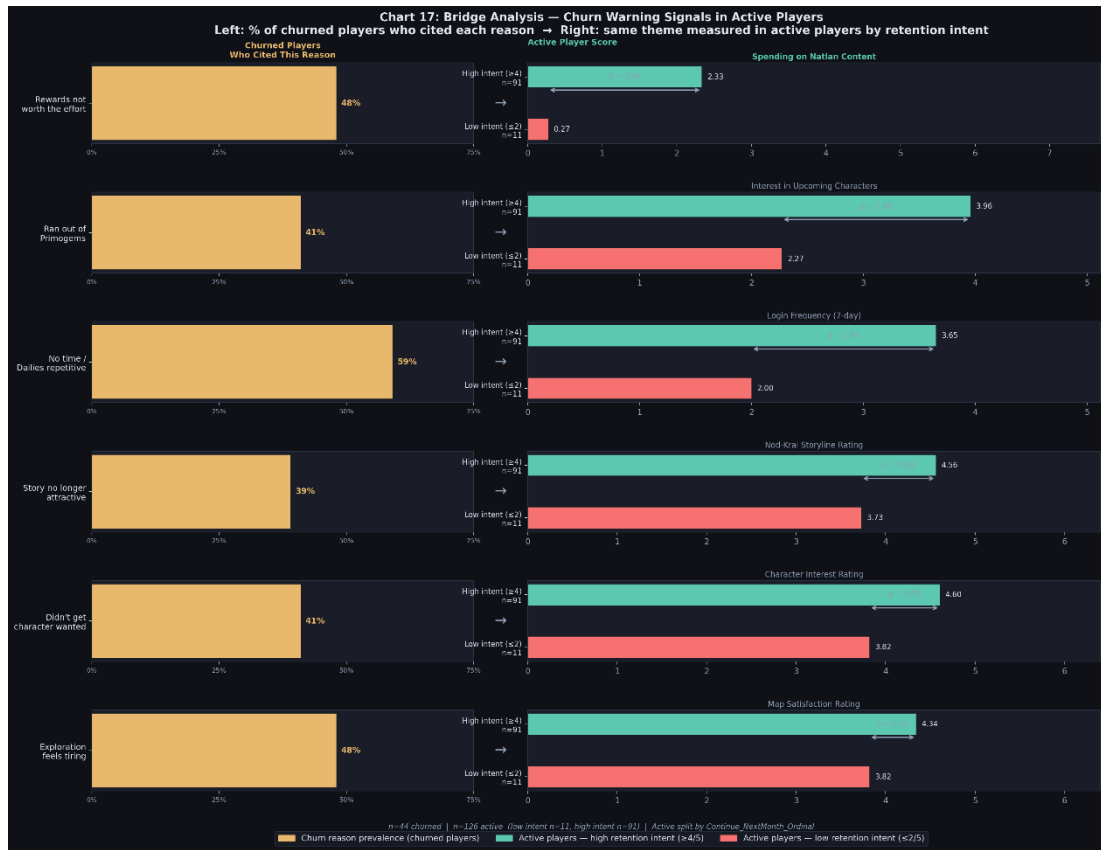


Figure 4.29: Bridge Analysis - Churn Warning Signal

Figure 4.29 presents the Bridge Analysis, which is the main methodological contribution of the churn prediction component. The analysis connects the churn reasons reported by *churned players* (*left panel*) to the corresponding behavioural variables measured among active players, split by retention intent: High Intent (≥ 4 , $n=91$) versus Low Intent (≤ 2 , $n=11$).

The chart shows that the behavioural patterns linked to each stated churn reason are already visible within the active player dataset, before any churn happens. For example, among churned players who cited "*No Time / Dailies Repetitive*" as a churn reason (59%), active players with low retention intent report a *Login Frequency* (7-day) of 2.00 — well below the 3.65 recorded for high-intent active players. Similarly, for the churn reason "*Rewards Not Worth the Effort*" (48% of churned players), active

players with low retention intent report Spending on Natlan Content scores of 0.27, compared to 2.33 for high-intent players.

The Bridge Analysis shows that early signs of churn are already present within the active player population, appearing as differences between high-intent and low-intent groups on the same dimensions that churned players later cite as their reasons for leaving. This provides a way to identify players at risk of quitting before they actually leave, linking behavioural measurement to early-stage churn detection..

4.6 Community Analysis

This section presents the results of the secondary data analysis across four platforms : *Google Play*, *Bilibili*, *YouTube*, and *HoYoLAB*, covering game versions 4.8 through 6.4. The community analysis independently supports and adds context to the survey findings by looking at what players say naturally and at scale. A key advantage of using data from multiple platforms is that it is less likely to be affected by the culture of one platform, when the same patterns appear across different platforms with different users, the findings become much more reliable.

Community text data was processed and analysed using Term Frequency–Inverse Document Frequency (TF-IDF), then grouped using K-Means clustering to find the main discussion topics automatically. The topic labels were added afterward using AI to help interpret the results.

4.6.1 Google Play Ratings by Version

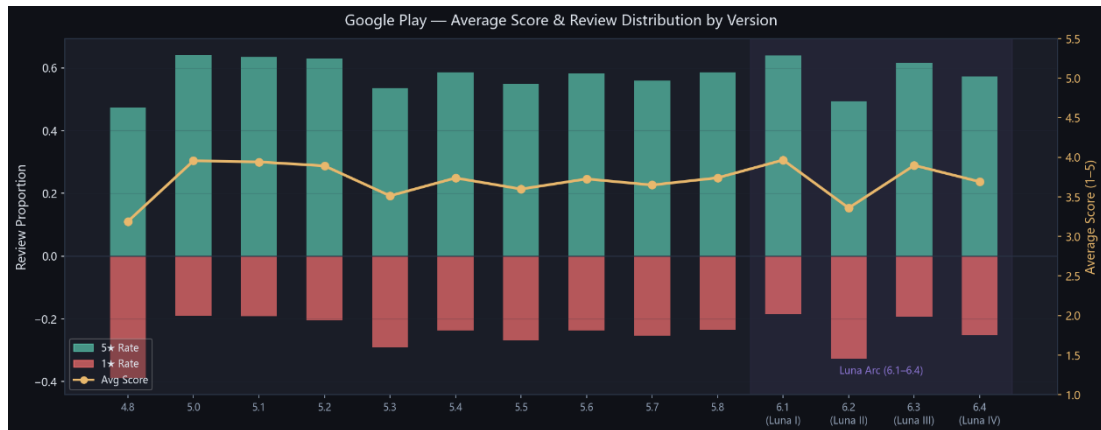


Figure 4.30: Google Play Average Rating by Version

Figure 4.30 tracks the average Google Play store rating across each major game version from 4.8 to 6.4. Ratings change a lot across versions, with the lowest point at **Luna Arc v6.2** (average rating: 3.36). Recovery to 3.97 at v6.1 and 3.90 at v6.3 shows that ratings respond quickly to content quality and reward generosity. Higher-rated versions consistently match major free-pull events or anniversary updates, while rating drops align with rerun banners and less new story content. This supports the survey finding that reward generosity is the most important factor in both player retention and overall game satisfaction.

4.6.2 Cross-Platform Negative Sentiment Signal

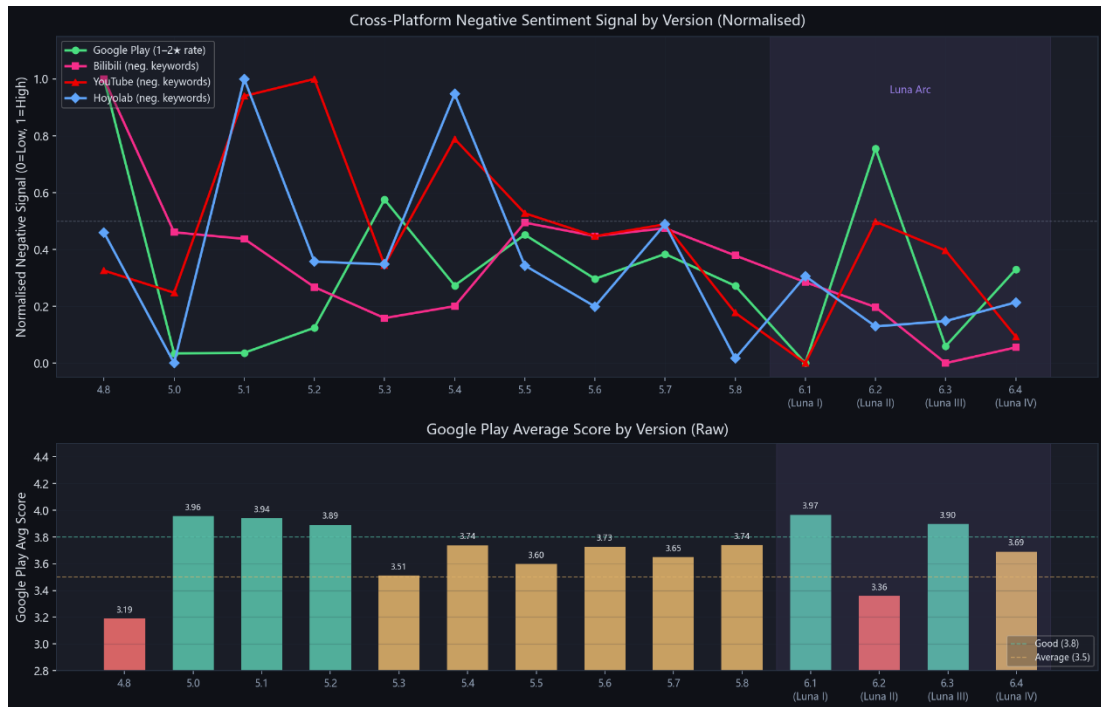


Figure 4.31: Cross-Platform Negative Sentiment Signal by Version

Figure 4.31 presents the normalised negative sentiment signal across all four platforms over the full version range. The chart overlays platform-specific negative indicators, *Google Play* 1–2 star rate, and negative keyword density from *Bilibili*, *YouTube*, and *HoYoLAB* to show a multi-source timeline of player dissatisfaction.

Several version-specific spikes appear across multiple platforms at the same time, showing that player dissatisfaction appeared across multiple platforms simultaneously. The Luna Arc period shows a consistent rise in the *Google Play* negative signal, matching the v6.2 rating low identified in Section 4.6.1. The fact that negative signals appear across platforms at the same time, rather than independently on each platform, confirms that these events reflect genuine community-level disengagement rather than something caused by the culture of one platform. This is consistent with the survey data showing that reward scarcity and content repetition are the most commonly cited friction points.

4.6.3 Engagement Volume by Platform

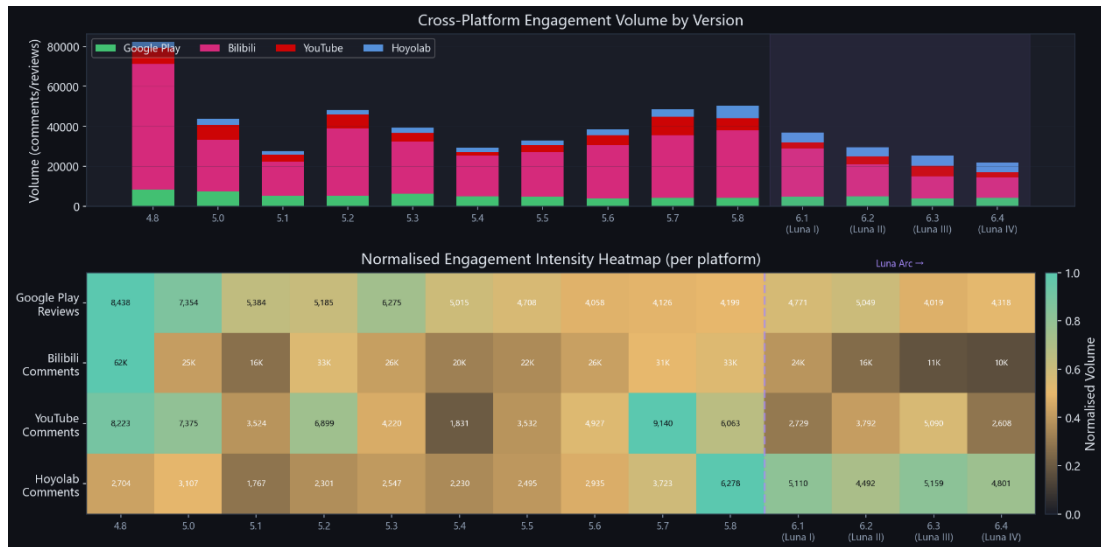


Figure 4.32: Engagement Volume Heatmap

Figure 4.32 presents the *engagement volume heatmap* across platforms and version periods, where darker cells indicate higher engagement. Two patterns stand out. First, the brightest cells consistently align with new region launches (notably Natlan v5.0) and anniversary events, confirming that new characters and exploration content are the main things that bring players back at the community level, directly consistent with the SHAP finding that *Interest in Upcoming Characters* is the strongest individual-level retention predictor. Second, low engagement cells cluster consistently in the final weeks of each patch cycle, when rerun banners are active and no new story content is available. This maps directly onto the top stated churn reason: *exploration fatigue* and a feeling that there is *nothing new to do*.

4.6.4 Negative Keyword Analysis

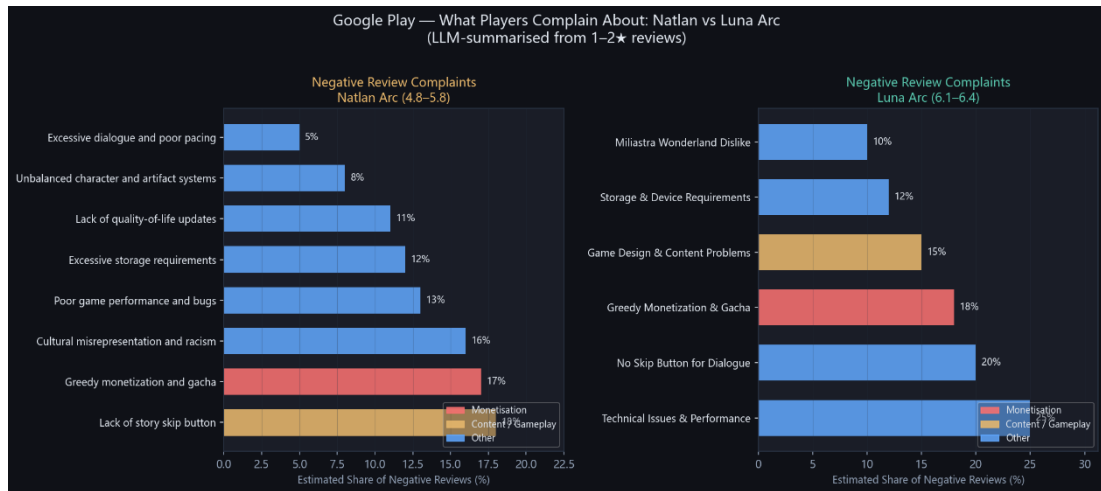


Figure 4.33: Google Play Top Negative Keywords

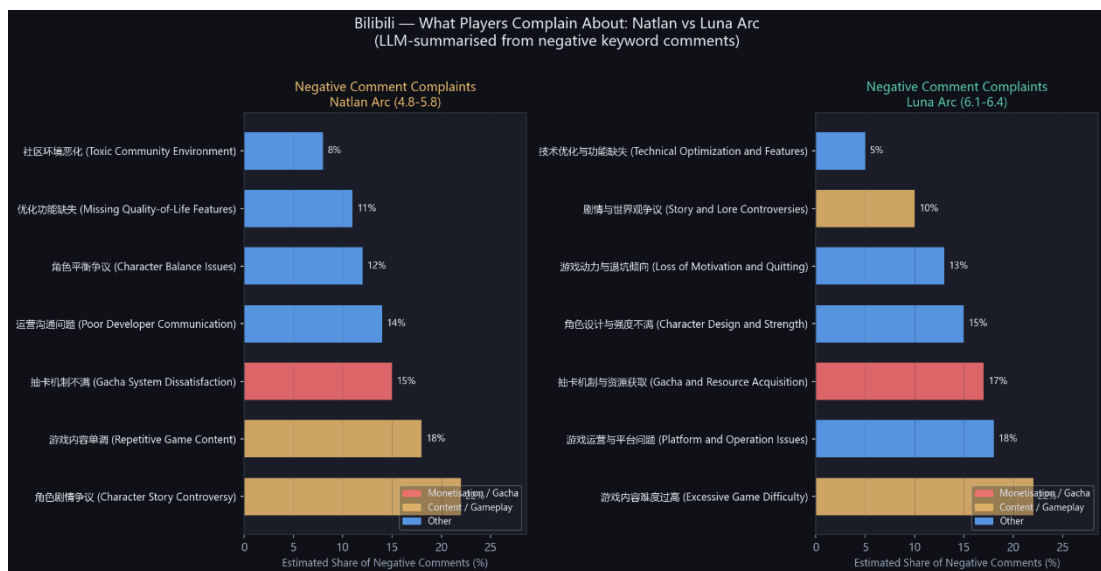


Figure 4.34: Bilibili Top Negative Comment Themes

Figures 4.33 and 4.34 present the top negative keywords from Google Play reviews and Bilibili comments respectively. The most common negative terms in English-language Google Play reviews fall into three themes: *gacha cost* (pull rates, Primogem scarcity), *progression barriers* (resin system, Adventure Rank grinding), and *content repetition* (daily commission fatigue, lack of endgame variety). These map directly onto the survey's top churn reasons, **story not engaging** and **spending pressure**.

Bilibili (Chinese-language) negative comment analysis surfaces the same themes independently: *gacha costs*, *resin system restrictions*, and *banner character value dominate*. The fact that similar findings appear across different languages is particularly worth noting, despite coming from different linguistic communities with separate platform cultures, the friction points are the same. This shows this is not just a regional issue, and confirms the **reward system** as a universal pain point.

4.6.5 Community Discussion Themes by Arc

Figures 4.35 through 4.40 present the K-Means discussion theme clusters for three platforms (YouTube, Bilibili, HoYoLAB) across two story arcs, the Natlan Arc (v4.8–v5.8) and the Luna Arc (v6.1–v6.4). Reading these two arcs side by side reveals not just a shift in what players were talking about, but a signal that HoYoverse's design decisions in the Luna Arc were responding to feedback the community had been giving throughout Natlan.

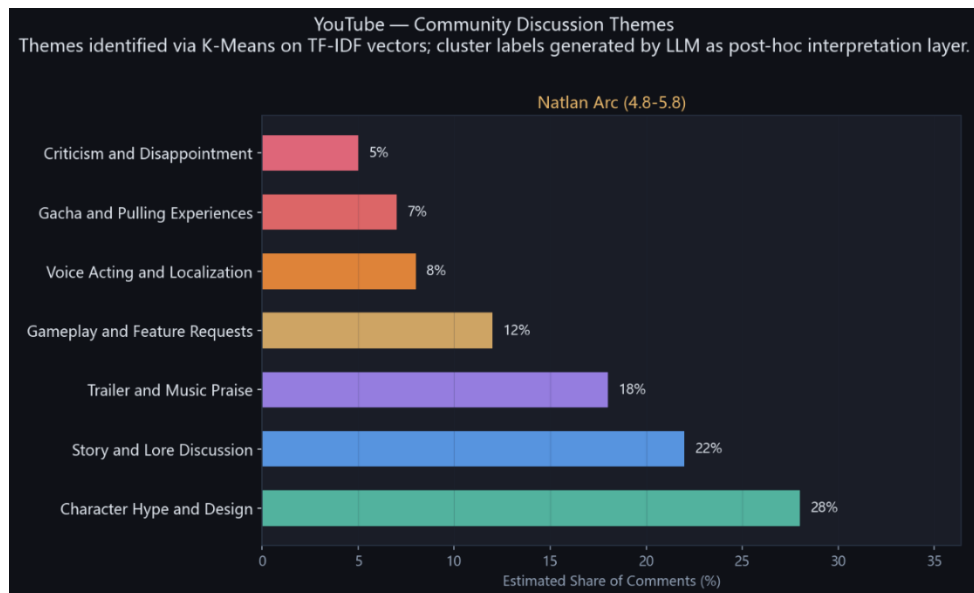


Figure 4.35: YouTube - Natlan Arc Discussion

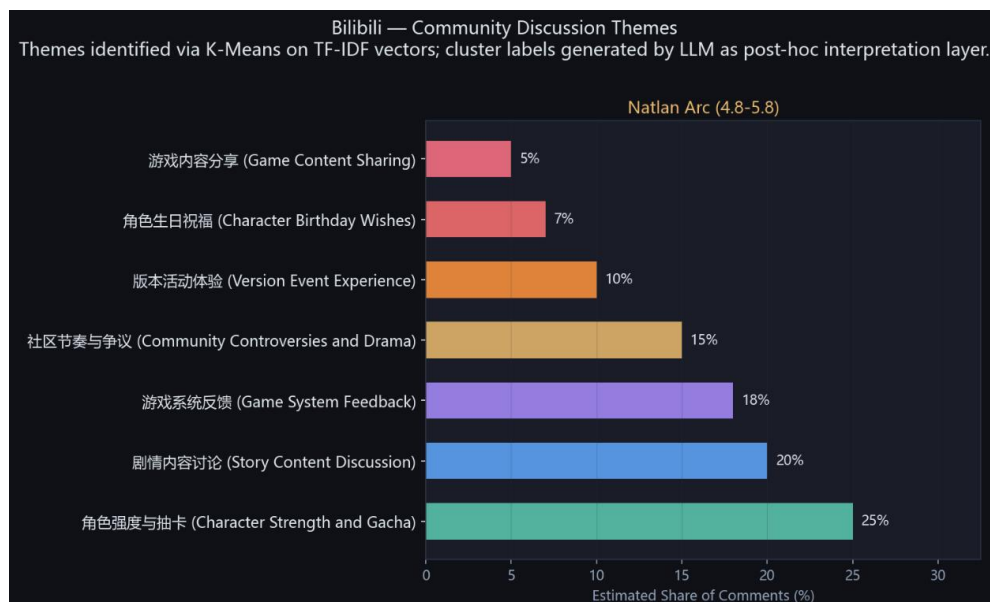


Figure 4.36: Bilibili - Natlan Arc Discussion

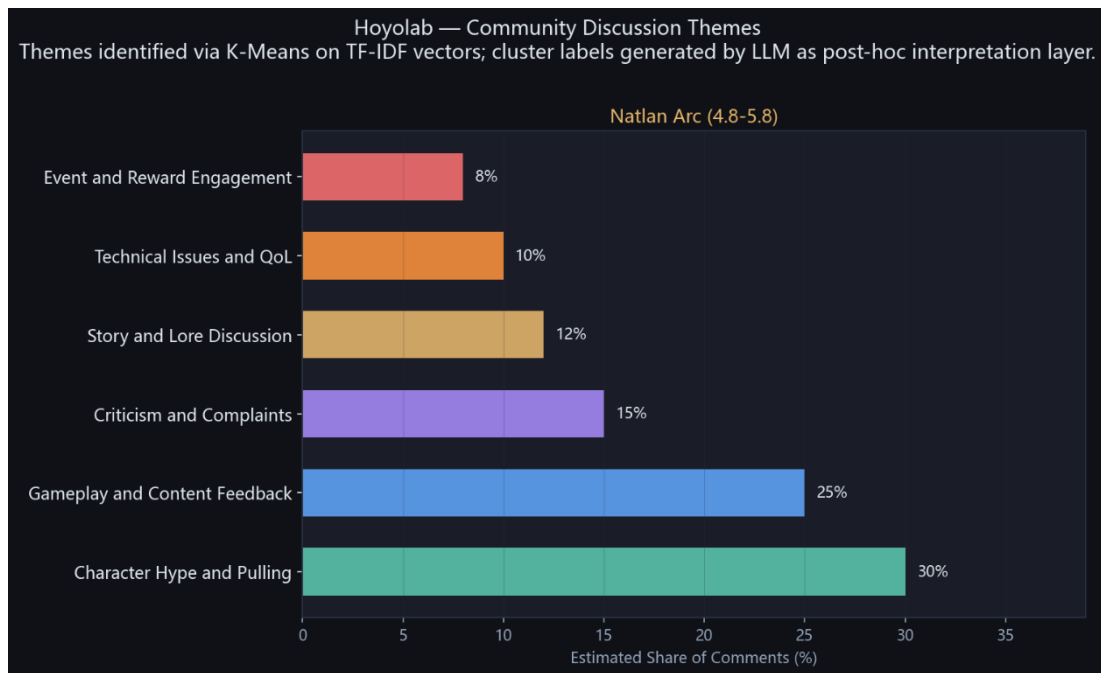


Figure 4.37: HoYoLAB - Natlan Arc Discussion

During the Natlan Arc, discussion across all three platforms follows a recognisable pattern. On YouTube and Bilibili, the dominant clusters centre on character hype and pull decisions, particularly around major releases such as Mavuika and Citlali, with engagement peaking around new banner announcements. These positive clusters, however, exist alongside persistent negative threads. On Bilibili, a distinct cluster around resin restrictions and daily cap frustration appears consistently across the arc, reflecting players who enjoyed the content but felt limited in how much they could actually play. On HoYoLAB, discussions lean toward lore analysis, storyline breakdowns, and fan content, pointing to a player segment that is deeply invested in the narrative but whose engagement is self-sustaining rather than event-driven.

A second thread running through the Natlan Arc is exploration fatigue. As the arc progressed through large, densely layered regions, community commentary increasingly referenced the time commitment required to fully explore each area and collect available rewards. This mirrors the survey finding that **Exploration Feels Tiring** was the second most cited churn reason among former players, a signal that the

spacing of regional exploration had become a friction point even for players who were still active.

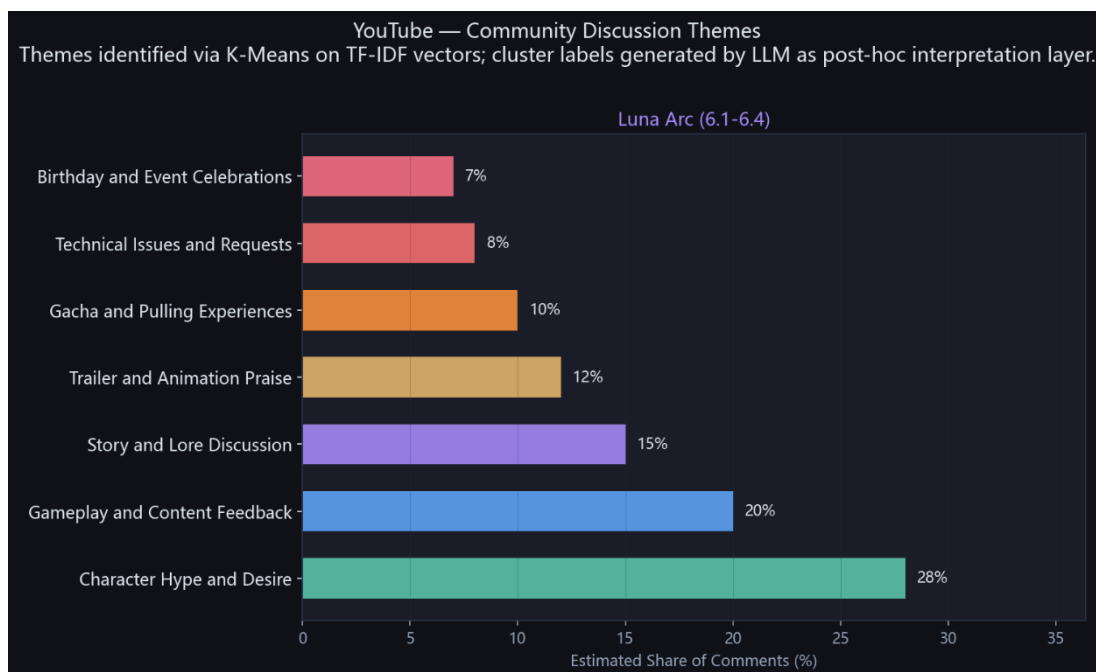


Figure 4.38: YouTube - Luna Arc Discussion

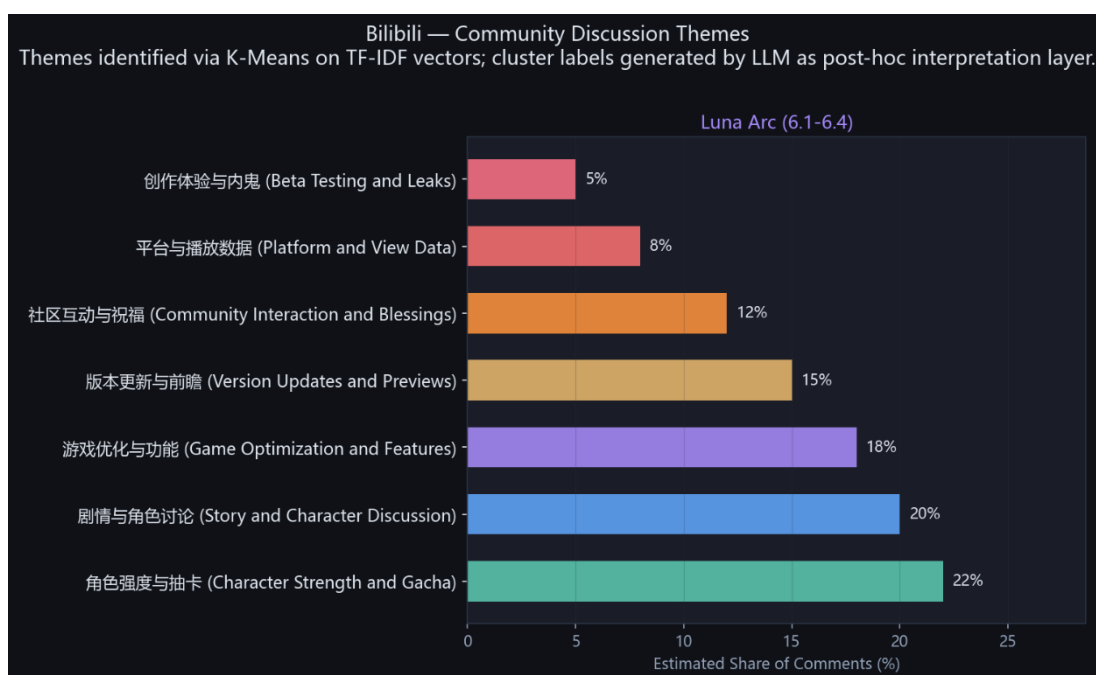


Figure 4.39: Bilibili - Luna Arc Discussion

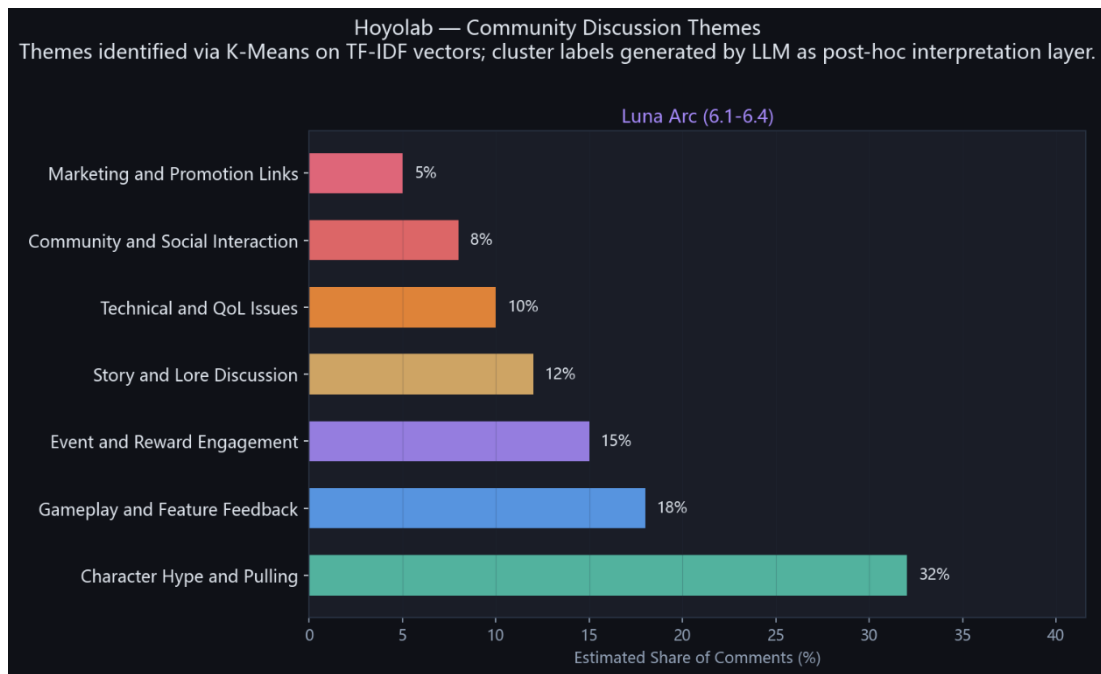


Figure 4.40: HoYoLAB - Luna Arc Discussion

The Luna Arc introduces several shifts that are worth examining carefully, as they suggest the community's complaints during Natlan were not going unnoticed.

The most notable design change in the Luna Arc is the Nod-Krai region itself. Community discussions across YouTube and Bilibili from this period consistently describe the region's map as more compact and manageable compared to the sprawling areas of Natlan. Where earlier regions required extended sessions to fully explore, community feedback around Nod-Krai noted that players could complete regional exploration in a much shorter time while still receiving a substantial Primogem reward from exploration chests. This design decision directly addresses the exploration fatigue complaint that defined the Natlan Arc discussion; the region is not asking less of players in terms of content, but is delivering it more efficiently.

This shift is also visible in the survey data. The Nod-Krai satisfaction heatmap (Figure 4.6) shows that MapDesign ratings sit at 3.87 for 1 to 2 year players and above 4.10 for players with three or more years of playtime, which are relatively positive ratings compared to what the community was saying about Natlan-era exploration during the same period. Long-term players in particular appear to appreciate the design

restraint, possibly because they have spent enough time in larger regions to recognise the difference.

Beyond map design, the Luna Arc also introduces two other notable patterns in the community data. First, anniversary and birthday event discussions form a visible new cluster on YouTube that was not present during Natlan. This reflects a growing community awareness of the anniversary reward cycle, consistent with the survey finding in Section 4.3.6 that anniversary rewards are among the top conditions churned players would need to see improved before returning. The fact that this is becoming a standalone discussion topic suggests players are watching the anniversary period as a signal of whether HoYoverse is listening.

Reading the Two Arcs Together

The shift from Natlan to Luna Arc in community data tells a more nuanced story than simply whether things got better or worse. What it shows is a game that received a sustained wave of feedback around exploration pacing and reward scarcity throughout Natlan, and then made a visible design adjustment in the following arc, one that the community noticed and responded to positively, at least in terms of the regional design.

This does not mean the underlying reward and spending concerns have been resolved. The anniversary discussion cluster and the Primogem-related threads in the Luna Arc data show those conversations are still very much ongoing. However, the Nod-Krai map design change represents a concrete example of HoYoverse moving in a direction that aligns with what players asked for, specifically reducing the time cost of exploration without reducing the reward available from it. Whether this design direction continues into future arcs will be a key factor in whether the survey's Watching and Waiting segment (56.8% of churned players) eventually decides to return or moves on permanently.

4.6.6 Integrated 4-Platform Dashboard

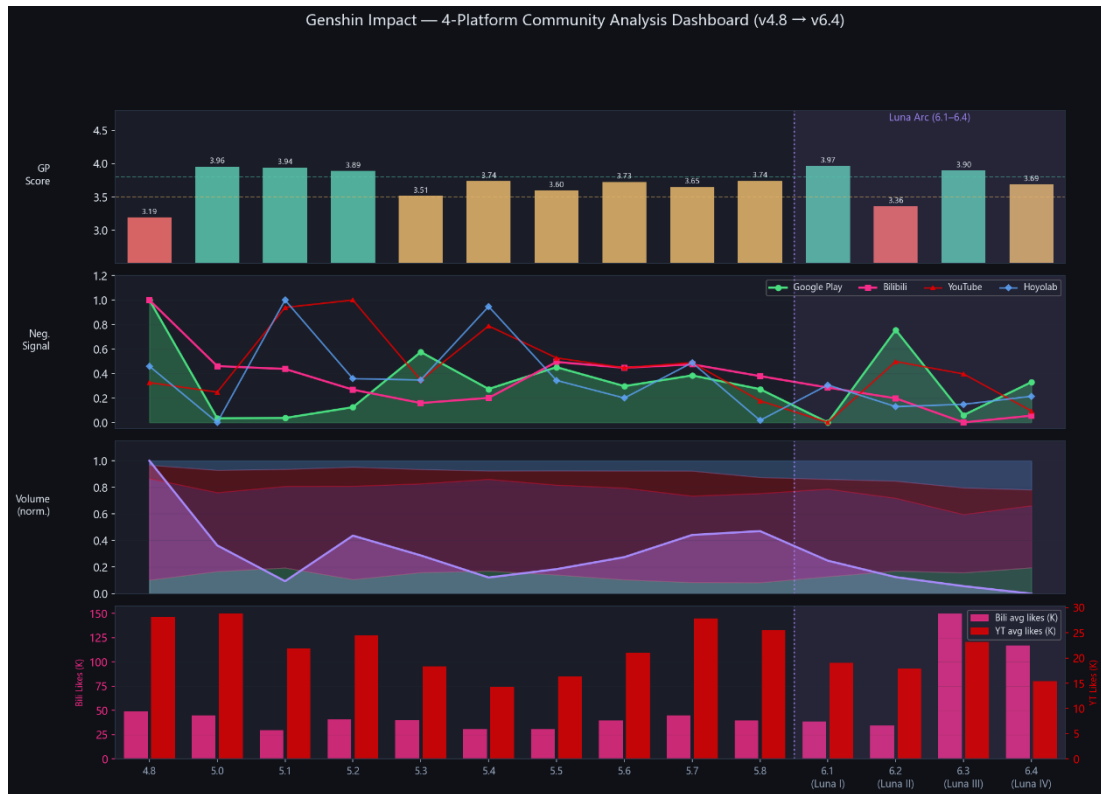


Figure 4.41: Integrated 4-Platform Community Analysis Dashboard

Figure 4.31 presents the integrated community analysis dashboard, combining Google Play rating trends, YouTube comment volume, and Bilibili and HoYoLAB engagement metrics into a single view spanning versions 4.8 through 6.4. The dashboard shows that the most consistent cross-platform pattern is a drop in activity at the end of each patch cycle, when new content runs out and rerun banners take over. This confirms that *Genshin Impact*'s player retention is driven mainly by a content-refresh cycle rather than players naturally wanting to stay: when content slows down, player activity drops across all four platforms at the same time.

The dashboard further confirms that the strongest engagement peaks match new region launches and anniversary events, the exact events that active players said they want most (Section 4.3.3) and that churned players said are most likely to bring them back (Section 4.3.6), which shows provides strong cross-method support for the study's main retention findings.

4.7 Business Action Recommendations

Based on the findings across all five analytical components, eight business action recommendations are proposed for HoYoverse, each directly linked to evidence from this study. The recommendations are grouped into three priority levels.

High Priority — Address Immediately

The first three recommendations address the reward system, which emerged as the most consistent friction point across both survey and community data. These include reducing the daily Resin requirement from 160 to 120 or lower to ease time pressure, introducing a free 10-pull per version in line with what other HoYoverse titles already offer, and upgrading anniversary rewards to a level compelling enough to re-engage the Watching and Waiting segment.

Medium Priority — Address Within 2–3 Versions

The next four recommendations address content and engagement design. These cover replacing patch-end filler events with replayable content to reduce exploration fatigue, investing in combat design beyond damage numbers, expanding co-op with non-combat multiplayer activities to strengthen social retention, and committing to visible quality-of-life updates every other version to rebuild trust with the player base.

Long-Term Strategic

The final recommendation focuses on story accessibility, specifically adding a skip system and condensed recap formats to reduce the barrier for returning players, while preserving narrative as a core retention driver.

Competitive Context

These recommendations are supported by the competitive signals identified in the community analysis. Wuthering Waves competitor comparison clusters appeared for the first time during the Luna Arc, and survey data confirms it as the top destination for churned players. With approximately 63% of churned players still potentially

recoverable, the window for re-engagement remains open but narrows with each version that delivers no meaningful improvement.

The full details of each recommendation, including supporting evidence and implementation suggestions, are presented in the prototype website discussed in Section 4.8.

4.7.1 Framework Integration and Automated Retention Pipeline

To transition the developed system from an academic prototype into an actionable industry tool, the machine learning toolchain must be integrated into a standardised, automated player retention pipeline. Rather than treating prediction models as isolated offline processes, this integration framework establishes a real-time, closed-loop monitoring and intervention system within live-service environments. The integrated tool is designed to connect the data preprocessing, Random Forest classification, and SHAP explainability layers directly to the studio's central data pipeline. The system automatically executes feature construction to track dynamic player patterns, such as drops in login frequency or event participation. When an active player's predicted churn risk crosses a predefined threshold, the framework flags the account and uses localised SHAP values to trigger target-specific automated interventions.

Content Fatigue Trigger: If the risk is driven by exploration fatigue, the system automatically interfaces with the game client to push prompts highlighting time-efficient, compact regional content (such as the optimised gameplay pacing in the Nod-Krai region).

Reward Scarcity Trigger: If the risk is tied to reward dissatisfaction, the pipeline automatically triggers targeted retention incentives via the in-game mail system, delivering version-specific free pulls or tailored resource packages.

By synchronising micro individual actions with macro community dashboard insights, this automated tool transforms predictive outputs into real-time decision-support assets, enabling proactive intervention before permanent churn occurs.

4.8 Prototype Website

As part of this study, a prototype website was developed to present the research findings in an interactive and accessible format. The website can be accessed at <https://kelvinportfolio.onrender.com/> and is structured around eight main sections: Home, Research, Methodology, Customer Profile, Results, Community, Recommendations, and Limitations.

The website also features a built-in AI chatbot powered by the DeepSeek API. The chatbot is context-aware, reading the content of the current page to answer questions related to the study's findings, methodology, and results. This allows non-technical readers to interact with the research directly without needing to read through the full report.

Screenshots of the Home page, Community Analysis page, and Business Action Recommendations page are shown in Figures 4.42, 4.43, and 4.44 respectively.



Figure 4.42: Prototype Website - Home Page

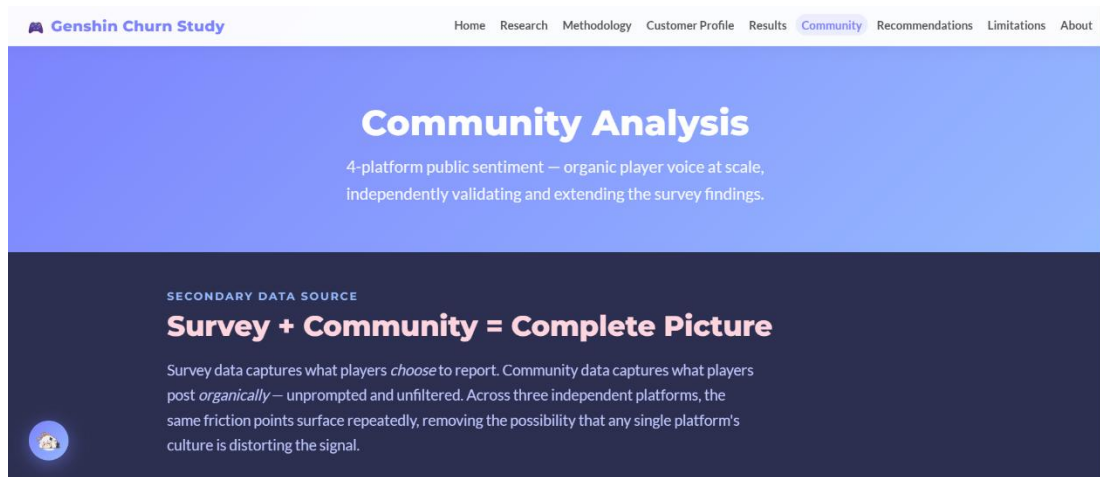


Figure 4.43: Prototype Website - Community Analysis Page

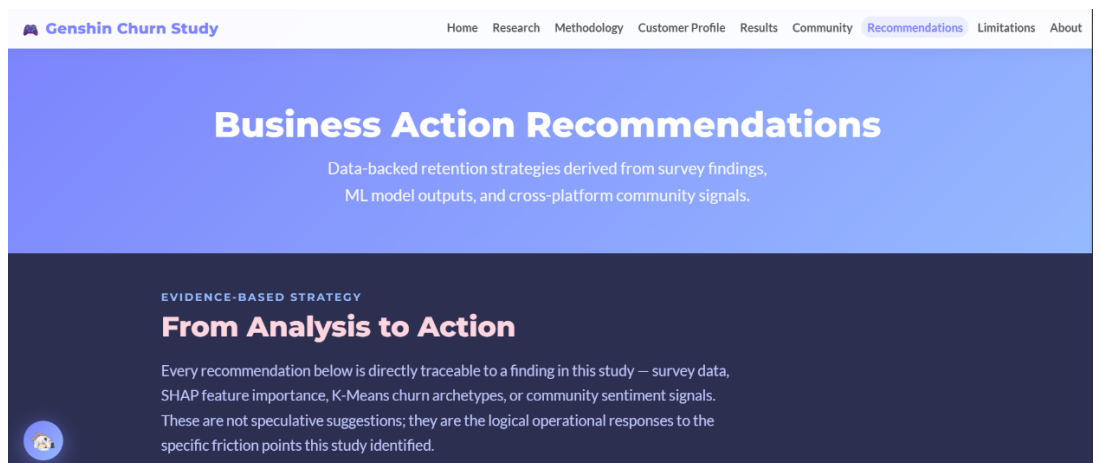


Figure 4.44: Prototype Website - Business Action Recommendations Page

4.9 Summary

This chapter presented the results and analysis of the study across five analytical components: customer profile analysis, player segmentation, churn prediction and SHAP explainability, community analysis, and business action recommendations.

Across all survey components, two consistent patterns stood out. Both active and churned players share the same core frustrations around reward generosity and exploration pacing, and long-term players show a fundamentally different engagement profile from newer players, being more self-motivated, less friend-dependent, and more sensitive to content quality. The machine learning models confirmed these patterns at the statistical level, with interest in upcoming characters, event participation, and login frequency emerging as the strongest predictors of retention. The Bridge Analysis further showed that early behavioural signals among active players mirror the same dimensions that churned players later cited as their reasons for leaving.

The community analysis added a temporal dimension to these findings. Throughout the Natlan Arc, exploration fatigue and reward scarcity dominated community discussions across all four platforms. The Luna Arc, however, brought a visible design shift, with Nod-Krai's compact map receiving broadly positive feedback and higher satisfaction ratings in the survey data, suggesting HoYoverse has begun responding to accumulated player feedback. Whether reward generosity follows the same direction will likely determine the fate of the 56.8% of churned players currently categorised as Watching and Waiting. The implications of these findings are discussed in Chapter 5.

CHAPTER 5

CONCLUSION AND FUTURE WORK

5.1 Introduction

This chapter brings together the key findings from the study and evaluates the extent to which the research objectives have been met. It begins by reviewing each objective against the results presented in Chapter 4, followed by an honest discussion of the study's limitations. Directions for future work are then identified based on the gaps that remain, and the chapter closes with a final conclusion on the overall contribution of the study.

5.2 Objectives Achieved

This study was guided by three research objectives, each of which is reviewed below in light of the findings.

Objective 1:

To develop a player behaviour analysis and churn prediction system using Python and machine learning techniques that can automatically identify hidden patterns from the data.

This objective was met through the development of a full analytical pipeline implemented in Python, covering data preprocessing, clustering, classification, and SHAP-based explainability. The system successfully identified patterns that would not

have been visible through manual analysis, including the distinction between High Engagement and Lower Engagement player segments, and the early churn warning signals captured in the Bridge Analysis. The prototype website further demonstrates how these outputs can be presented to both technical and non-technical audiences.

Objective 2:

To develop data-driven player segmentation and churn prediction models using clustering and classification algorithms.

This objective was met through the application of K-Means clustering and Self-Organising Maps for player segmentation, and Logistic Regression and Random Forest for churn and retention prediction. K-Means produced two meaningful active player segments, and the models achieved a 5-fold cross-validation accuracy of 0.746 for retention prediction and an AUC of 0.708 for churn classification. The cluster-level SHAP analysis further connected segmentation results to prediction outputs, showing that different player groups carry different churn risk profiles.

Objective 3:

To evaluate the accuracy and effectiveness of the system using standard evaluation metrics.

This objective was met through a consistent evaluation framework applied across all models, covering Accuracy, Precision, Recall, F1-Score, AUC, and Stratified 5-Fold Cross-Validation. The results were interpreted with appropriate caution given the sample size, and the limitations of each model were acknowledged, particularly the churn model's reliance on five demographic features and the class imbalance in the retention model. The SHAP outputs provided an additional layer of evaluation by confirming that the model's feature priorities align with what players themselves report as reasons for staying or leaving.

5.3 Limitations

Several limitations should be noted when interpreting the findings of this study.

First, the survey sample (n=170) is relatively small for supervised machine learning modelling, particularly for the churned player group (n=44). This limited sample size affects the reliability of some classification metrics and prevented the churned player K-Means from producing meaningful sub-clusters (K=1). Findings should therefore be treated as indicative rather than statistically conclusive.

Second, participants were recruited through gaming community platforms and personal networks, which may introduce self-selection bias. Players who responded are likely to be more engaged or vocal than the average player, meaning the results may not fully represent the broader player population, particularly less engaged or silent churners.

Third, the community data collected from Bilibili, YouTube, and HoYoLAB is also subject to platform bias. Users who actively post or comment on these platforms tend to be more engaged than the average player. As a result, community data is treated as supplementary evidence rather than a statistically representative sample.

Fourth, the survey-based design does not capture real-time behavioural data such as actual login timestamps or in-game activity logs. Behavioural variables were based on self-reported responses, which may not always reflect actual gameplay patterns.

Finally, the study focuses specifically on Genshin Impact and HoYoverse's ecosystem. While the framework may be applicable to other free-to-play games, the findings are most directly relevant to games with similar exploration-driven and gacha-based designs.

5.4 Future Work

Several directions are identified for future work building on this study.

The most immediate priority is expanding the survey sample to improve the reliability of the machine learning models, particularly for churn prediction and multi-class retention modelling. A larger and more diverse sample would allow for more stable clustering results and more confident SHAP interpretations across player segments. Future survey designs could also include retrospective behavioural questions specifically targeting churned players, to capture more precise information about the timing and context of disengagement.

Future work could also incorporate actual in-game behavioural logs rather than relying solely on survey self-report. Access to real-time login data, quest completion records, and spending history would allow for more precise and dynamic churn prediction, and would support the application of temporal models such as survival analysis.

The Bridge Analysis introduced in Section 4.5.8 represents a promising direction for building a real-time early warning system. Building upon the integrated framework discussed in Section 4.7.1, future iterations could focus on transitioning the current static prototype into an online, real-time early warning dashboard. By fully bridging the predictive pipeline with real-time in-game log APIs and automated cross-platform sentiment scrapers, the system could enable instantaneous risk flag triggering and dynamic SHAP value updating, thereby improving the adaptability and real-time responsiveness of the player retention pipeline.

Additionally, the community analysis could be extended to cover a longer time range and more platforms, including X (formerly Twitter) and Reddit, which were excluded from the current study due to data collection constraints. This would provide a more complete picture of cross-platform sentiment trends over time.

Finally, the prototype website developed in this study could be further enhanced into a fully interactive analytical dashboard, allowing game developers and

analysts to explore player segments, SHAP outputs, and community signals in real time, supporting more informed and responsive retention strategies.

5.5 Conclusion

This study developed a data-driven framework for analysing player behaviour and predicting churn in Genshin Impact, combining survey data (n=170) with community-generated content from four platforms across two major content arcs.

The findings consistently show that player retention in Genshin Impact is driven by a content-refresh cycle rather than passive loyalty. Character design and interest in upcoming characters are the strongest retention signals, while reward scarcity and exploration fatigue are the dominant reasons players leave. The Bridge Analysis demonstrates that these signals are detectable within the active player population before churn occurs, pointing toward the feasibility of a proactive early warning system.

The community analysis added an important dimension by showing that HoYoverse has begun responding to player feedback, most visibly through the compact map design of the Nod-Krai region in the Luna Arc, which received broadly positive community reception. Whether this responsiveness extends to the reward system will likely determine whether the 56.8% of churned players currently in the Watching and Waiting segment eventually return or move on permanently.

Beyond Genshin Impact, the framework developed in this study offers a transferable approach for any free-to-play game seeking to understand and reduce player churn through interpretable, data-driven methods.

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APPENDIX A: QUESTIONNAIRE

Introduction

This survey is part of a Final Year Project conducted at the University of Technology Sarawak (UTS). The purpose of this study is to examine player behaviour and identify factors influencing player retention and churn in *Genshin Impact*. Participation is voluntary, and all responses are anonymous and used strictly for academic research purposes only. No personal or sensitive information is collected, and participants may withdraw at any time.

Section 1: Respondent Information

Q1: **Age** (*Single choice*)

- < 18
- 18–24
- 25–34
- 35–44
- 45+

Q2: Gender (*Single choice*)

- Male
- Female
- Prefer not to say

Q3: How long have you been playing Genshin Impact? (Single choice)

- < 1 year
- 1–2 years
- 3–4 years
- 5 years or more

Q4: Which platform(s)/device(s) did you mainly use to play Genshin Impact?
(Checkboxes)

- PC
- Mobile (Android / iOS)
- PlayStation (PS4 / PS5)

Section 2: Recent Gameplay Status

Q1: Have you continued playing Genshin Impact in the past 30 days? (Single choice)

- Yes
- No

[If “Yes”, proceed to Section 13: Active Player Section. (Active Player Section)]

[If “No”, proceed to Section 3: Reasons for Not Playing (Churn Player Section).]

Churn Player Section

Section 3: Reasons for Not Playing

Q1: Why have you not played Genshin Impact recently? (Checkboxes)

- The story is no longer attractive
- No time to play (study/work pressure)
- Could not keep up with game difficulty
- The story feels childish
- Did not get the character I wanted / ran out of Primogems
- In-game benefits are not as good as other HoYoverse games
- Exploration feels tiring and rewards are not worth the effort
- Quests are not fully voiced
- Other: _____\

Section 4: Do you still watch PVs or livestreams related to Genshin Impact?

Q1: Have you watched the recent “Nod-Krai” version preview? (Single choice)

- Yes, I still watch them.
- Sometimes, but not regularly.
- No, I no longer watch previews.

[If "No", proceed to Section 5: Reasons for Not Watching PVs/Livestreams.

Otherwise, proceed to Section 6: Why the Preview Didn’t Change Your Mind.]

Section 5: Reasons for Not Watching PVs/Livestreams

Q1: Why did you decide to stop watching Genshin Impact PVs or livestreams? (Checkboxes)

- I am not interested in watching previews
- I don't follow Genshin Impact news anymore
- I don't play the game anymore, so I didn't bother to watch.
- Other: _____

[After completing this section, proceed to Section 7: Friends & Social Influence.]

Section 6: Why the Preview Didn't Change Your Mind

Q1: If you watched the Nod-Krai preview but don't plan to return, why not? (Checkboxes)

- The preview story/content is not attractive
- Rewards and benefits are not appealing compared to the effort
- I feel the gameplay looks repetitive or boring
- I don't like the upcoming characters or designs
- I don't have time to return even if the preview looks good
- I have already lost long-term interest in the game
- Other: _____

[After completing this section, proceed to Section 7: Friends & Social Influence.]

Section 7: Friends & Social Influence

Q1: Do you have friends who still play Genshin Impact? (Single choice)

- Yes, most of my friends still play
- Yes, but only a few
- No, most of my friends have quit
- I usually play alone

Q2: Do your friends' activities influence your intention to return? (Single choice)

- Yes, I might return if my friends become active again
- Maybe a little, but not the main reason
- No, I decide independently

Section 8: Have you play other HoYoverse Games

Q1: Do you also play other HoYoverse / miHoYo games? (Single choice)

- Yes
- No, I only play Genshin Impact
- I have stopped playing HoYoverse games

[If "Yes", proceed to Section 9: Other HoYoverse Games. Otherwise, proceed to Section 10.]

Q2: Do you also play non-HoYoverse games? If yes, which ones? (*Checkboxes*)

- Wuthering Waves
- Tower of Fantasy
- Honkai-like games (e.g., Punishing: Gray Raven)
- Blue Archive
- Arknights
- Fate/Grand Order
- Goddess of Victory: NIKKE
- Gacha RPGs from other developers
- Open-world ARPGs

Section 9: Other Hoyoverse Games.

Q1: What other HoYoverse/miHoYo games are you playing now? (Checkboxes)

- Honkai Impact 3rd
- Honkai: Star Rail
- Zenless Zone Zero
- Tears of Themis

Q2: If you are still playing other HoYoverse games, why do you continue playing them?
(Checkboxes)

- The story in those games is easier to follow and interesting.
- The gameplay and combat feel different from Genshin, so it's refreshing.
- I like the art style, character designs, or music.
- The games feel more relaxing or convenient to play (daily tasks, events, QoL).
- My friends are playing them, so I joined too.
- I enjoy trying out different worlds and themes besides Genshin.
- The rewards in other HoYoverse games are more generous
- Other: _____

[After completing this section, proceed to Section 11: Conditions for Returning.]

Section 10: Reasons for Quitting or Not playing other HoYoverse Games

Q1: If you have stopped or are not playing HoYoverse/miHoYo games, what are the main reasons? (Checkboxes)

- Lost interest in the storylines of other HoYoverse games
- Game mechanics feel repetitive or boring
- Rewards are not appealing compared to effort
- Too time-consuming to keep up
- Switched to other games (e.g., Wuthering Waves or other non-HoYoverse games)
- Spending pressure (gacha, monetization)
- Disappointed with HoYoverse as a company (e.g., decisions, communication, player treatment)
- Other: _____

[After completing this section, proceed to Section 11: Conditions for Returning.]

Section 11: Conditions for Returning

Q1: Which of the following changes would most likely encourage you to return to Genshin Impact? (Checkboxes)

- Generous Rewards: More free pulls/Primogems (e.g., Anniversary rewards).
- Content Optimization: "Skip Button" for dialogue or faster daily dailies.
- Gacha Improvement: Better rates or weapon banner mechanism.
- Better Artifact System: Easier way to get good stats (Loadouts/Main stats selection).
- Main Story: Only when the main Archon Quest (Snezhnaya/Khaenri'ah) is updated.
- Provide more generous Anniversary rewards (e.g., giving out a free limited 5★ weapon).
- Give out free pulls each version (e.g., 10 Intertwined Fates per patch, similar to other HoYoverse games).
- Reduce the daily Resin consumption (e.g., from 160 to 120 or even lower).
- Nothing: I have completely lost interest and will not return regardless of changes.
- Other: _____

Q2: Do you have any other feedback or suggestions you would like to said to the Genshin Impact team? (Open-ended)

Section 12 : Closing Message (Churned Player)

Message shown:

- Thank you for your answers.
- We hope that one day you may consider returning to Genshin Impact.
- The developers will continue to improve the game experience, and the starry sky of Teyvat will always keep your place and memories.

Active Player Section

Section 13: Have You Watched Genshin Impact Livestreams or PVs?

Q1: Do you usually watch Genshin Impact version previews / livestreams? (Single choice)

- Yes, I always watch them.
- Sometimes, only if I'm interested.
- Rarely.
- No, I don't watch them.

[If “Yes, I always watch them” or “Sometimes, only if I’m interested”,

proceed to Section 14: What Makes You Watch Genshin Impact PVs.

Otherwise, proceed to Section 15: Reasons for Not Watching Genshin Impact PVs.]

Section 14: What Makes You Watch Genshin PVs

Q1 What motivates you to watch version previews? (Checkboxes)

- To learn new content early
- To see new characters
- To plan primogem usage
- To check story updates
- To see combat / map changes
- Out of habit
- Other: _____

Section 15: Reasons You Don't Watch Genshin PVs

Q1 Why don't you watch version previews? (Checkboxes)

- Content creators summarize everything anyway
- I prefer to experience updates directly in game
- Not interested in trailers
- No time to follow news
- focus only on gameplay
- Other: _____

Section 16: Your Recent Gameplay Status

Q1: In the past 7 days, on how many days did you log in to Genshin Impact? (Single choice)

- 0 days
- 1–2 days
- 3–4 days
- 5–6 days
- 7 days

Q2: What is your average daily playtime in the past 7 days? (Single choice)

- 1-10 minutes
- 10-30 minutes
- 30-60 minutes
- 60-120 minutes
- 120-180 minutes
- More than 180 minutes

Q3: In the past 30 days, on how many days did you log in to Genshin Impact?
(Single choice)

- 0 days
- 1-7 days
- 8-14 days
- 15-21 days
- 22-30 days

Section 17: Gameplay Progress & Preferences

Q1: How would you describe your overall exploration progress? (Single choice)

- Low (0–40%)
- Medium (41–70%)
- High (71–90%)
- Full (91% - 100%)

Q2: How would you describe your overall quest completion? (Single choice)

- Low (0–40%)
- Medium (41–70%)
- High (71–90%)
- Full (91% - 100%)

Q3: What type of content do you enjoy the most? (Single choice)

- Exploring new regions
- Story quests & characters
- Combat & challenges
- Collecting & completion
- Just casual play
- Character building & leveling
- Other: _____

Q4: How satisfied are you with the overall map and region design in Genshin Impact?

(5-point Likert scale: 1 = Not worth exploring, 5 = Worth exploring and enjoyable)

Q5: How interested are you in upcoming playable characters in future versions of Genshin Impact?

(5-point Likert scale: 1 = Not interested at all, 5 = Very interested)

Section 17: Nod-Krai Region Experience

Q1: What are your impressions of Nod-Krai?

Please indicate your level of agreement with the following statements:

(5-point Likert scale: 1 = Strongly disagree, 5 = Strongly agree)

- The map design is enjoyable to explore
- The atmosphere, music, and visuals feel immersive.
- The storyline in this region is engaging
- The new characters introduced in this region are attractive, or interesting
- The regional combat & mechanisms feel well-balanced.
- The overall experience of the region meets my expectations.

Q2: What did you like or dislike the most about Nod-Krai?

(Open-ended)

Section 18: Friends & Co-op Experience

Q1: When playing Genshin Impact, do you usually join Co-op for Domains, Boss fights, or Weekly bosses? (Single choice)

- Almost every week, I like playing together
- Sometimes, when I need help or feel like it
- Rarely, I mostly play solo
- Never, I don't like Co-op

Q2: Do you have fixed friends or a community you often play Genshin with? (Single choice)

- Yes, I have close friends that I always play with
- Yes, but we don't play together that often
- No, I usually play alone
- My friends have mostly quit, rarely online anymore

Q3: Does having friends who still play Genshin make you want to keep playing too? (Single choice)

- Yes, friends are a big reason I still play
- A little, but I mainly play for myself
- No, my play has nothing to do with friends

Section 19: Will You Keep Playing?

Q1: Do you expect to continue playing Genshin Impact in the next month? (Single choice)

- Definitely yes
- Probably yes
- Not sure
- Probably not
- Definitely not

Q2: How do you usually complete your Daily and Weekly tasks? (Single choice)

- I complete both Daily and Weekly tasks.
- I complete Daily tasks but not Weekly tasks.
- I don't complete Daily or Weekly tasks.

Q4: In recent versions, how often do you participate in limited-time events? (Single choice)

- I complete all major events
- I only complete events that give Primogems
- I sometimes skip events
- I often skip or ignore events
- I rarely or never do events

Q3: In the past few days or recent versions, have you challenged the Spiral Abyss or Imaginarium Theater? (Single choice)

- Yes
- No

[If “Yes”, proceed to Section 20: Spiral Abyss & Imaginarium Theater Challenge Experience.

If “No”, proceed to Section 21: Reasons for Not Challenging Spiral Abyss or Imaginarium Theater.]

Section 20: Spiral Abyss & Imaginarium Theater Challenge Experience

Q1: If **Yes**. Why did you challenge them? (Checkboxes)

- I had a team suitable for the challenge.
- For Primogem rewards.
- For self-challenge / I enjoy the combat.
- Other: _____

[After completing this section, proceed to Section 22: Spending & Pulling Behaviour.]

Section 21: Reasons for Not Challenging Spiral Abyss or Imaginarium Theater

Q1: If **No**. Why did you not challenge them? (Checkboxes)

- I didn't have a team suitable for the challenge.
- The Primogem rewards were too few.
- My character levels were too low, I didn't want to struggle.
- Other: _____

[After completing this section, proceed to Section 22: Spending & Pulling Behaviour.]

Section 22: Spending & Pulling Behaviour

Q1: During the Natlan version (Version 5.0 to 5.8) of Genshin Impact, how much money did you spend (e.g., Welkin Moon, Genesis Crystals, Character/Weapon Banners)? (Single choice, optional – respondents may skip this question)

- No spending
- Less than MYR20 ($\approx$ USD5)
- MYR20–49 ($\approx$ USD5–10)
- MYR50–99 ($\approx$ USD10–20)
- MYR100–199 ($\approx$ USD20–40)
- MYR200–499 ($\approx$ USD40–100)
- MYR500+ (USD100+)

Q2: Compared to the **previous version (5.0–5.8, Natlan)**, do you expect your spending in the **current or future versions (6.0–6.8, Nod-Krai region)** to be: (Single choice, optional – respondents may skip this question)

- Higher
- About the same
- Lower
- Depends on future characters or weapon banners
- Not decided yet (waiting to see upcoming content)

Q3: Do you still feel interested in the characters who have appeared in the story (eg. Columbina, Sandrone) but are not playable yet? (Single choice)

- Yes, I'm still interested
- A little interested
- Not really interested anymore

- I've lost interest because they take too long to release
- I don't know these characters well

Q4: What affects your interest in pulling for future playable characters?
(Checkboxes)

- Character design
- Personality & story role
- Combat performance
- Kit uniqueness
- Release timing (that are too slow)
- I'm saving for someone else
- I don't plan to pull anymore
- Do you (or in the upcoming Nod-Krai version) plan to pull characters (30+ wishes)?
- Other: _____

Q5: Do you (or in the upcoming Nod-Krai version) plan to pull characters (30+ wishes)?

(Single choice)

- Yes / 有
- No / 没有

Section 24: Annual Rewards & Long-term Incentives

Q1: Do the yearly Anniversary Rewards (e.g., free 10 pulls, primogems, events) motivate you to keep playing Genshin Impact? (Single choice)

- Yes, definitely
- A little, but not a major factor

- Not really
- Not at all
- I didn't know about Anniversary Rewards

Q2: If the game provides long-term progression rewards (e.g., completing 8 out of 12 weekly mission cycles to **redeem a specific 5★ character constellation from an official selection pool**), would this motivate you to complete weekly tasks?

- Yes
- No

Q3: Do you plan to participate in the “Milistar Wonderland UGC” mode, either by creating your own levels or actively playing levels designed by other players? (Single choice)

- Yes, I will create and play other's levels
- Yes, I will create levels
- Yes, I will mainly play others' levels
- Maybe, not sure yet
- No, I am not interested in UGC

Section 25: Current Play Status

Q1: How would you describe your current Genshin Impact play status? (Single choice)

- Active player — I play every week.
- Casual player — I play sometimes
- Returning player — I stopped playing before but recently came back.
- Potential return player — I rarely play now, but I might return in the future.

Q2: Do you also play non-HoYoverse games? If yes, which ones? (Checkboxes)

- Wuthering Waves
- Tower of Fantasy
- Honkai-like games (e.g., Punishing: Gray Raven)
- Blue Archive
- Arknights
- Fate/Grand Order
- Goddess of Victory: NIKKE
- Gacha RPGs from other developers
- Open-world ARPGs

Section 26: Active Player Insights

Q1: Do you also play other HoYoverse games? (Single choice)

- Yes
- No

[If “Yes”, proceed to Section 27: Cross-Game Behaviour & Player Positioning.

If “No”, proceed to Section 28: Why Only Genshin?]

Section 27: Cross-Game Behaviour & Player Positioning

Q1: If yes, please select the ones you are currently playing. (Checkboxes)

- Honkai: Star Rail
- Honkai Impact 3rd
- Zenless Zone Zero
- Tears of Themis

Q2: Do you consider Genshin Impact as a secondary game? (Single choice)

- Yes
- No

Q3: Why do you also play other HoYoverse games alongside Genshin Impact?
(Checkboxes)

- The story in those games is easier to follow and interesting.
- The gameplay and combat feel different from Genshin, so it's refreshing.
- I like the art style, character designs, or music.
- The games feel more relaxing or convenient to play (daily tasks, events, QoL).
- My friends are playing them, so I joined too.
- I enjoy trying out different worlds and themes besides Genshin.
- The rewards in other HoYoverse games are more generous (e.g., 10 free pulls each version).
- Other: _____

[After completing this section, proceed to Section 29: Suggestions and Motivation to Keep Playing.]

Section 28: Why Only Genshin?

Q1: Why do you only play Genshin Impact and not other HoYoverse games?
(Checkboxes)

- I prefer Genshin's story, art style, characters, and world.
- Other HoYoverse games don't match my gameplay preferences.
- Hardware / device / platform limitations.
- I don't have enough time / energy to play multiple games.
- The gacha system or spending in other games feels too stressful.
- I've already invested a lot of time and resources in Genshin, so I don't want to start over.
- Other HoYoverse games are harder to access or not very popular where I live.
- I'm just not interested in the other games.
- Other: _____

[After completing this section, proceed to Section 29: Suggestions and Motivation to Keep Playing.]

Section 29: Suggestions and Motivation to Keep Playing

Q1: Why do you think you are able to stay motivated to continue playing Genshin Impact? (Checkboxes)

- Attracted by well-designed characters, rich storylines, and the immersive world, enjoying collecting and developing characters.
- Motivated by the desire to explore the mysteries of Teyvat and immerse themselves in the beauty of the environment and world design.
- The steady flow of new content and updates keeps me engaged and curious.
- The game inspires creativity, such as writing fan stories, designing characters, or engaging in community creations.
- Enjoying the game itself, focusing less on gacha results.
- I don't expect too much, so I can enjoy the game for a long time.
- My friends are still playing the game, so I keep playing together.
- Other: ____

Q2: What improvements would you like to see in Genshin Impact? (Checkboxes)

- Generous Rewards: More free pulls/Primogems (e.g., Anniversary rewards).
- Content Optimization: "Skip Button" for dialogue or faster daily dailies.
- Gacha Improvement: Better rates or weapon banner mechanism.
- Better Artifact System: Easier way to get good stats (Loadouts/Main stats selection).
- Provide more generous Anniversary rewards (e.g., giving out a free limited 5★ weapon).
- Give out free pulls each version (e.g., 10 Intertwined Fates per patch, similar to other HoYoverse games).
- Reduce the daily Resin consumption (e.g., from 160 to 120 or even lower).

- Other: ____

Q3: Do you have any other feedback or suggestions you would like to share with the Genshin Impact team?

(Open-ended)

Section 30: Closing Message(Active Player)

Thank you for taking the time to complete this survey!

Your feedback is valuable for understanding player behaviour and for improving the long-term game experience in Genshin Impact.

I sincerely appreciate the valuable time you have dedicated to this questionnaire.

May every Traveler continue to discover new stories and enjoy the scenery of the journey in Nod-Krai and beyond.

Please click “Submit” below to finish.

APPENDIX B : Supplementary Data and Survey Evidence

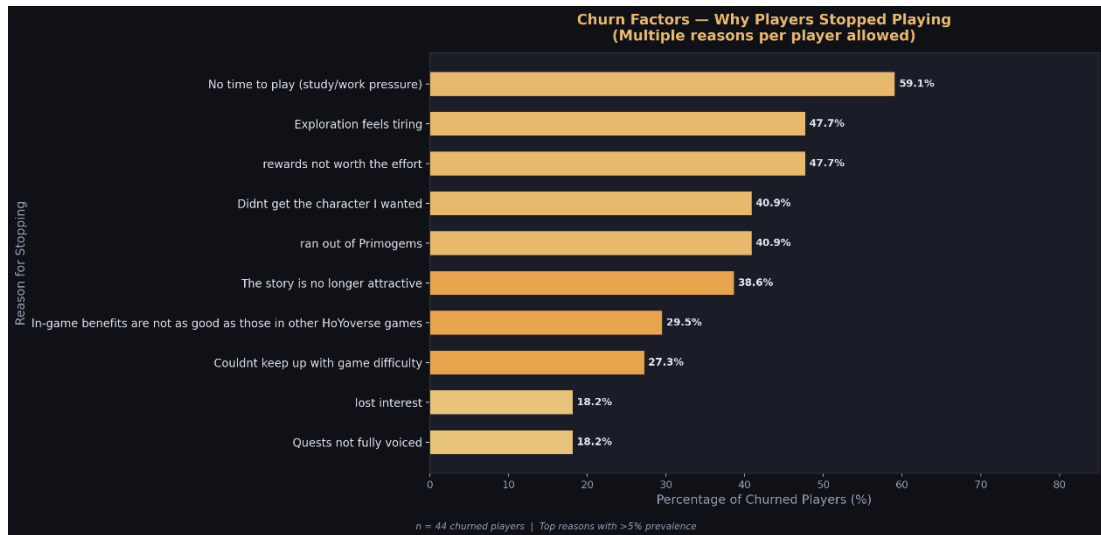


Figure B.1: Top Churn Reasons

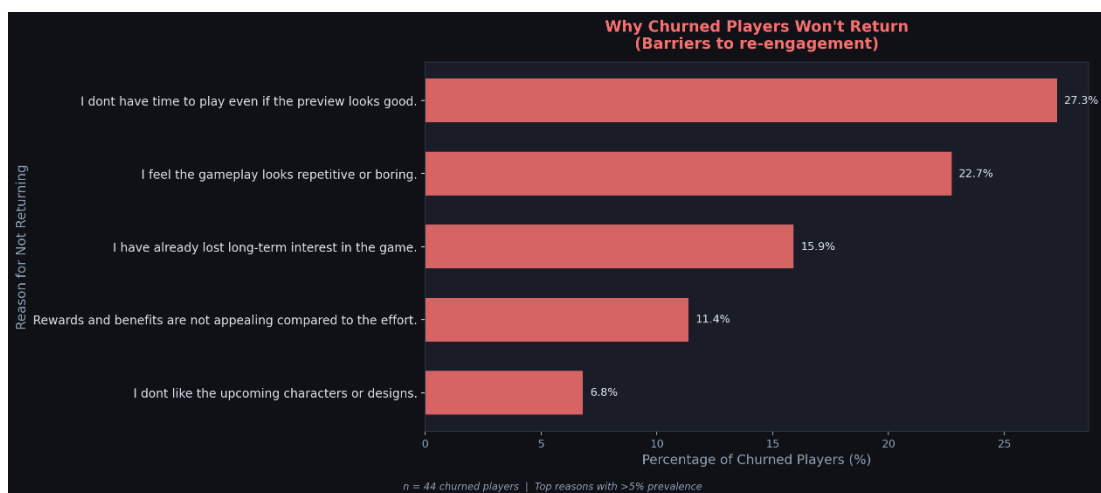


Figure B.2: Non-Return Reasons

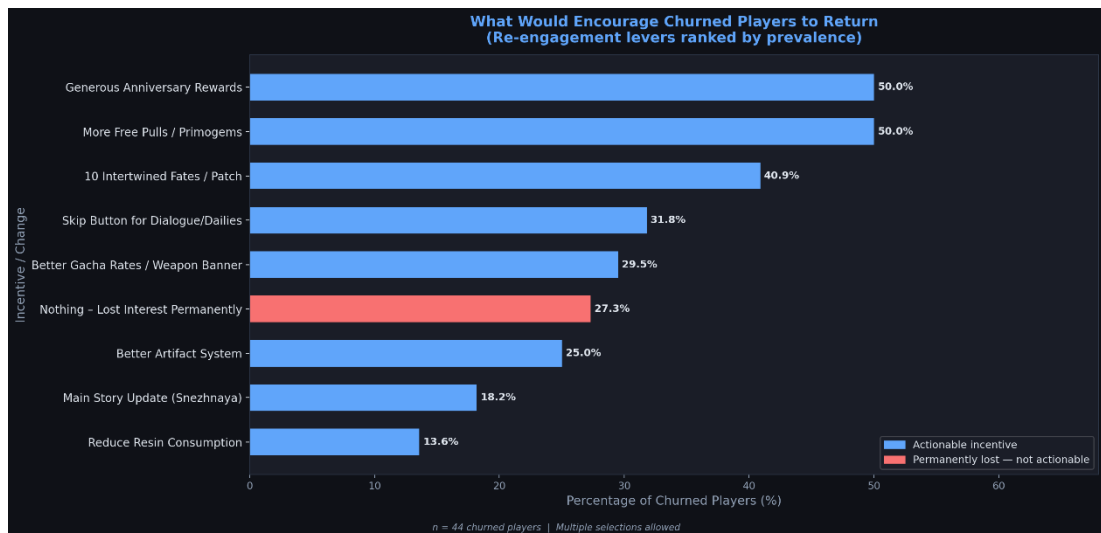


Figure B.3: Return Conditions

Support | SurveyCircle <support@surveycircle.com>
to me ▾

Fri, Feb 27, 3:23 PM ☆ ☺ ↶ ⋮

Hi Kelvin,

We've just posted your study on Twitter/X (with a social media image, of course). Have a look: <https://twitter.com/surveycircle/status/2027283388210520314>

If you want, you can easily increase the reach of the tweet by liking and retweeting it.

Another idea regarding reach: We can also post your study as a story on our Instagram account (<https://www.instagram.com/surveycircle>). If you'd like us to do this, simply post one of our social media images for your study (see my previous email) in your story and tag us @surveycircle. It would be good if your account is already set to public (or if you temporarily set it to public). We'll then repost your story for free in our story and also add it to our highlight as soon as possible.

Best wishes,

Dennis

PS: One more thing about Instagram: If your profile is set to private, you can also send us a DM asking us to follow you. We'll then send you a request and as soon as you accept it, you tag us and we can repost.

Figure B.4: SurveyCircle Distribution Confirmation Email

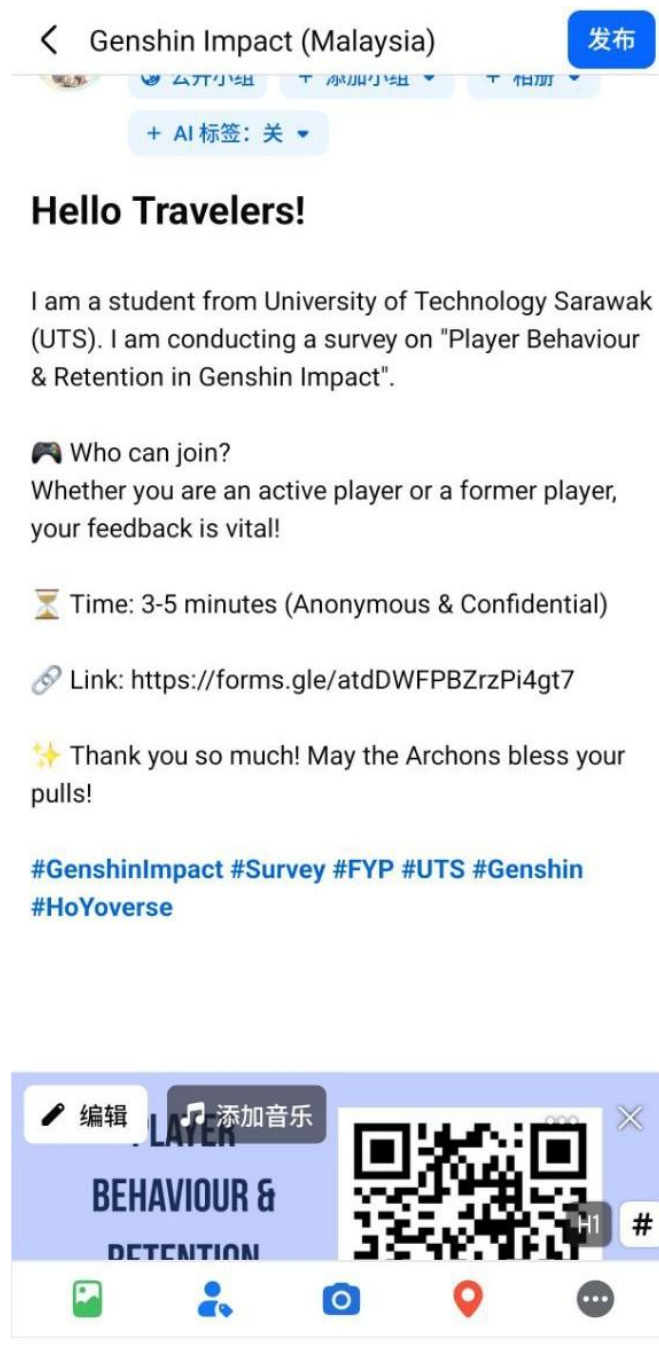


Figure B.5: Survey Post on Facebook — Genshin Impact (Malaysia) Group